

wagner
magnete

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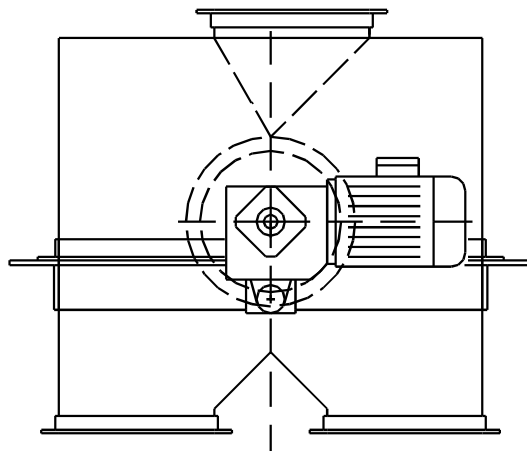
B (0)445e

03.04

Operating instructions

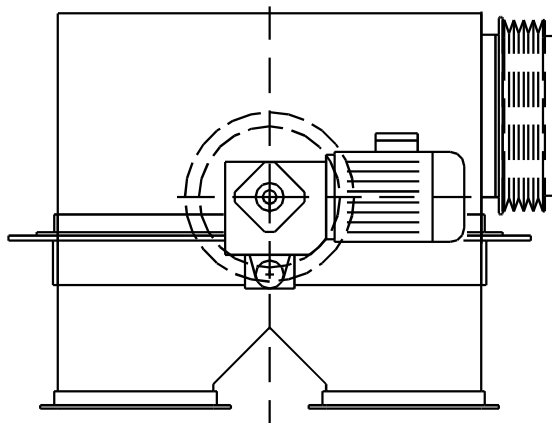
Magnetic Machine

Type 0446-30/70 K G2:380 S T DN150, No. 516924



Structural Shape A

Feeding of stock from above



Structural Shape B

Lateral feeding of stock

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1. Safety

All persons involved with the erection, startup, operation and maintenance of the machinery are strictly advised to read and adhere to the following instructions very carefully.

Please take note of all items in the operating instructions prior to startup.

1.1 Safety instructions

Please note that this WAGNER magnetic machine was subject to a safety test.

However, the following dangers are possible in the event of faulty operation or misuse:

- Danger to adjacent machinery or other property of the proprietor
- Risk of inefficient operation
- Danger to the operation personnel

Legend:

Danger:

Below mentioned instructions must be followed in order to prevent the user from damage.



Danger

Warning:

Please stick to the following instructions in order to prevent the device from being damaged.

**Note:**

This symbol marks important notes for the proper use of the device. If the notes are ignored, the machine can be damaged.



1.2 Operating personnel

The proprietor of the machine is advised to make the operating instructions accessible to the operator and to make absolutely sure that they are read and understood completely. Only then can the device be taken into operation.

The operator must make sure that authorized persons work on the equipment only. He takes the responsibility with regard to third persons within the operating area of the equipment.

The operator is obliged to ensure that only persons work at the machine who:

- know the basic industrial safety precautions and accident prevention precautions and who were introduced to the handling of the machine.
- read the chapter about safety precautions and warnings of this manual and who understood and accepted it by signature.

The magnetic machine emits magnetic fields. In accordance with today's level of technology, it is possible that they can have an effect on persons with cardiac pacemakers, for example. Unfortunately, no clear medical statements concerning this matter are known up until now.



Warning!

When working within the magnet's sphere of action (see chapter 5), magnetizable parts, as for example tools, are attracted with intense force. Risk of bruises!



Danger

Storage media working on a magnetic basis like cheque cards and credit cards, hard disks, video tapes etc. can be impaired by the magnetic field's impact. Therefore there should be a distance of at least 2 m between such items and the separator.



1.3 Stipulated use

Magnetic machines are used for the automatic separation of ferromagnetic materials from a feed current.

The dimensions of the smallest detectable metal particle depend on the following factors:

- The extracted particles, themselves:
 - Material
 - Substance of particles
 - Shape of particles
 - Position of particles in feed current
- The given feed current
 - Composition of feed current (which materials, dry or moist material, etc.)
 - Grain size of all monitored materials
 - Layer height of feed current
 - Apparent density of material
 - Flotation of material
- Speed of feed current

We point out that this machine is not capable of separating plastics, wood, glass, nonferrous metals, nonferrous alloys or any other non-ferromagnetic materials!

For safety reasons, any conversions or modifications are only permitted on the express consent of the manufacturers!

Otherwise, the operability of the machine can be jeopardized!



Danger

1.4 Safety precautions at site of installation

The supporting construction for this magnetic machine must be dimensioned for static and dynamic loading so that absolute stability is guaranteed with the adequate safety. Please take into consideration that the machine generates vibrations during operation. Take all necessary precautions in order to make sure that the machine is reliably and permanently fixed in place!

Make sure that no persons are located within the material exit zone when the machine is in operation.

Within the magnet's sphere of action (see chapter 4) ferromagnetic material must not be used. Danger of pinching

Assembly and maintenance works must only be carried out when the machine stands still.

The proprietor of the drum must arrange for the appropriate protection of assembly and maintenance personnel (e.g. erection platform in the event of an elevated erection site, in order to prevent the danger of falling).

The proprietor of the entire plant is responsible for an emergency switch into which the magnetic drum (including drive) is integrated!

In case of damage of the bearing or the motor they can heat up considerably. Direct contact between user and heated parts is partly avoided by the casing of the device. Maintenance works should only be carried out after the device has cooled down.

2. Technical data

Magnetic Machine

Type:	0446-30/70 K G2:380 S T DN150
Structural shape.:	B
Serial no.	516924
Continuous duty:	100 %
Weight:	422 kg
Max. ambient temperature:	+45 °C

Magnetic system

(for electric drums only)

Type	--
Protective system:	IP--
Voltage:	-- V DC
Current consumption: (cold)	-- A
Current consumption: (warm)	-- A
Power consumption: (cold)	-- kW
Power consumption: (warm)	-- kW

Motor data

Type:	SA47DT80K4
Make:	SEW
Connection voltage:	380 V
Frequency:	50 Hz
Consumption:	0,37 kW
Protective system:	IP 55
Drive speed:	36 r.p.m.

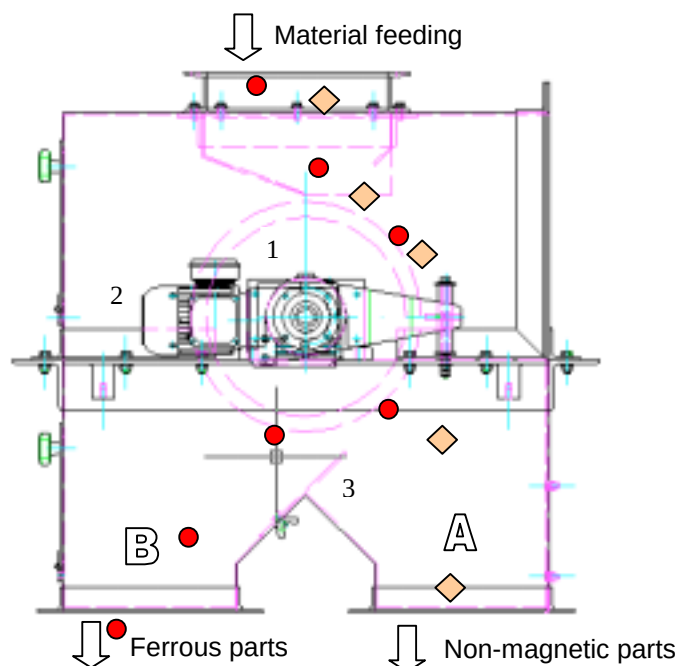
Special accessories

- ☐ Vibrating chute type:--
- ☒ Overspeed monitor type: Bi5-G18-Y1x

3. Product description

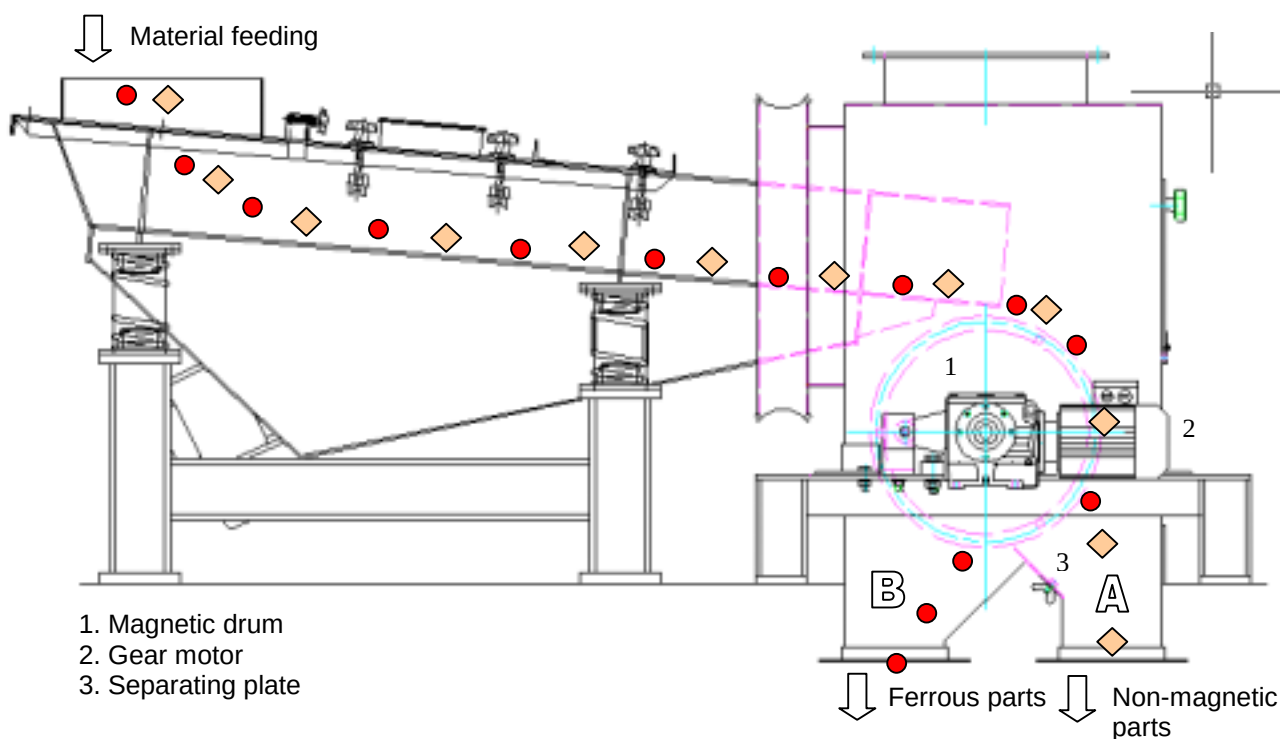
Magnetic machines are compact units. They are mainly used for processing dry bulk materials. There are two types available:

Type A (without vibrating chute): material is fed from the top



The bulk material is fed onto the magnet drum. Non-magnetic parts fall into hopper A. Magnetic parts are attracted by the magnetic field in the drum and ejected into hopper B at the end of the magnetic zone.

Type B (with vibrating chute): material is fed from the side



4. Transport and Storage

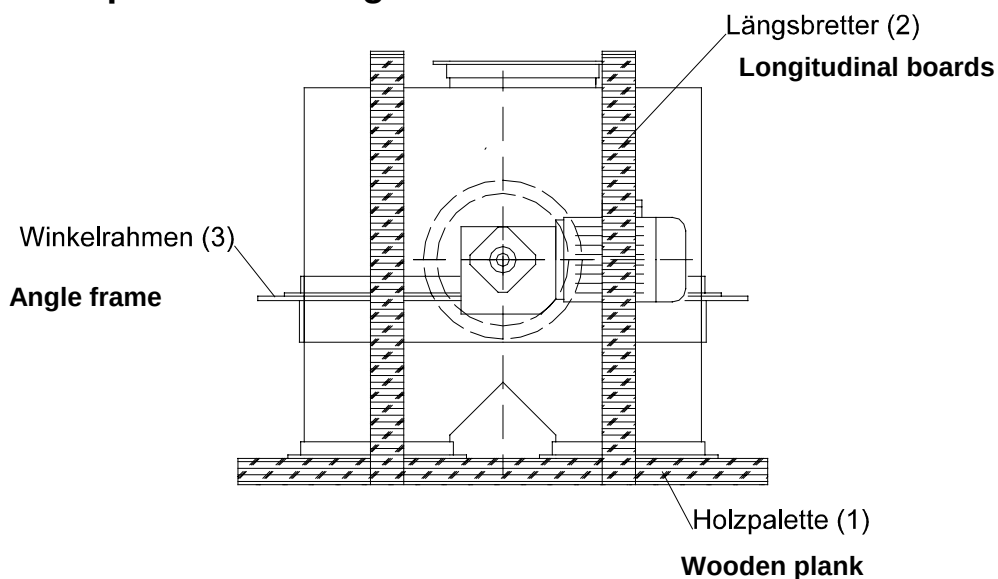


Fig. 1

Please take note of the total weight of the magnetic machine when selecting your means of transport (see chapter 2, „Technical data“).



The machine is fixed to a wooden plank (1) by means of tightening straps. Pick up the whole device with your fork lift truck below the plank.

Storage:

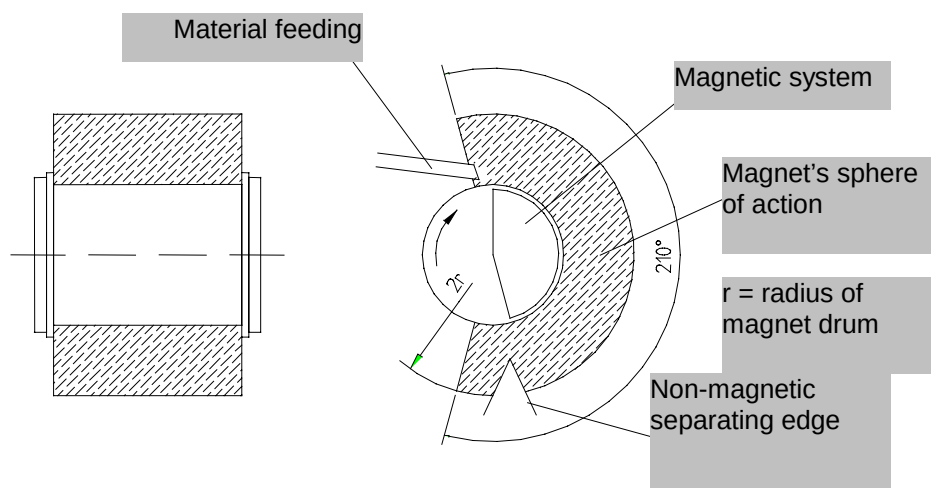
Please observe the guidelines given in chapter 2 “Technical data”. Protect the machine from humidity when storing the machine. Warning: Do not store outdoors!



5. Assembly

5.1 Precautions at erection site

- Make sure that erection and maintenance personnel have free access to the device and that they can work on it safely.
- When selecting the erection site please note that lifting gears require access to the machine.
- Do not use magnetizable construction parts within the operating area of the magnetic separator!



It does not suffice to cover up construction parts with non-magnetic materials. Any parts within the magnetic field will be magnetized and will attract other ferrous parts accordingly.
The deposited material can lead to interferences in the feed current.



5.2 Assembly of the machine

Release the tightening straps and remove the longitudinal boards (2). Take the machine with appropriate hoists from the wooden plank.

Fix the machine at the angle frame (3). Usually, borings for holding down bolts are not made into the frame before delivery. Please see to it that appropriate borings are made on the spot.

Attention:

Do not burden the drum casing of the magnetic system during transport or assembly because otherwise the device can be damaged.



Attention:

Both bearings require enough clearance at the side. They should not be braced with the frame!



Caution:

Permanent magnetic drums will attract normal tools such as wrenches or hammers. Risk of injury! The same applies to electro-magnets, which are activated.



Danger

5.3 Connection of electric system

Caution:

The electric connection must be carried out according to the electro documentation by skilled and instructed electro experts (in the sense of VDE 1000 part 10). Furthermore, the regulations of standard VDE 0100 must be followed.



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Connection of drive motor:

(Please refer to operating instructions for motors in the appendix)

Connection of magnetic system:

(only for series 421 - electro drums)

Prior to connection, please pay attention to the nominal values specified on the output plate!
The electromagnet must be supplied with direct-current.



The output voltage of the current supply must correspond with the nominal voltage of the magnet! Please take into consideration that the maximum current-carrying capacity of your current supply is at least as big as the current consumption of the electromagnet in cold condition! (See chapter 2, „Technical data“).

Open the lid of the contact box for a direct connection. Make sure that the connection wire is long enough since the re-adjustment of the magnetic system is necessary during the startup and this can affect the wire length.

Wagner electromagnetic drums of the series 421-...S-0 are supplied with integrated rectifier including fuses. They can be connected to the network directly.

6. Startup

Pay attention that nobody is standing within the danger zone of the machine!

Within the sphere of action ferromagnetic material must not be used. Danger of pinching!

Setting the magnetic system of the magnet drum inside the machine:



Danger



Danger

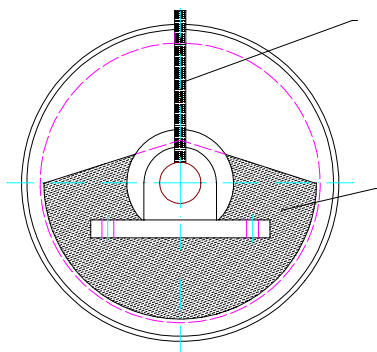


Fig. 2

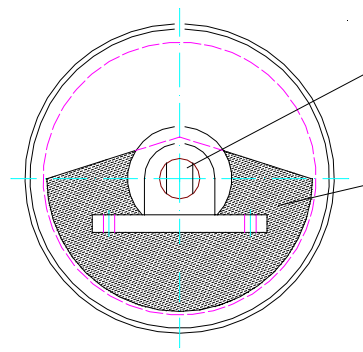


Fig. 3

Wagner magnetic drums with a diameter of up to 400 mm are supplied with a fixed adjusting lever according to fig. 1. Larger drums have a wrench surface on one side of the shaft. In the latter case, either a fork wrench or a separate adjusting lever are supplied in addition.

Before you can move the magnetic system inside the drum from transport position to working position it is necessary to remove the clamping devices attached.

For drums up to a diameter of 400 mm the magnetic system is, by default, fixed to plunger block using clamping screws. For drums with diameters from 500 mm, clamping inserts are used instead of clamping screws (see the manufacturer's work instructions attached – observe the torques required).

The drum is fixed from the side opposite the drive side..

Warning:

Please observe the weight of the magnetic system installed. If larger drums are used, it is essential to use appropriate hoisting gears before opening the clamping devices. The magnetic system turns back into transport position if it is not adjusted correspondingly.



Danger

After loosening the clamping devices you may now adjust the magnetic system using the adjusting lever or fork wrench provided.

Position the magnetic system of your drum according to fig. 4. The position of the magnetic system can vary from the position as shown in fig. 4. Depending on the material to be separated, the position needs to be determined on site. The separating quality depends on the settings made here.

The feeding of the material onto the drum case should take place slightly staggered behind the upper dead centre of the drum.

The magnetic system cannot be identified on the outside of the drum.

The best way of determining the exact position of the magnetically active zone is by means of a small metal object.

In the event that our works have specified a certain direction of rotation for the drum, please feed the material and set the system in accordance with this direction.

Adjusting the separating edge: Depending on the material to be separated, the position needs to be determined on site. The separating quality depends on the settings made here. For the magnetic machine there are two ways of adjusting the separating edge: the inclined adjustment or the horizontal adjustment. In both cases, loosen the screws slightly, shift the separating edge, and re-tighten the screws.

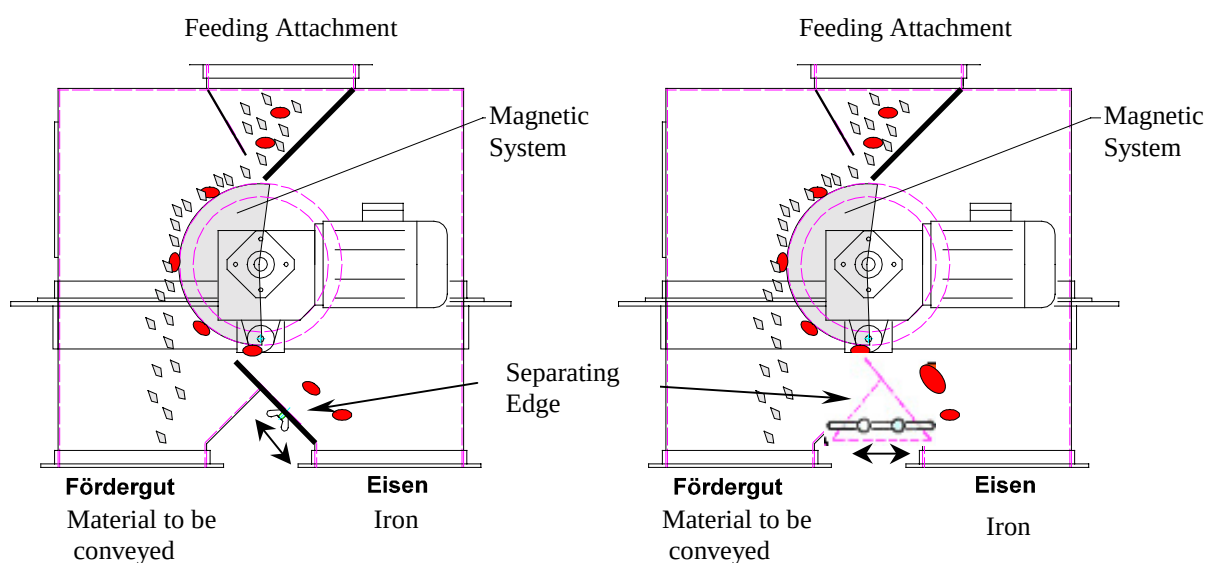


Fig. 4

7. Maintenance

Maintenance works must only be carried out when the machine stands still.

Please stick to the instructions of the maintenance plan. Warranty claims and liability claims personal injury and damage to property are excluded if they are ignored.

Please regularly verify the condition of the drum casing. In the event of any damage you should exchange it on time, otherwise the integrated coil and magnetic systems can be damaged.



Danger



7.1 Maintenance schedule

POINT OF ACTION	WHEN	DESCRIPTION
Drive motor		See manufacturer's operating instructions attached
Condition of bearings (This is particularly necessary if magnetic tapes or comparable goods are within the material to be conveyed)	Daily	Make a visual check If necessary remove wrapped-up tapes and fibres
Condition of magnet	Daily	Check and clean magnet. Remove any impurities and deposits at regular intervals. To do this, turn off the machine, open inspection window, and clean the magnet. If the magnet is soiled heavily, dismount the cover.
Lubricate the bearings on the magnetic drum	After 1500 working hours during normal operation	You can use grease Mobil Mobilux EP2 Aral Aralup HL2, BP Energrease LS2 or a comparable one. Proceeding: Bearing with lubricating nipple: Clean force-feed lubrication nipple insert approx. 2,5 g fat (approx. 2 liftings) by means of a grease gun.

7.2 Checking the bearings

All bearings must be checked at regular intervals. Prevent dirt, impurities and moisture from penetrating the bearings to avoid premature failures. When lubricating the bearing there is always the risk of impurities penetrating the bearing interior. Always ensure that the lubricant reservoir and lubricating devices are clean and that the lubricant is not soiled when decanting it. Always clean the lubricating nozzles before lubricating the bearings!

Beware of excessive lubrication! If too much lubricant is used, this may lead to damage inside the bearing. If excessive lubricant cannot escape, high temperatures can be generated which can damage or destroy the functioning of the lubricant.

Always observe the time intervals for smaller lubrication processes (see maintenance schedule). Beware of excessive lubrication since this may lead to high temperatures. If the housing is almost filled after several lubrication actions, the lubricant needs to be exchanged.



8. Trouble-shooting

Problem	How to proceed
No magnetic power (for electromagnets only)	• Check switch setting and supply voltage at rectifier. The rectifier on the models 420...S-0 is located in the connection box at the drum. • Check fuses at rectifier. • Check feed cable to magnet. • Measure voltage and current consumption directly at the magnet connection. Compare the values with the specification on the output plate. If the current differs distinctly, the coil is damaged. Please contact our works. • Check whether the drum casing is damaged. This can be the cause of the damage to the magnet or coil system. • Check insulation resistance to earth. ($R > 1 \text{ M}\Omega$)
Too little separation	• Feeding speed of bulk material is too high. • Quantity of bulk material is too high. • Piece size of bulk material is too large. • CD current supply is too weak. Cross section of feed wires may be too small (see VDE 0100) • Spherical, small or weak magnetic ferrous parts are located amongst the bulk material. • Compare operating width of the magnetic drum with the width of the feed material. • Check position of magnetic drum (see chapter 4.2)
Drum casing is blocked	• Check whether roller bearings run smoothly. • Check whether drum casing is damaged.



Konformitätserklärung

(Declaration of conformity)

im Sinne der EG-Richtlinie Maschinen 98/37/EG, Anhang II A

(within the meaning of the EG-stipulation Maschinen 98/37/EG, Appendix II A)

Wir, die Firma

(We, the company)
(address)

WAGNER Magnete GmbH & Co. KG
Spann- und Umwelttechnik
Obere Straße 15
D-87751 Heimertingen

erklären, daß die Bauart von

(declare, that the type of construction)

Bezeichnung: (name)	Magnetic-Machine
Type: (type)	0446-30/70 K G2:380 S T DN150
Seriennummer: (serial number)	516924

in der gelieferten Ausführung folgenden einschlägigen Bestimmungen entspricht:

(is manufactured in the delivered design according to the following appropriate regulations)

EG-Richtlinie Maschinen 98/37/EG, Anhang I
(Machinery Directive 98/37/EG, Appendix I)
EG-Niederspannungsrichtlinie 73/23/EWG
(Low voltage Directive 73/23/EWG)
EMV-Richtlinie 89/336/EWG i.d.F. 92/31/EWG
(EMC-Directive 98/336/EWG based on 92/31/EWG)

Angewendete harmonisierte Normen, insbesondere

(Applicable harmonized standards, particularly)

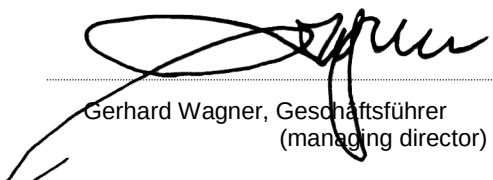
DIN EN 60204.1	Elektrische Ausrüstung für Industriemaschinen (Safety of machinery. Electrical Equipment of machines)
EN 50081-2	Fachgrundnorm Störaussendung (Generic Emission Standard (Emission))
EN 50082-2	Fachgrundnorm Störfestigkeit (Immunity)

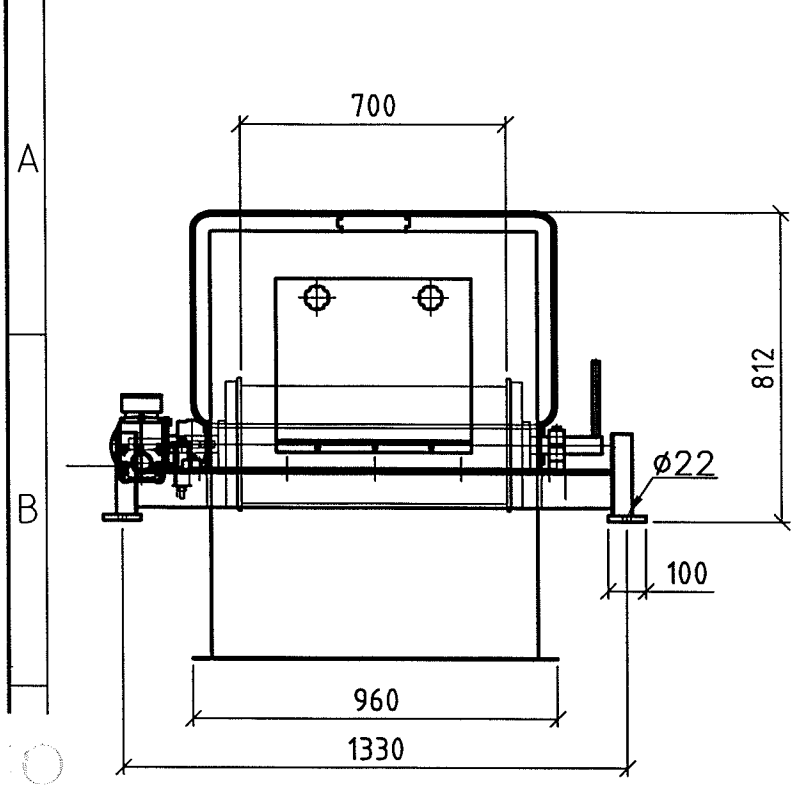
Angewendete nationale Normen und technische Spezifikationen, insbesondere

(Applicable national standards and technical specifications, particularly)

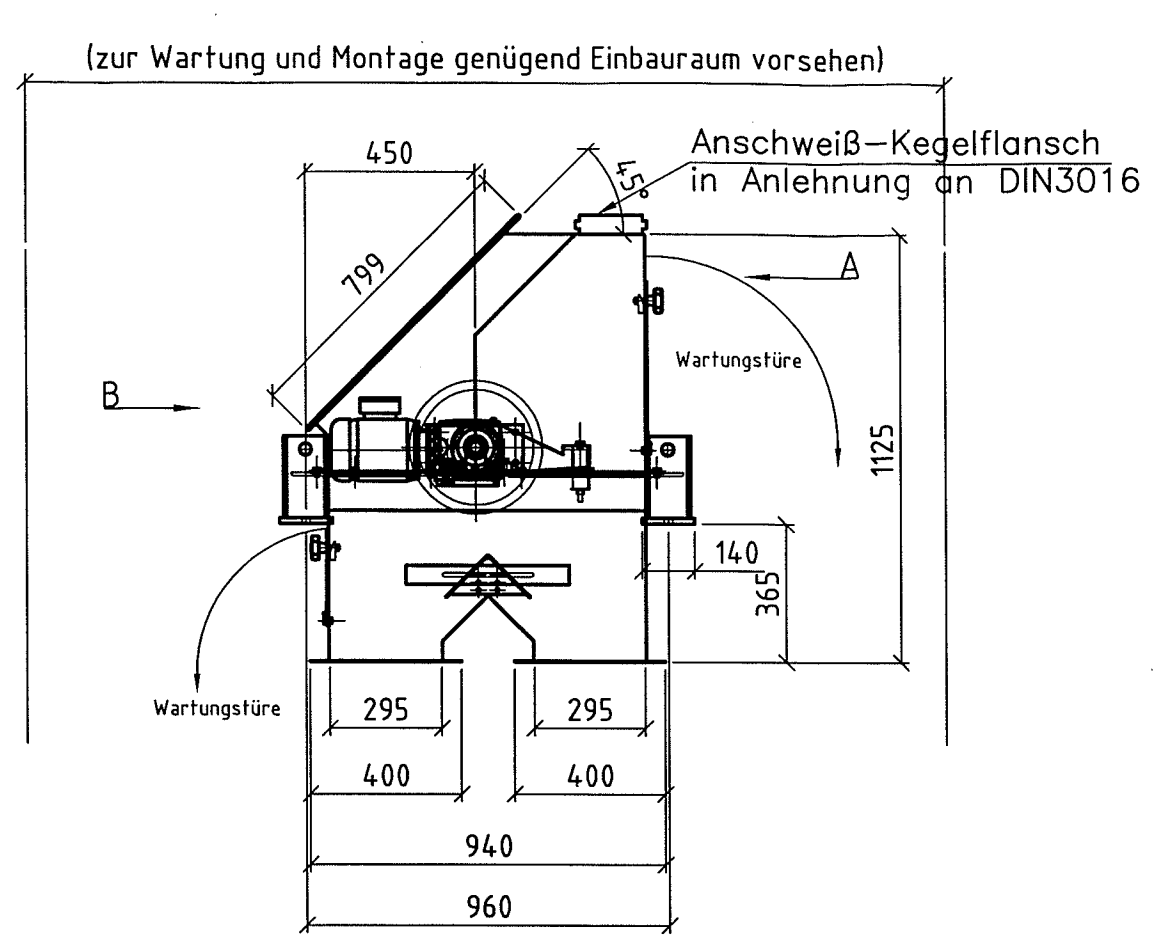
VDE 0580	Elektromagnetische Geräte (Electromagnetic devices)
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Heimertingen, den 06.11.02
(Place and date of issue)

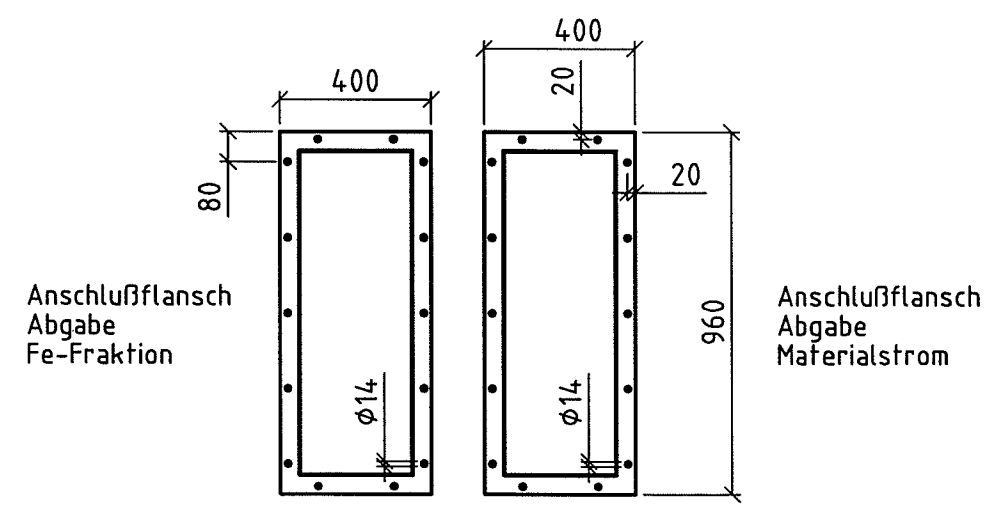
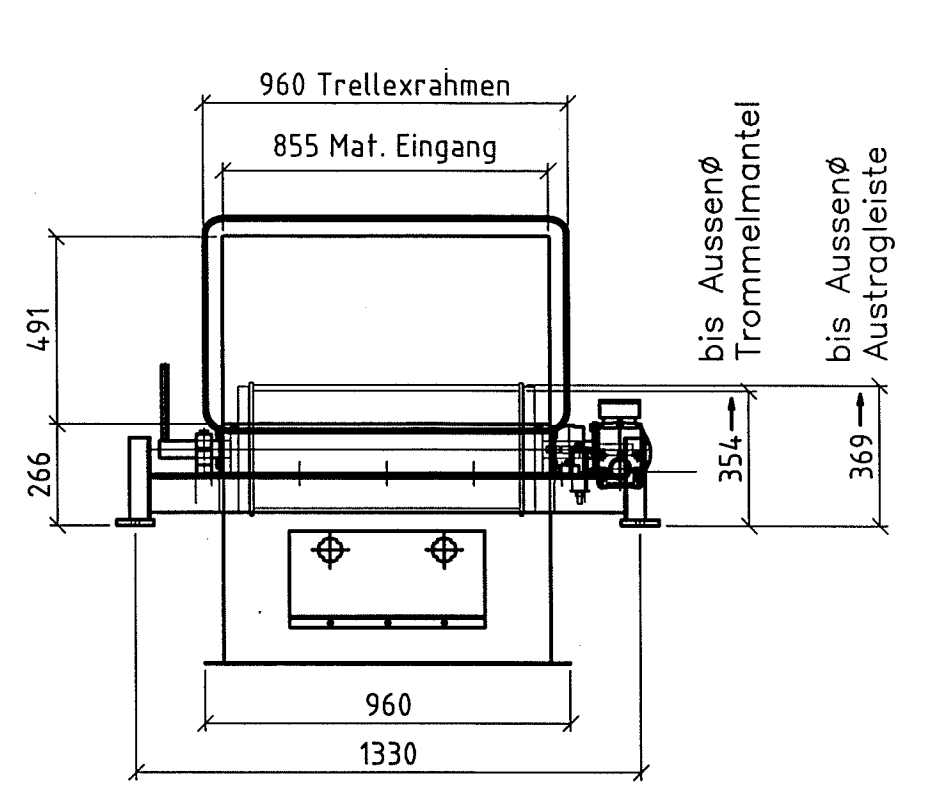

Gerhard Wagner, Geschäftsführer
(managing director)



Ansicht A



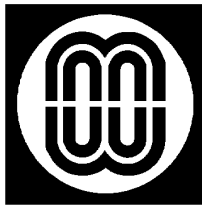
Ansicht B



Projekt:
07316.4352 Narayangonj Clinker (Bangla Desh)

Konstruktionsbedingte
Änderungen vorbehalten

 wagner magnete		DIN7168 fein mittel grob		Oberfläche	Maßstab 1:20	Position -	Menge -
					-		
					-		
				Datum	Name	Magnettrommel-Scheider 0446(N)-30/70 K S DN150 staubdicht	
			Bearb.	05.11.2004	Frommel		
			Gepr.				
			Norm				
						ZK4-1805	
Zust.	Änderungen	Datum	Name	Dateiname	Falle Projekt Losse der Wagners / 04.05.2009	Blatt - 81	



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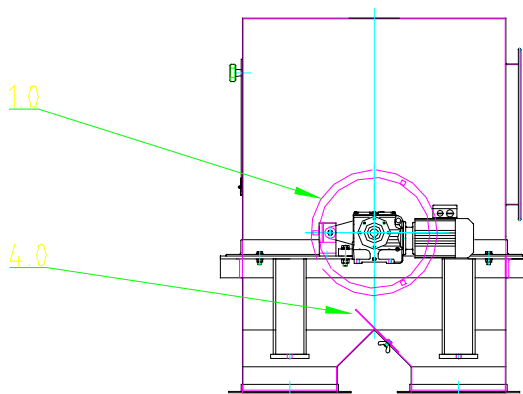
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Ersatzteilliste Magnet-Maschine

List of spare parts magnetic machine

Liste des pièces de rechange pour une machine magnétique

Type **0446-30/70 K G2:380 S T DN150** Nr./No. **516924**

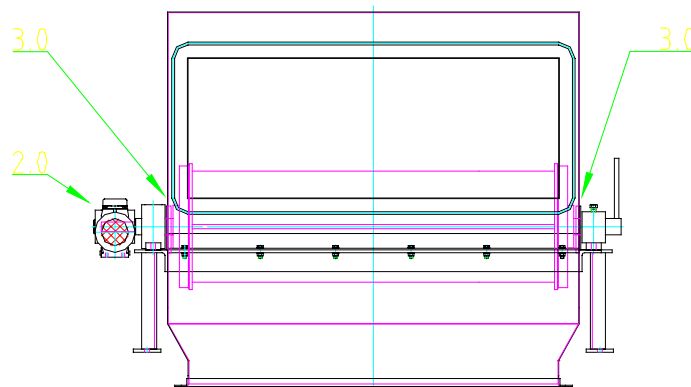


1.0 Magnettrommel
vgl. LE (0)420

Magnetic drum
Tambour magnétique

2.0 Getriebemotor

Gear motor/
Moto-réducteur

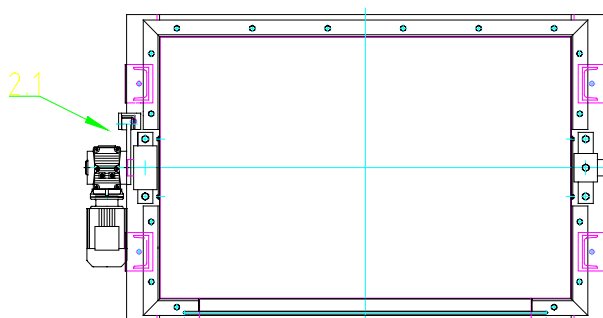


2.1 Drehmomentstütze

Turning moment
support/
Support de moment
d'un couple

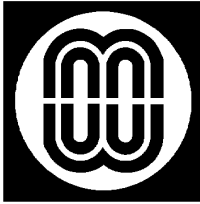
3.0 Druckplatte und
Dichtgummi

pressure plate and
sealing rubber /
Plaque de pression et
joint en caoutchouc



4.0 Trennblech

Partition plate /
Tôle de séparation



wagner
magnete

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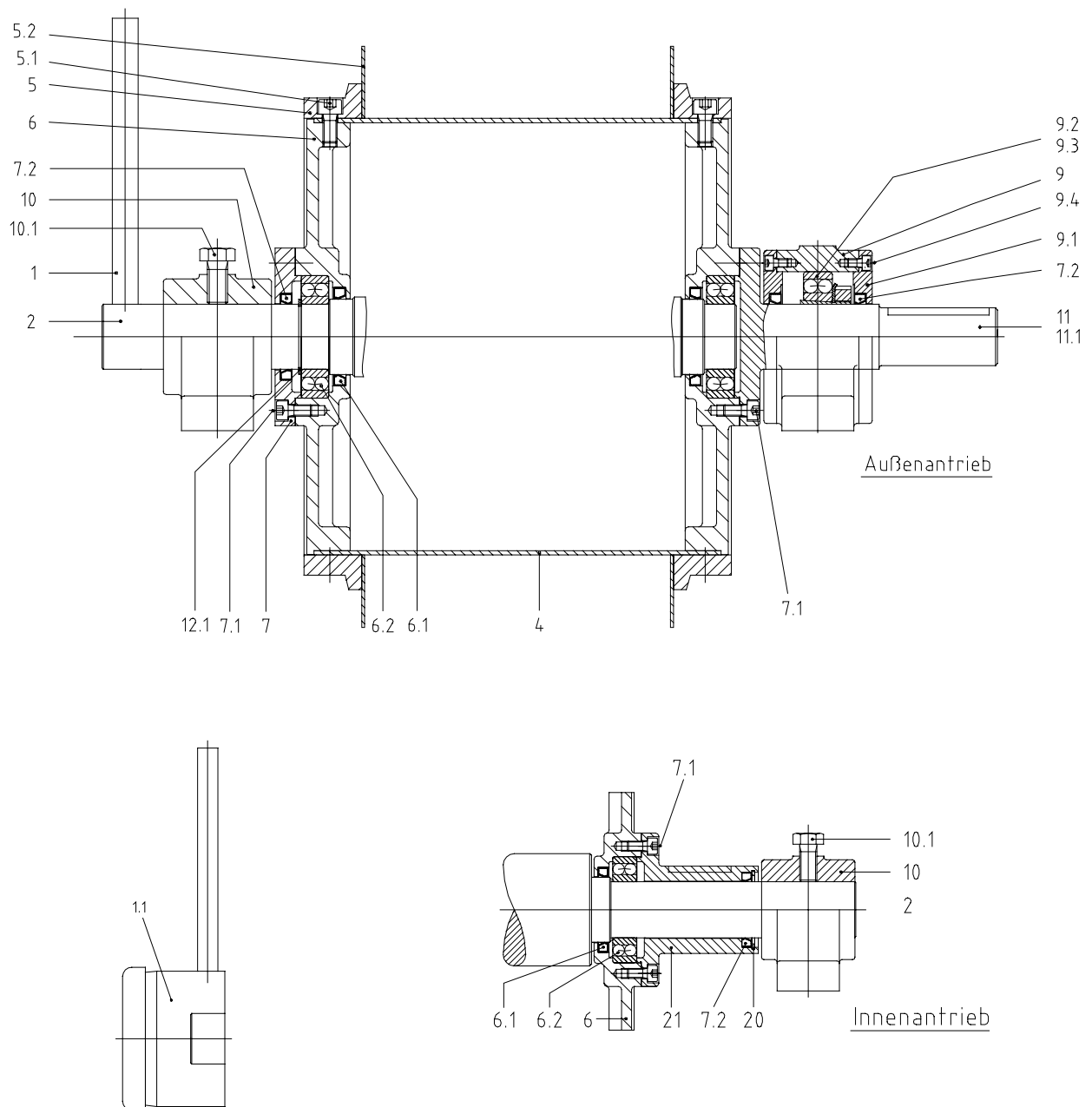
Ersatzteilliste für Trommeln

Spare part list for drums
Liste des pièces de rechange pour tambours

Type 0421-30/70-3 A 3AL 15x15

Nr./No. 516925

Übersichtszeichnung / drawing / Dessin d' ensemble



Ersatzteilstückliste Trommeln (0)420

Spare part list for drums / Liste des pièces de rechange pur tambours

Type 0421-30/70-3 A 3AL 15x15**Nr./No. 516925**

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 E-Mail info@wagner-magnete.de

Pos.	Bezeichnung	Description	Désignation	Stück/pcs/Quantité	V / E
1	Stellhebel	Operating handle	Levier de commande	1	E
1.1	Anschlußdose (bei Elektroausführung)	Junction box	Boîte de commande	1	E
4	Trommelmantel	Drum casing	Enveloppe de tambour	1	V
5	Trommelring	Drum ring	Anneau de tambour	2	E
5.1	Schraube	Screw	Vis	12	E
5.2	Materialbegrenzungsring (Option)	Material limiting ring (option)	Bague limiteuse de matière (option)	2	E
6	Seitenschild (Stellhebelseite)	Side shield (Operating handle side)	Lame latéral (côté de levier de commande)	1	E
6	Seitenschild (Antriebsseite)	side shield (drive side)	Lame latéral (côté d'entraînement)	1	E
6.1	Wellendichtring innen	Shaft sealing ring – inside	Bague à lèvres avec ressort – à l'intérieur	2	E
6.2	Kugellager	Ball bearing	Roulement à billes	2	E
7	Lagerdeckel (Stellhebelseite)	Bearing cover (operating handle side)	Chapeau de palier (côté de levier de commande)	1	E
7.1	Schraube	Screw	Vis	8	E
7.2	Wellendichtring außen	Radial seal ring – outside	Bague à lèvres avec ressort – à l'extérieur	3	E
9	Stehlagergehäuse	Pillow block housing	Chaise pour paliers à roulement	1	E
9.1	Stehlagergehäusedeckel	Pillow block housing cover	Chapeau de chaise pour paliers à roulement	2	E
9.2	Kugellager	Ball bearing	Roulement à billes	2	E
9.3	Spannhülse	Locking sleeve	Manchon	2	E
9.4	Schraube	Screw	Vis	1	E
10	Festlager	Fixed bearing	Palier fixé	2	E
10.1	Schraube	Screw	Vis	1	E
11	Antriebswelle	Drive shaft	Arbre menant	1	E
12.1	Sicherungsring	Locking ring	Anneau d'étanchéité	1	E

E = Ersatzteil / spare part / Elément de réserve

V = Verschleißteil / expendable part / Partie résistante à l'usure

Bei Ersatzteil-Bestellung sind Nummer und Type der Trommel anzugeben. With orders for spare parts both number and the type of drum is to be given.

Lors de la commande de pièces de rechange, nous vous prions d'indiquer le numéro et le type du tambour



SEW
EURODRIVE

Gear Units

R..7, F..7, K..7, S..7 Series, Spiroplan® W

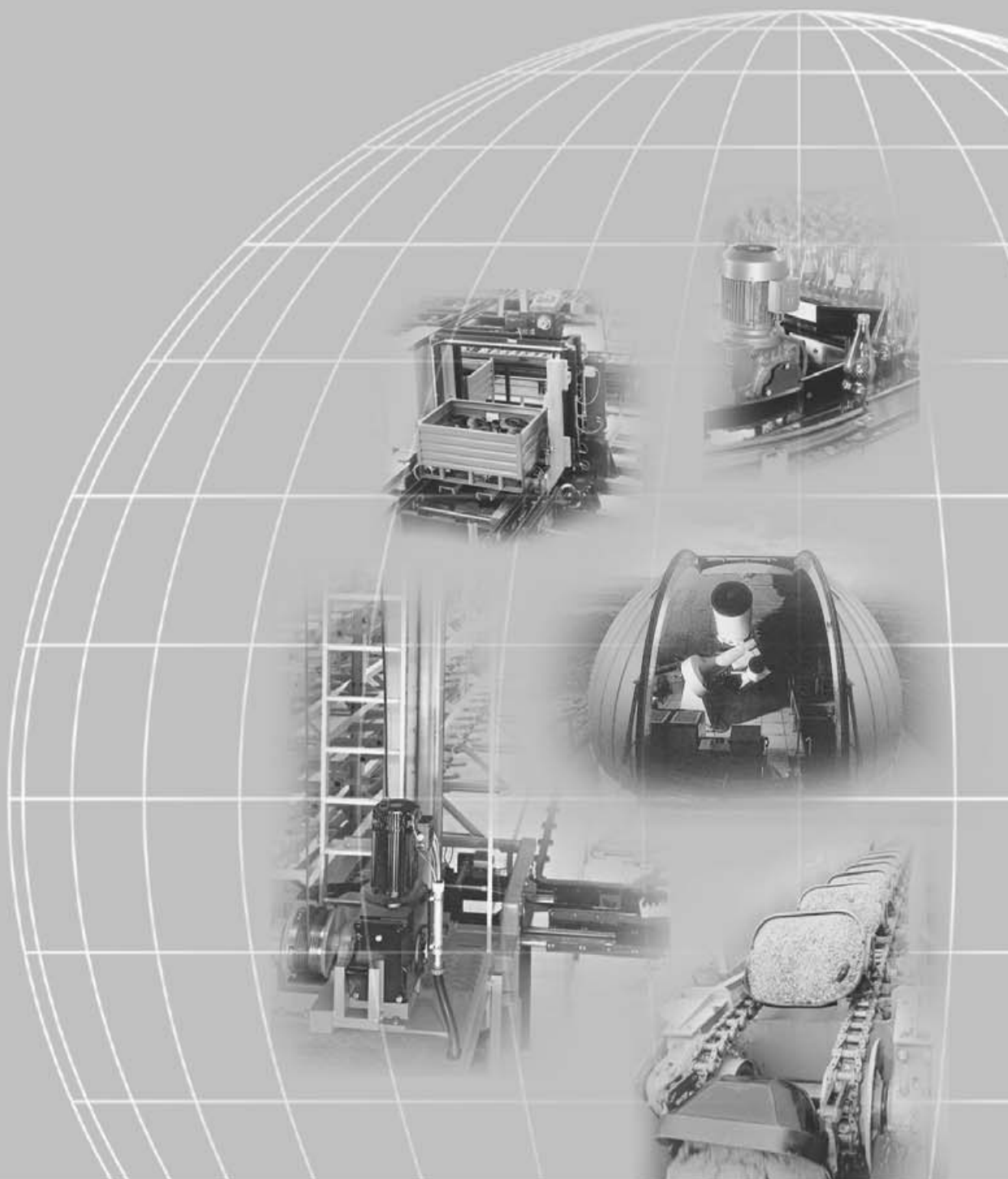
Edition

07/2002



Operating Instructions

1055 2715 / EN



SEW-EURODRIVE





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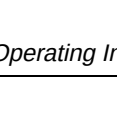


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1 Important Notes

Safety and warning instructions

Always follow the safety and warning instructions contained in this publication!



Electrical hazard

Possible consequences: Severe or fatal injuries.



Hazard

Possible consequences: Severe or fatal injuries.



Hazardous situation

Possible consequences: Slight or minor injuries.



Harmful situation

Possible consequences: Damage to the drive and the environment.



Tips and useful information.



A requirement of fault-free operation and fulfillment of any rights to claim under guarantee is that you adhere to the information in the operating instructions. Consequently, read the operating instructions before you start operating the gear unit!

The operating instructions contain important information about servicing; as a result, they should be kept in the vicinity of the gear unit.



- **Adjust the lubricant fill volumes and position of the breather valve accordingly in case you change the mounting position (see Sec. 'Lubricants' and 'Mounting Positions').**
- **Please follow the instructions in Sec. 'Mechanical Installation' / 'Installing the gear unit'!**

Waste disposal



Please follow the current instructions:

- Dispose of housing parts, gears, shafts and anti-friction bearings of the gear units as scrap steel. The same applies to gray cast iron castings unless there are separate collection arrangements.
- Some worm gears are made of non-ferrous metals and must be disposed of accordingly.
- Collect waste oil and dispose of it correctly.



2 Safety Notes

Preliminary remarks

The following safety notes are principally concerned with the use of gear units. If using **geared motors**, please also refer to the safety notes for motors in the corresponding operating instructions.

Please also take account of the supplementary safety notes in the individual sections of these operating instructions.

General information

During and after operation, geared motors, gear units and motors have live and moving parts and their surfaces may be hot.

All work related to transport, putting into storage, setting up/mounting, connection, startup, maintenance and repair should only be performed by trained personnel observing

- the corresponding detailed operating instructions booklet(s) and wiring diagrams,
- the warning and safety signs on the gear unit/geared motor,
- the specific regulations and requirements for the system and
- national/regional regulations governing safety and the prevention of accidents.

Severe injuries and damage to property may result from

- incorrect use,
- incorrect installation or operation,
- removal of required protective covers or the housing when this is not permitted.

Designated use

These geared motors/gear units are intended for industrial systems. They correspond to the applicable standards and regulations.

The technical data and the information about permitted conditions are to be found on the nameplate and in the documentation.

It is essential that you observe all specified information!

Transportation

Inspect the delivery for any damage in transit as soon as you receive the delivery. Inform the shipping company immediately. It may be necessary to preclude startup.

Tighten screwed in transport lugs firmly. They are only designed for the weight of the geared motor/gear unit; do not attach any additional loads.

The installed lifting eyebolts are in accordance with DIN 580. The loads and regulations specified in that document must always be observed. If the geared motor is equipped with two suspension eye lugs or lifting eyebolts, then both of the suspension eye lugs should be used for transport. In this case, the tension force vector of the slings must not exceed a 45° angle in accordance with DIN 580.

Use suitable, sufficiently rated handling equipment if necessary. Remove any transportation fixtures prior to startup.



Extended storage of gear units

Gear units of the 'extended storage' type have

- an oil fill suitable for the mounting position so the unit is ready to run (mineral oil CLP and synthetic oil CLPHC). You must still check the oil level before startup (see Sec. 'Inspection and Maintenance' / 'Inspection and maintenance').
- a higher oil level in some cases (synthetic oil CLP PG). Correct the oil level before startup (see Sec. 'Inspection and Maintenance' / 'Inspection and maintenance').

Comply with the storage conditions specified in the following table for extended storage:

Climate zone	Packaging ¹⁾	Storage location	Storage time
Temperate (Europe, USA, Canada, China and Russia, excluding tropical zones)	Packed in containers, with desiccant and moisture indicator sealed in the plastic film.	With roof, protected against rain and snow, no shock loads.	Max. 3 years with regular checks on the packaging and moisture indicator (rel. atmospheric humidity < 50 %).
	Open	With roof, enclosed, at constant temperature and atmospheric humidity (5 °C < ϑ < 60 °C, < 50 % relative atmospheric humidity). No sudden temperature fluctuations and controlled ventilation with filter (free from dirt and dust). No aggressive vapors and no shock loads.	2 years or more given regular inspections. Check for cleanliness and mechanical damage as part of the inspection. Check the corrosion protection is intact.
Tropical (Asia, Africa, Central and South America, Australia, New Zealand excluding temperate zones)	Packed in containers, with desiccant and moisture indicator sealed in the plastic film. Protected against insect damage and mildew by chemical treatment.	With roof, protected against rain, no shock loads.	Max. 3 years with regular checks on the packaging and moisture indicator (rel. atmospheric humidity < 50 %).
	Open	With roof, enclosed, at constant temperature and atmospheric humidity (5 °C < ϑ < 60 °C, < 50 % relative atmospheric humidity). No sudden temperature fluctuations and controlled ventilation with filter (free from dirt and dust). No aggressive vapors and no shock loads. Protection against insect damage.	2 years or more given regular inspections. Check for cleanliness and mechanical damage as part of the inspection. Check the corrosion protection is intact.

1) Packaging must be performed by an experienced company using the packaging materials which have been expressly qualified for the particular application.

Installation/mounting

Comply with the instructions in Sec. 'Installation' and Sec. 'Assembly/removal'!

Startup/operation

Check the direction of rotation is correct in **decoupled** status (also listen out for unusual grinding noises as the shaft rotates).

Secure the shaft keys for test mode without drive components. Do not render monitoring and protection equipment inoperative even for test mode.

Switch off the geared motor if in doubt whenever changes occur in relation to normal operation (e.g. increased temperature, noise, vibration). Determine the cause; contact SEW if necessary.

Inspection and maintenance

Follow the instructions in Sec. 'Inspection and Maintenance'!



3.1 Basic structure of helical gear units

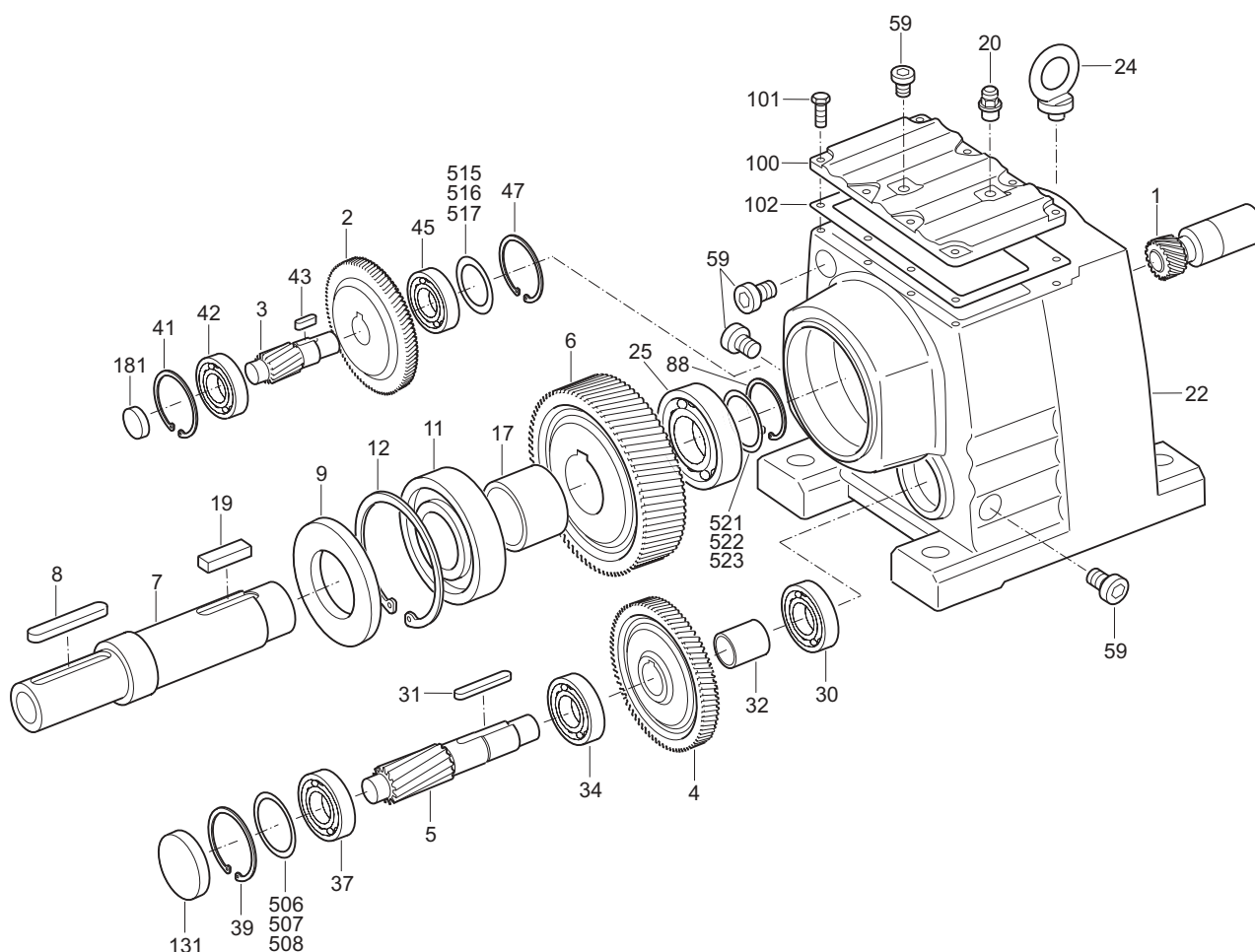


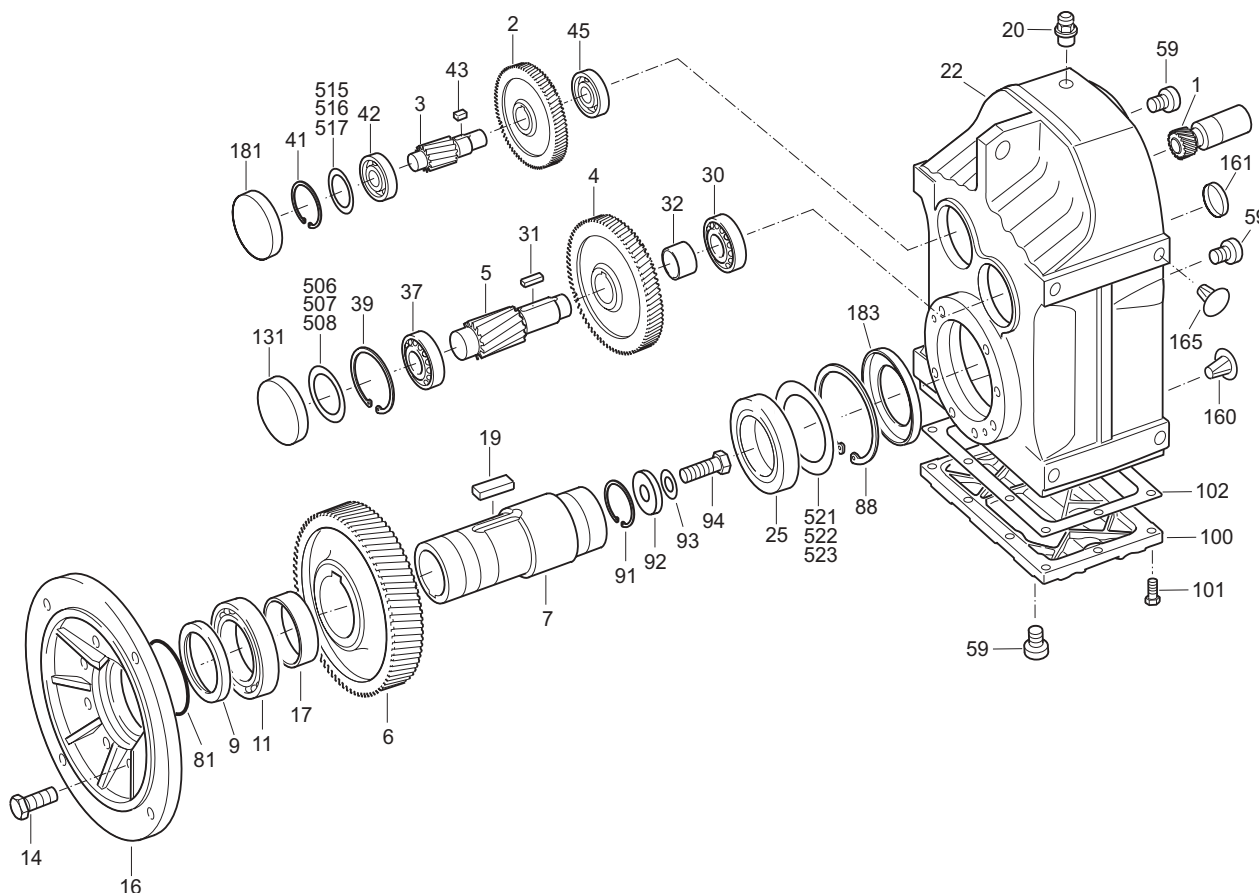
Figure 1: Basic structure of helical gear units

Key

1	Pinion	19	Key	42	Anti-friction bearing	507	Shim
2	Gear	20	Breather valve	43	Key	508	Shim
3	Pinion shaft	22	Gearcase	45	Anti-friction bearing	515	Shim
4	Gear	24	Lifting eyebolt	47	Circlip	516	Shim
5	Pinion shaft	25	Anti-friction bearing	59	Screw plug	517	Shim
6	Gear	30	Anti-friction bearing	88	Circlip	521	Shim
7	Output shaft	31	Key	100	Gearcase cover	522	Shim
8	Key	32	Distance piece	101	Hex head screw	523	Shim
9	Oil seal	34	Anti-friction bearing	102	Seal		
11	Anti-friction bearing	37	Anti-friction bearing	131	Closing cap		
12	Circlip	39	Circlip	181	Closing cap		
17	Distance piece	41	Circlip	506	Shim		



3.2 Basic structure of parallel shaft helical gear units



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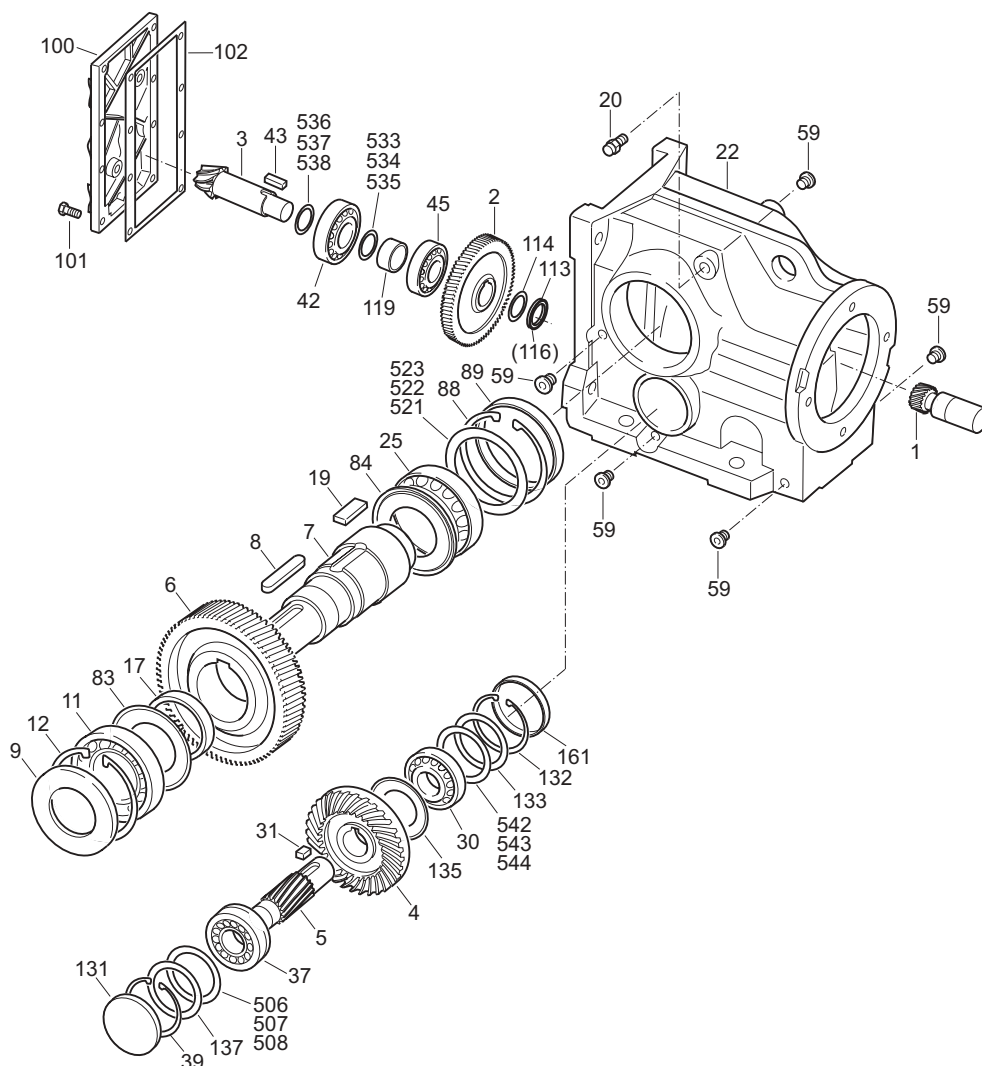
Figure 2: Basic structure of parallel shaft helical gear units

Key

1 Pinion	22 Gearcase	91 Circlip	506 Shim
2 Gear	25 Anti-friction bearing	92 Washer	507 Shim
3 Pinion shaft	30 Anti-friction bearing	93 Lock washer	508 Shim
4 Gear	31 Key	94 Hex head screw	515 Shim
5 Pinion shaft	32 Distance piece	100 Gearcase cover	516 Shim
6 Gear	37 Anti-friction bearing	101 Hex head screw	517 Shim
7 Hollow shaft	39 Circlip	102 Seal	521 Shim
9 Oil seal	41 Circlip	131 Closing cap	522 Shim
11 Anti-friction bearing	42 Anti-friction bearing	160 Closing plug	523 Shim
14 Hex head screw	43 Key	161 Closing cap	
16 Output flange	45 Anti-friction bearing	165 Closing plug	
17 Distance piece	59 Screw plug	181 Closing cap	
19 Key	81 O-ring	183 Oil seal	
20 Breather valve	88 Circlip		



3.3 Basic structure of helical-bevel gear units



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Figure 3: Basic structure of helical-bevel gear units

Key

1 Pinion	25 Anti-friction bearing	102 Adhesive and sealant	522 Shim
2 Gear	30 Anti-friction bearing	113 Slotted round nut	523 Shim
3 Pinion shaft	31 Key	114 Multi-tang washer	533 Shim
4 Gear	37 Anti-friction bearing	116 Thread lock	534 Shim
5 Pinion shaft	39 Circlip	119 Distance piece	535 Shim
6 Gear	42 Anti-friction bearing	131 Closing cap	536 Shim
7 Output shaft	43 Key	132 Circlip	537 Shim
8 Key	45 Anti-friction bearing	133 Spacer	538 Shim
9 Oil seal	59 Screw plug	135 Nilos ring	542 Shim
11 Anti-friction bearing	83 Nilos ring	161 Closing cap	543 Shim
12 Circlip	84 Nilos ring	506 Shim	544 Shim
17 Distance piece	88 Circlip	507 Shim	
19 Key	89 Closing cap	508 Shim	
20 Breather valve	100 Gearcase cover	521 Shim	
22 Gearcase	101 Hex head screw	521 Shim	

3.4 Basic structure of helical-worm gear units

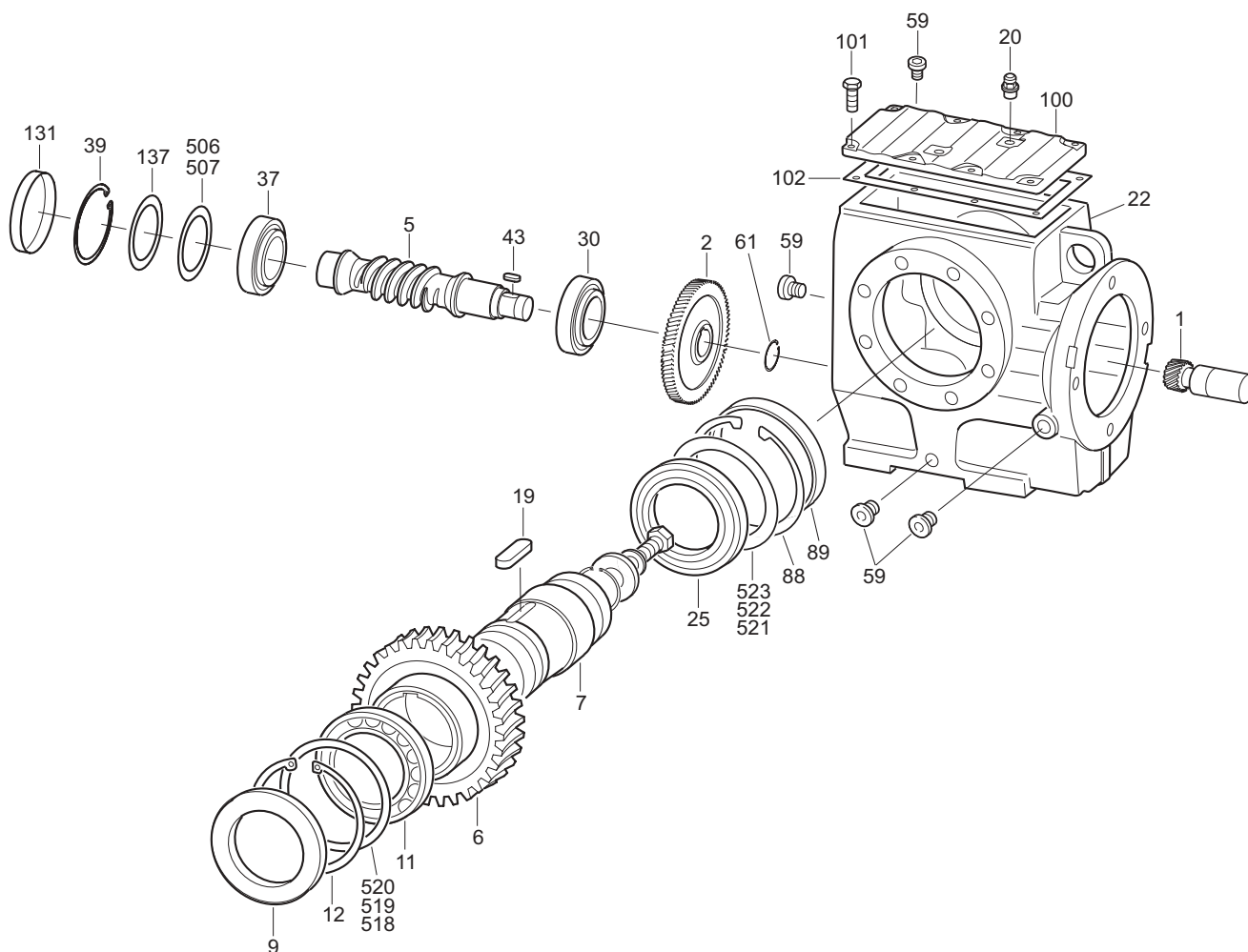


Figure 4: Basic structure of helical-worm gear units

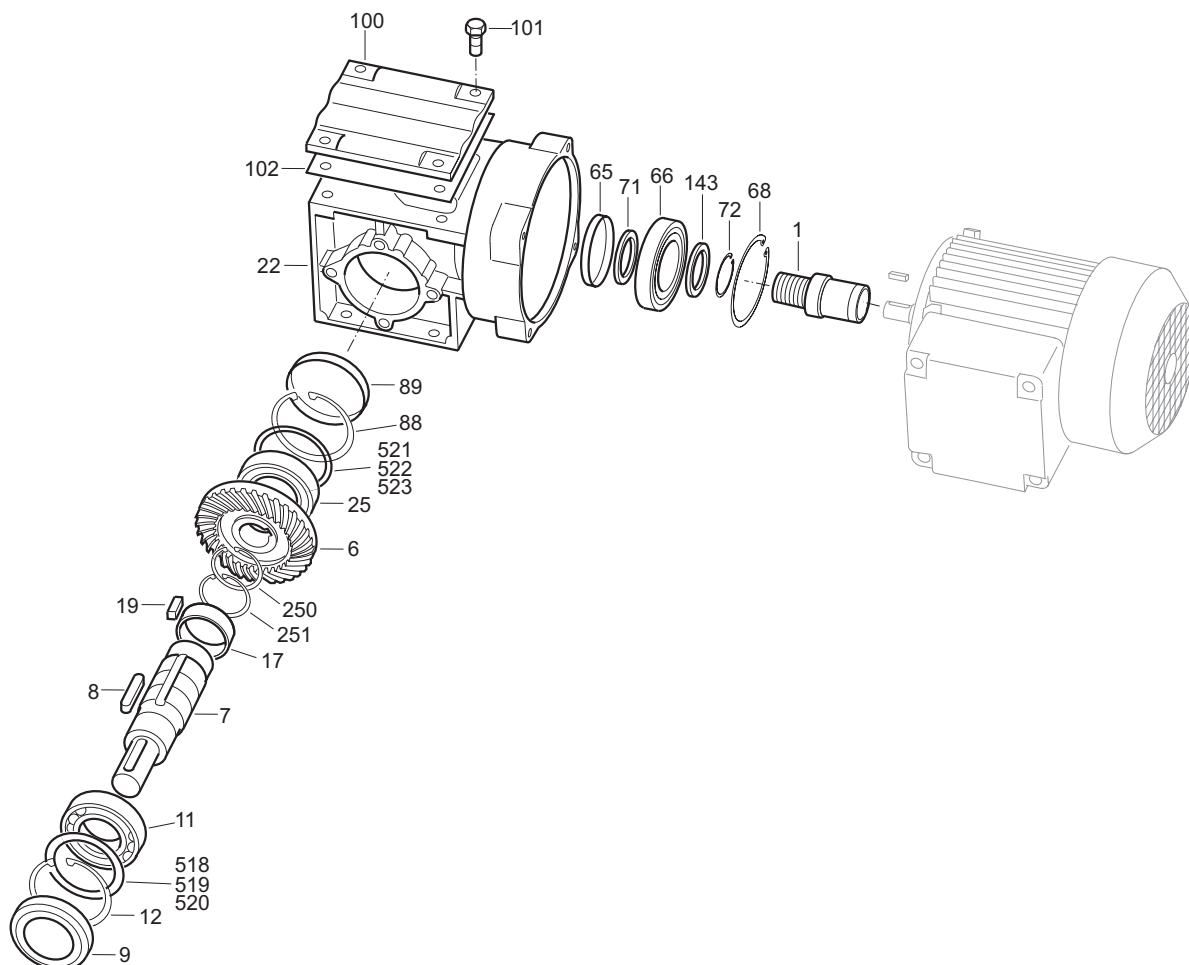
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Key

1	Pinion	20	Breather valve	88	Circlip	518	Shim
2	Gear	22	Gearcase	89	Closing cap	519	Shim
5	Worm	25	Anti-friction bearing	100	Gearcase cover	520	Shim
6	Worm gear wheel	30	Anti-friction bearing	101	Hex head screw	521	Shim
7	Output shaft	37	Anti-friction bearing	102	Rubber seal	522	Shim
9	Oil seal	39	Circlip	131	Closing cap	523	Shim
11	Anti-friction bearing	43	Key	137	Spacer		
12	Circlip	59	Screw plug	506	Shim		
19	Key	61	Circlip	507	Shim		



3.5 Basic structure of SPIROPLAN® gear units



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Figure 5: Basic structure of SPIROPLAN® gear units

Key

1 Pinion	19 Key	88 Circlip	251 Circlip
6 Gear	22 Gearcase	89 Closing cap	518 Shim
7 Output shaft	25 Anti-friction bearing	100 Gearcase cover	519 Shim
8 Key	65 Oil seal	101 Hex head screw	520 Shim
9 Oil seal	66 Anti-friction bearing	102 Seal	521 Shim
11 Anti-friction bearing	71 Spacer	132 Circlip	522 Shim
12 Circlip	72 Circlip	183 Oil seal	523 Shim
17 Distance piece	143 Spacer	250 Circlip	



4 Mechanical Installation

4.1 Required tools / resources

- Set of spanners
- Torque wrench (for shrink disks, AQH motor adapter, input shaft assembly with centering shoulder)
- Mounting device
- Shims and distance rings if necessary
- Fastening devices for input and output elements
- Lubricant (e.g. NOCO® Fluid)
- Bolt locking compound (for input shaft assembly with centering shoulder), e.g. Loctite 243

Installation tolerances

Shaft end	Flanges
Diametric tolerance in accordance with DIN 748 • ISO k6 for solid shafts with $\varnothing \leq 50$ mm • ISO m6 for solid shafts with $\varnothing > 50$ mm • ISO H7 for hollow shafts • Center hole in accordance with DIN 332, shape DR..	Centering shoulder tolerance in accordance with DIN 42948 • ISO j6 with $b1 \leq 230$ mm • ISO h6 with $b1 > 230$ mm

4.2 Before you begin

The drive may only be installed if

- the entries on the nameplate of the geared motor match the voltage supply system,
- the drive is undamaged (no damage caused by transportation or storage) and
- it is certain that the following requirements have been met:
 - **with standard gear units:**
ambient temperature according to the lubricant table in Sec. 'Lubricants' (see standard), no oil, acid, gas, vapors, radiation, etc.
 - **with special designs:**
drive configured in accordance with the ambient conditions
 - **with helical-worm / Spiroplan® W gear units:**
no large external mass moments of inertia which could exert a retrodriving load on the gear unit
[at η' (retrodriving) = $2 - 1/\eta < 0.5$ self-locking]

4.3 Preliminary work

The output shafts and flange surfaces must be thoroughly cleaned of anti-corrosion agents, contamination or such like (use a commercially available solvent). Do not let the solvent come into contact with the sealing lips of the oil seals since it could damage the material!



4.4 Installing the gear unit

The gear unit or geared motor may only be mounted or installed in the specified mounting position (Spiroplan® gear units are not dependent on the mounting position) on a level¹, vibration damping and torsionally rigid support structure. Do not tighten the housing legs and mounting flanges against one another and comply with the permitted overhung and axial loads!

Always use bolts of quality 8.8 for mounting geared motors.

Bolts of **quality 10.9** must be used for fastening the flange to the customer unit in order to transmit the rated torque specified in the catalog in case of the following flange-mounted helical geared motors (RF..) and foot/flange-mounted helical geared motors (R..F):

- RF37, R37F with flange Ø 120 mm
- RF47, R47F with flange Ø 140 mm
- RF57, R57F with flange Ø 160 mm



The oil level plug and drain screws as well as the breather valves must be freely accessible!

At the same time, also check that the oil fill is as specified for the mounting position (see Sec. 'Lubricants' / 'Lubricant fill quantities' or refer to the information on the nameplate). The gear units are filled with the required oil volume at the factory. There may be slight deviations at the oil level plug as a result of the mounting position, which are permitted within the manufacturing tolerances. **Adjust the lubricant fill volumes and the position of the breather valve accordingly in case you change the mounting position.**

Please contact our customer service if you change the mounting position of K gear units to M5 or M6 or between M5 and M6.

Please contact customer service if you change the mounting position of size S47...S97 S gear units to mounting position M2.

Use plastic inserts (2 – 3 mm thick) if there is a risk of electrochemical corrosion between the gear unit and the driven machine (connection between different metals such as cast iron and high-grade steel)! Also fit the bolts with plastic washers! Ground the housing additionally – use the grounding bolts on the motor.

Installation in damp areas or in the open

Drives are supplied in corrosion-resistant versions for use in damp areas or in the open. Any damage to the paint work (e.g. on the breather valve) must be repaired.

1. Maximum permitted flatness error for flange mounting (approximate values with reference to DIN ISO 1101): with → flange 120...600 mm max. error 0.2...0.5 mm

**Gear unit venting**

No breather plug is required for the following gear units:

- R07 in mounting positions M1, M2, M3, M5 and M6
- R17, R27 and F27 in mounting positions M1, M3, M5 and M6
- Spiroplan® W gear units

All other gear units are delivered by SEW-EURODRIVE ready for the respective mounting position with the breather valve fitted and activated.

Exception:

Gear units for extended storage, with pivoting mounting positions and for mounting at an angle are delivered with a screw plug on the vent hole. Prior to startup, the customer must replace the highest screw plug by the supplied breather valve on each individual gear unit

- **With geared motors** for extended storage, with pivoting mounting positions and for mounting at an angle, the supplied breather valve is located in the **terminal box of the motor**.
- **With gear head units** requiring venting on the input end, a breather valve is supplied in a plastic bag.
- **Enclosed gear units** are delivered without a breather valve.

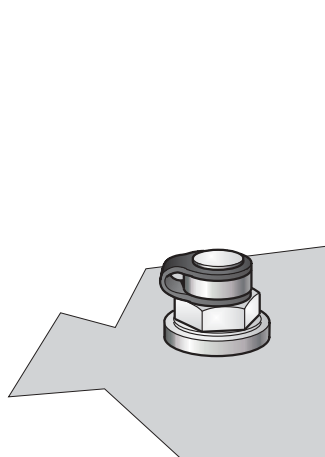
Activating the breather valve

As a rule, the breather valve is already activated at the factory. If this is not the case, remove the transportation fixture from the breather valve before starting up the gear unit!

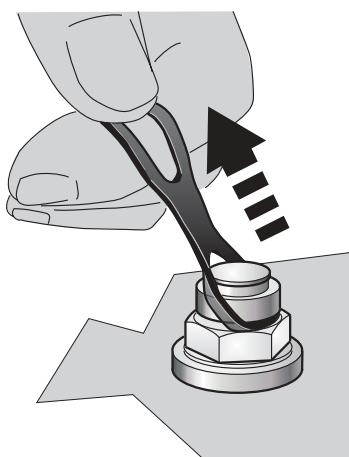
1. Breather valve with transportation fixture

2. Remove the transportation fixture

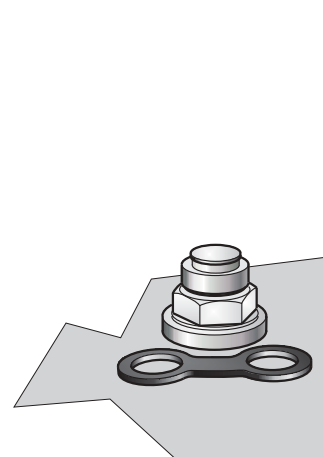
3. Breather valve activated



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Painting the gear unit

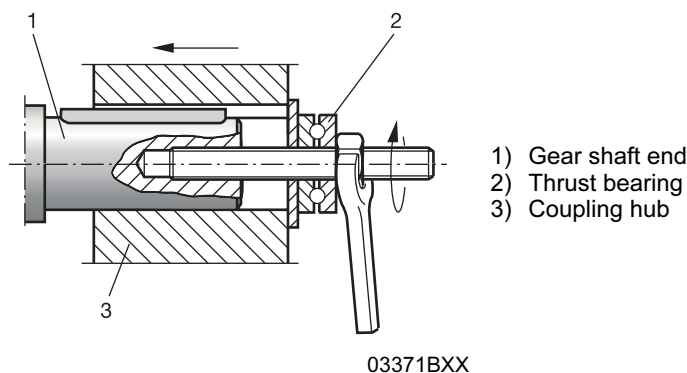
If all or some of the drive surface is to be painted over, make sure that you carefully mask over the breather valve and the oil seals. Remove the strips of tape after completing the paint job.



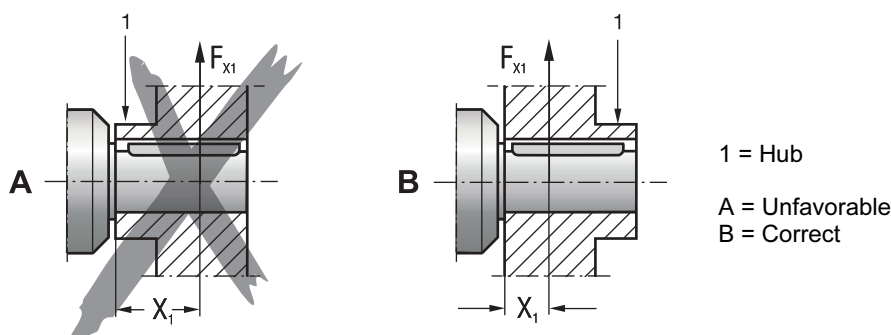
4.5 Gear units with solid shafts

Installing input and output elements

The following illustration shows an example of a mounting device for installing couplings or hubs on gear unit or motor shaft ends. It may be possible to dispense with the thrust bearing on the mounting device.



The following illustration shows the correct mounting arrangement **B** of a gear or chain sprocket for avoiding impermissibly high overhung loads.



- Only use a mounting device for installing input and output elements. Use the center bore and the thread on the shaft end for positioning purposes.
- **Never drive belt pulleys, couplings, pinions, etc. onto the shaft end by hitting them with a hammer (damage to bearings, housing and the shaft!).**
- **In the case of belt pulleys, make sure the belt is tensioned correctly (in accordance with the manufacturer's instructions).**
- Power transmission elements should be balanced after fitting and must not give rise to any impermissible radial or axial forces (see the 'Geared Motors' catalog for permitted values).



Note:

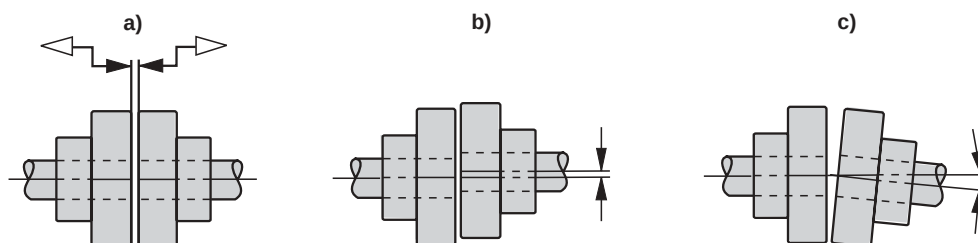
Mounting is easier if you first apply lubricant to the output element or heat it up briefly (to 80 – 100 °C).



Mounting of couplings

Couplings must be mounted and balanced according to the information provided by the coupling manufacturer:

- a) Maximum and minimum clearance
- b) Axial misalignment
- c) Angular misalignment



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Figure 6: Clearance and misalignment for clutch mounting



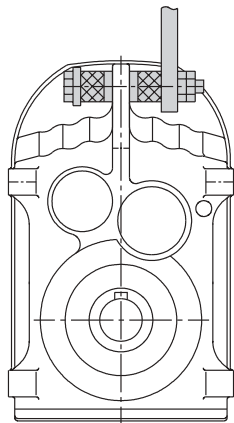
Input and output elements such as belt pulleys, couplings, etc. must be equipped with touch guards!



4.6 Installing torque arms for shaft-mounted gear units

Do not place torque arms under strain during installation!

Parallel shaft helical gear units

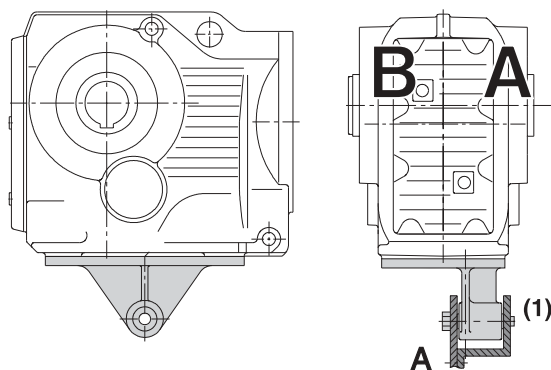


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Figure 7: Torque arm for parallel shaft helical gear units

Helical-bevel gear units

- Bushing with bearings on both ends → (1)
- Install connection end B as a mirror image of A



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Figure 8: Torque arm for helical-bevel gear units


**Helical-worm
gear units**

- Bushing with bearings on both ends → (1)

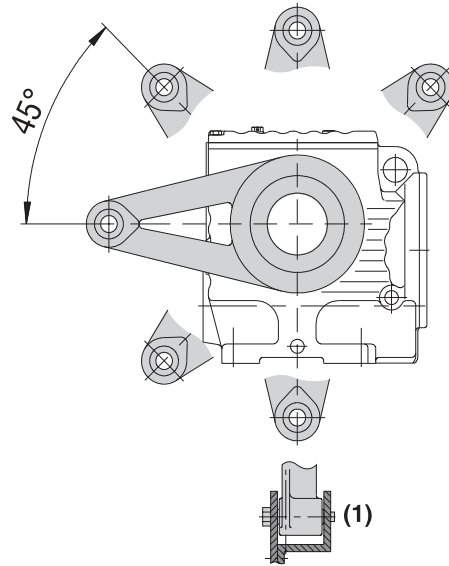


Figure 9: Torque arm for helical-worm gear units

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**SPIROPLAN® W
gear units**

- Bushing with bearings on both ends → (1)

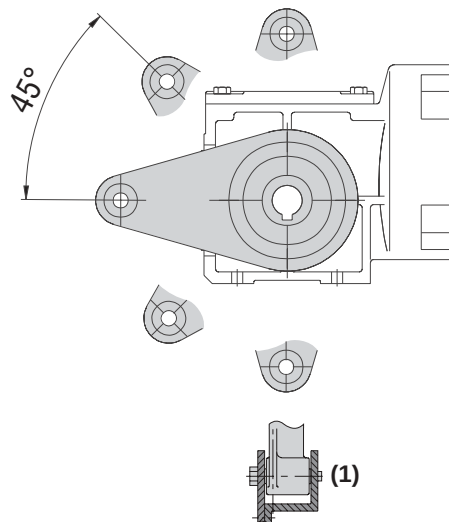


Figure 10: Torque arm for SPIROPLAN® W gear units

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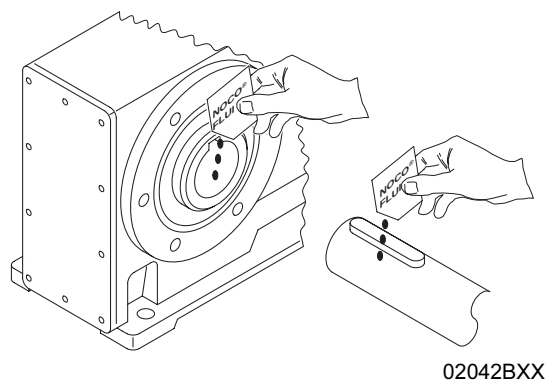
4.7 Assembly/removal of shaft-mounted gear units with keyway or splined hollow shaft



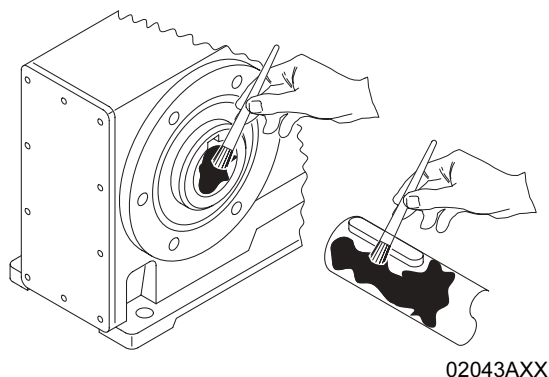
Please also refer to the design notes in the Geared Motors catalog concerning configuration of the customer shaft!

Installation notes

1. Apply NOCO® fluid.

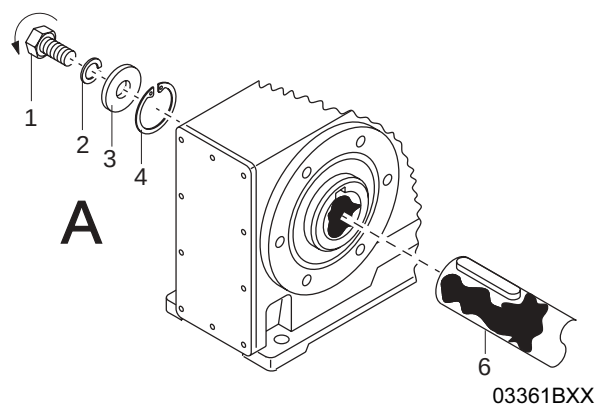


2. Distribute the NOCO® fluid carefully.



3. Install the shaft and secure it axially.
(mounting is facilitated by using a mounting device).

3A: Mounting with standard scope of supply

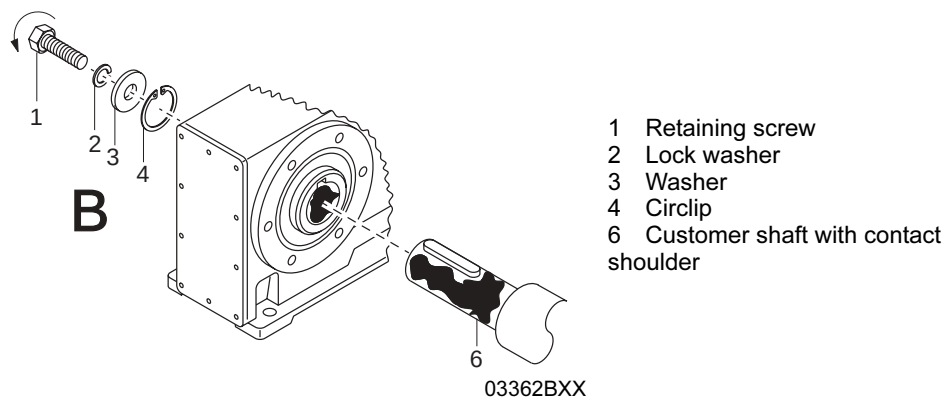


- 1 Short retaining screw
(standard scope of supply)
- 2 Lock washer
- 3 Washer
- 4 Circlip
- 6 Customer shaft



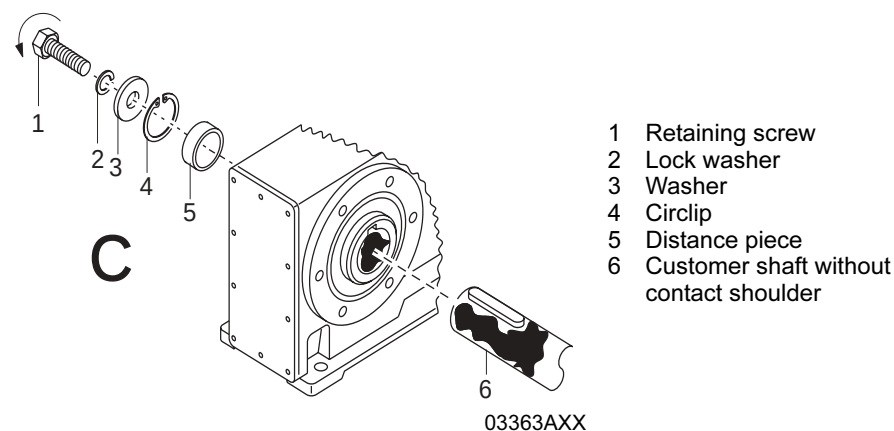
3B: Mounting with SEW-EURODRIVE assembly/removal kit (→ page 22)

– Customer shaft **with** contact shoulder

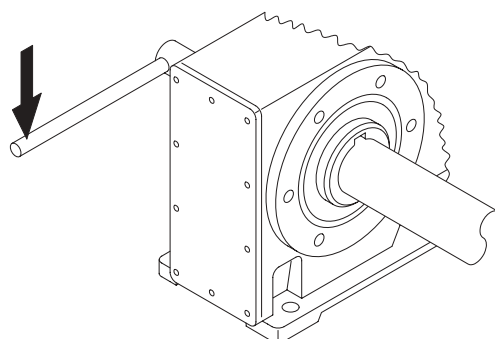


3C: Mounting with SEW-EURODRIVE assembly/removal kit (→ page 22)

– Customer shaft **without** contact shoulder



4. Tighten the retaining screw to the appropriate torque (see table).



Bolt	Tightening torque [Nm]
M5	5
M6	8
M10/12	20
M16	40
M30	80
M24	200



Note:

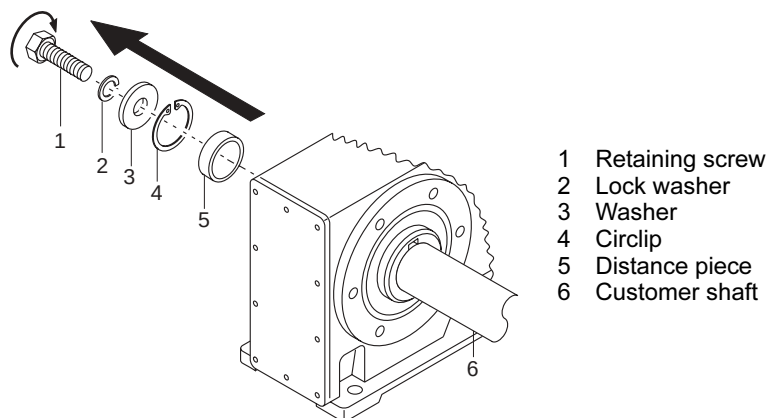
To avoid contact corrosion, we recommend that the customer shaft should additionally be recessed between the two contact surfaces!



Information about removal

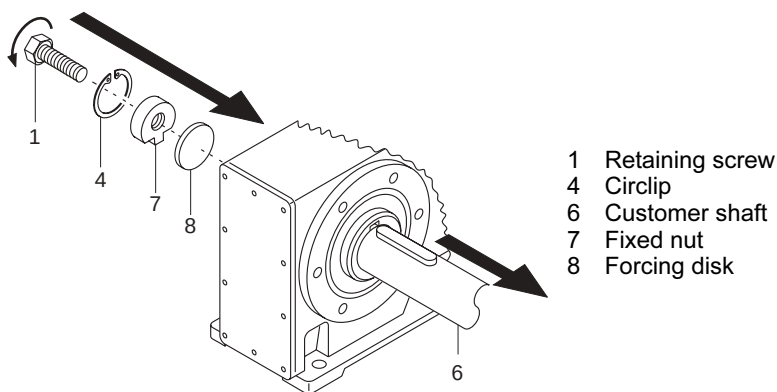
This description only applies if the gear unit was mounted using the SEW-EURODRIVE assembly/removal kit (→ page 22 and the description above, points 3B or 3C).

1. Loosen the retaining screw 1.
2. Remove parts 2 to 4 and, if fitted, distance piece 5.



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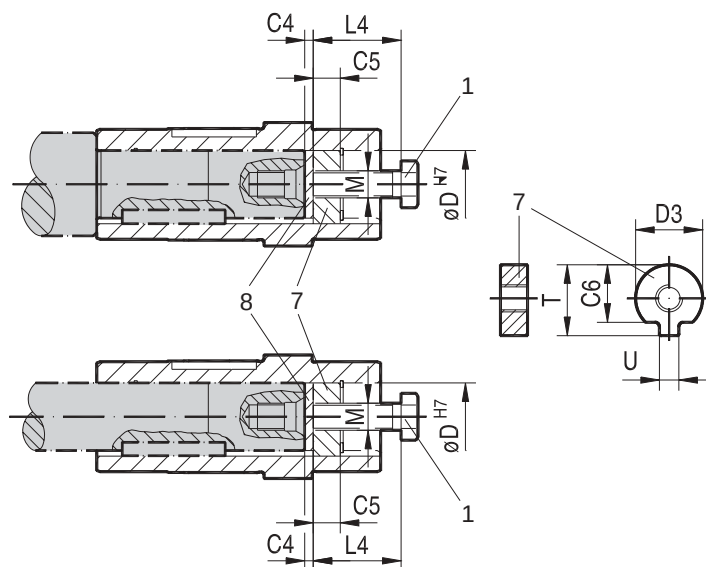
3. Insert the forcing disk 8 and the fixed nut 7 from the SEW-EURODRIVE assembly/removal kit between the customer shaft 6 and the circlip 4.
4. Re-insert the circlip 4.
5. Screw the retaining screw 1 back in. Now you can force the gear unit off the shaft by tightening the screw.



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**SEW assembly/
removal kit**

The SEW-EURODRIVE assembly/removal kit can be ordered by quoting the specified part number.



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Figure 11: SEW-EURODRIVE assembly/removal kit

- 1 Retaining screw
- 7 Fixed nut for removal
- 8 Forcing disk

Type	D ^{H7} [mm]	M ¹⁾	C4 [mm]	C5 [mm]	C6 [mm]	U ^{-0.5} [mm]	T ^{-0.5} [mm]	D3 ^{-0.5} [mm]	L4 [mm]	Assembly/ removal kit part number
WA..10	16	M5	5	5	12	4.5	18	15.7	50	643 712 5
WA..20	18	M6	5	6	13.5	5.5	20.5	17.7	25	643 682 X
WA..20, WA..30, SA..37	20	M6	5	6	15.5	5.5	22.5	19.7	25	643 683 8
FA..27, SA..47	25	M10	5	10	20	7.5	28	24.7	35	643 684 6
FA..37, KA..37, SA..47, SA..57	30	M10	5	10	25	7.5	33	29.7	35	643 685 4
FA..47, KA..47, SA..57	35	M12	5	12	29	9.5	38	34.7	45	643 686 2
FA..57, KA..57, FA..67, KA..67, SA..67	40	M16	5	12	34	11.5	41.9	39.7	50	643 687 0
SA..67	45	M16	5	12	38.5	13.5	48.5	44.7	50	643 688 9
FA..77, KA..77, SA..77	50	M16	5	12	43.5	13.5	53.5	49.7	50	643 689 7
FA..87, KA..87, SA..77, SA..87	60	M20	5	16	56	17.5	64	59.7	60	643 690 0
FA..97, KA..97, SA..87, SA..97	70	M20	5	16	65.5	19.5	74.5	69.7	60	643 691 9
FA..107, KA..107, SA..97	90	M24	5	20	80	24.5	95	89.7	70	643 692 7
FA..127, KA..127	100	M24	5	20	89	27.5	106	99.7	70	643 693 5
FA..157, KA..157	120	M24	5	20	107	31	127	119.7	70	643 694 3

1) Retaining screw



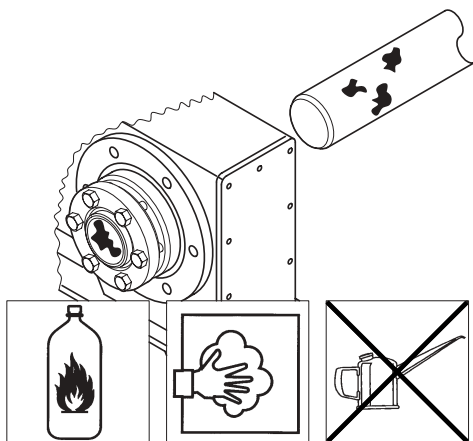
4.8 Assembly/removal of shaft-mounted gear units with shrink disk

Installation notes

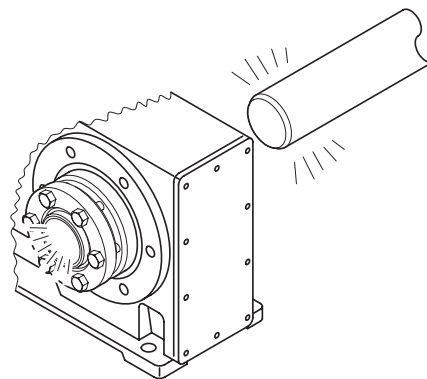
- Do not tighten the locking screws unless the shaft is installed since the hollow shaft could become deformed!

- Carefully degrease the hollow shaft bore and the input shaft.

- Hollow shaft/input shaft after degreasing.



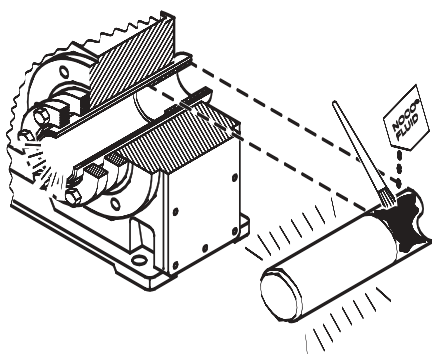
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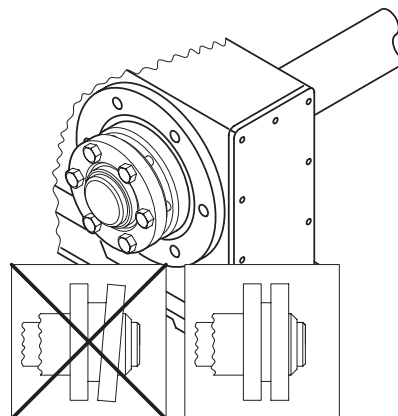
01816AXX

- Apply NOCO® fluid to the input shaft¹⁾ in the area of the bushing.

- Install the shaft, making sure that the locking collars of the shrink disk are equally spaced²⁾.



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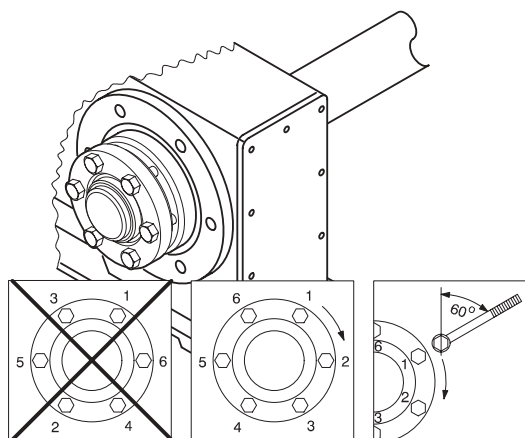
01818AXX



- Make sure that the clamping area of the shrink disk is free from grease!** As a result, never apply NOCO® fluid directly to the bushing. The paste may get into the clamping area of the shrink disk when the input shaft is installed.
- After assembly**, grease the outside of the hollow shaft in the area of the shrink disk. This is in order to prevent corrosion.



5. Tighten the locking screws by working clockwise several times from one screw to the next (not in opposite sequence). See the table for tightening torques.



01819AXX

Gear unit type			Bolt	Nm	max. ¹⁾
FH27	SH37		M5	5	60°
KH37...77	FH37...77	SH47...77	M6	12	
KH87/97	FH87/97	SH87/97	M8	30	
KH107	FH107		M10	59	
KH127/157	FH127		M12	100	

1) Maximum tightening angle per cycle

Notes on removal of the shrink disk

1. Unscrew the locking screws evenly one after the other. Each locking screw may only be unscrewed by about one quarter turn in the initial cycle to avoid tilting and jamming of the locking collars. Do not fully unscrew the locking screws!
2. Remove the shaft or pull the hub off the shaft. (It is first necessary to remove any rust which may have formed between the hub and the end of the shaft.)
3. Pull the shrink disk off the hub.



Important!

Risk of injury if the shrink disk is not removed correctly!

Cleaning and lubricating the shrink disk

There is no need to disassemble and re-grease the removed shrink disks before they are installed once again.

The shrink disk only needs to be cleaned and re-greased if it is contaminated.

Use one of the following solid lubricants for the tapered surfaces:

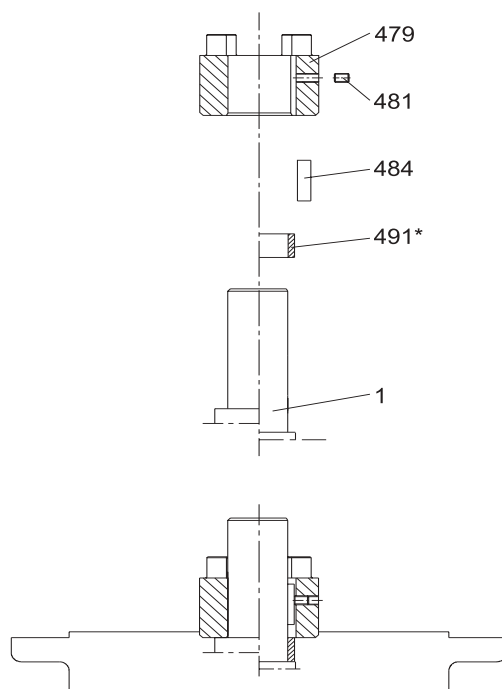
Lubricant (Mo S2)	Sold as
Molykote 321 (lube coat)	Spray
Molykote spray (powder spray)	Spray
Molykote G Rapid	Spray or paste
Aemasol MO 19P	Spray or paste
AemasolDIO-sétral 57 N (lube coat)	Spray

Grease the locking screws with a multipurpose grease such as Molykote BR 2 or similar.



4.9 Mounting the coupling of adapter AM

IEC adapter
AM63...225 /
NEMA adapter
AM56...365



* = NEMA adapter only
 1 = Motor shaft

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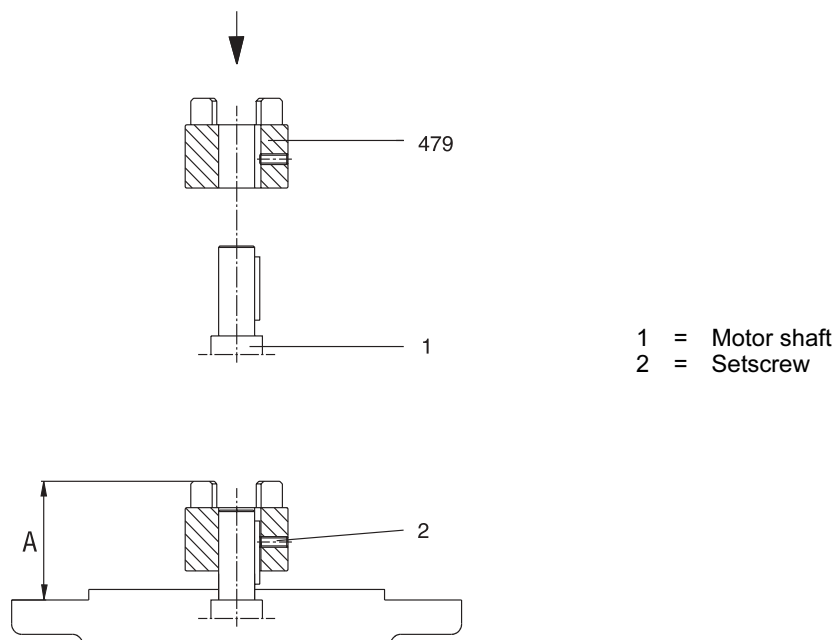
1. Clean the motor shaft and flange surfaces of the motor and the adapter.
2. **IEC adapter:** Remove the key from the motor shaft and replace it with the supplied key (484).
NEMA adapter: Remove the key from the motor shaft, push the distance piece (491) onto the motor shaft and insert the supplied key (484).
3. Heat the coupling half (479) to approx. 80 – 100 °C, push the coupling half onto the motor shaft.
IEC adapter: Insert up to the stop against the motor shaft collar.
NEMA adapter: Insert up to the stop against the distance piece.
4. Use the setscrew (481) to secure the key and coupling half on the motor shaft.
5. Mount the motor on the adapter. When doing this, make sure the coupling claws of the adapter shaft engage in the plastic cam ring.



Note: To avoid contact corrosion, we recommend applying NOCO® fluid to the motor shaft before mounting the coupling half.



**IEC adapter
AM250/AM280**



02047CXX

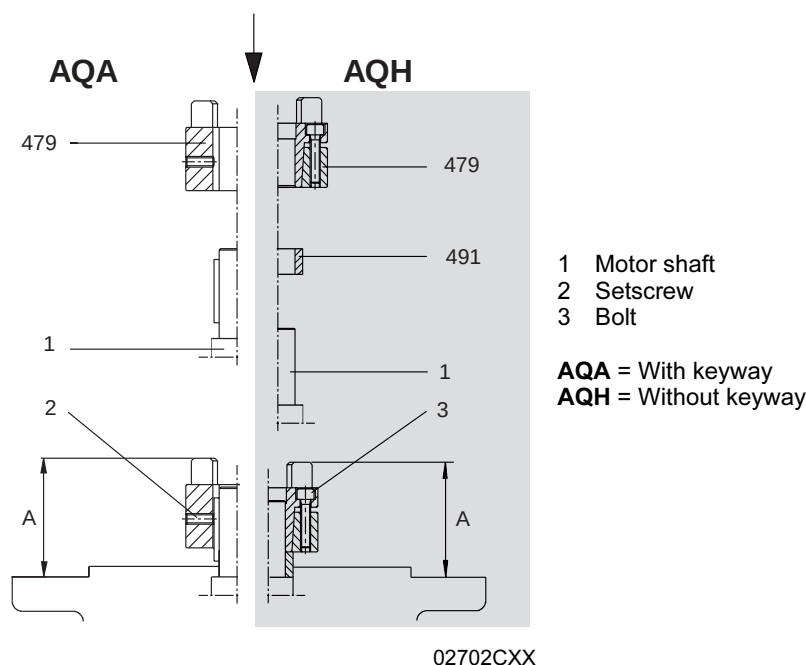
1. Clean the motor shaft and flange surfaces of the motor and the adapter.
2. Remove the key from the motor shaft and insert the supplied key (**size AM280 only**).
3. Heat the coupling half (479) to 80°C – 100°C) and push it onto the motor shaft (A = 139 mm).
4. Secure the coupling half with a setscrew and check its position (clearance 'A').
5. Mount the motor on the adapter. When doing this, make sure the claws of both coupling halves engage in the plastic cam ring.



Note: To avoid contact corrosion, we recommend applying NOCO[®] fluid to the motor shaft before mounting the coupling half.



4.10 Mounting the coupling of adapter AQ



1. Clean the motor shaft and flange surfaces of the motor and the adapter.
2. **Type AQH:** Push distance piece (491) onto the motor shaft.
3. **Type AQH:** Unscrew the bolts of the coupling half (479) and loosen the conical connection.
4. Heat the coupling half (80 °C 100 °C) and push it onto the motor shaft.
Type AQH: Insert up to the stop against the distance piece (491).
Type AQA: Insert up to clearance 'A' (see table).
5. **Type AQH:** Tighten the bolts on the coupling half (work round several times tightening the bolts evenly one after the other) until all bolts reach the tightening torque TA specified in the table.
Type AQA: Use a setscrew to secure the coupling half.
6. Check the position of the coupling half (clearance 'A', see table).
Fit the motor onto the adapter, making sure that the claws of the two coupling halves engage in one another. The force which must be applied when joining the two coupling halves is dissipated after final assembly, so there is no risk of any axial load being applied to adjacent bearings.

Setting dimensions, tightening torques

Type	Coupling size	Clearance 'A' [mm]	Bolts DIN 912 ¹⁾	Tightening torque TA ¹⁾ [Nm]
AQA /AQH 80 /1/2/3	19/24	44.5	M4	3
AQA /AQH 100 /1/2		39		
AQA /AQH 100 /3/4		53		
AQA /AQH 115 /1/2		62		
AQA /AQH 115 /3	24/28	62	M5	6
AQA /AQH 140 /1/2		62		
AQA /AQH 140 /3	28/38	74.5	M5	6
AQA /AQH 190 /1/2		76.5		
AQA /AQH 190 /3	38/45	100	M6	10

1) only type without keyway (AQH)

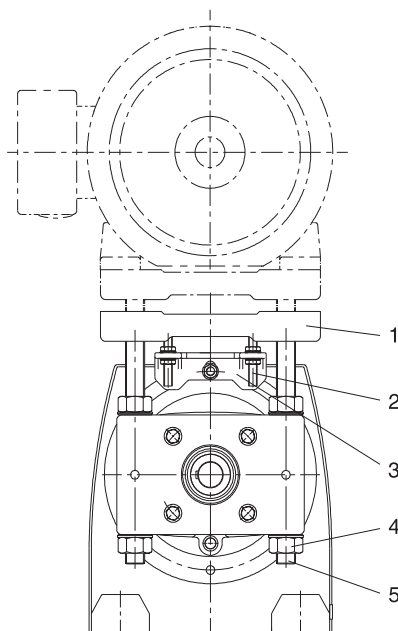


4.11 Mounting on input shaft assembly AD

Please refer to Sec. 'Installing input and output shafts' concerning the mounting of input elements.

Cover with motor mounting platform AD../P

Mounting the motor and adjusting the motor mounting platform



- 1 Motor mounting platform
- 2 Stud bolt (only AD6/P / AD7/P)
- 3 Support (only AD6/P / AD7/P)
- 4 Nut
- 5 Threaded column

03519BXX

1. Set the motor mounting platform to the required mounting position by evenly tightening the adjusting nuts. It may be necessary to remove the lifting eyebolt from helical gear units in order to achieve the lowest adjustment position. Touch up any damage to the paint work.
2. Align the motor on the motor mounting platform (shaft ends must be in alignment) and secure it.
3. Mount the input elements on the input shaft end and the motor shaft. Line them up with one another. Correct the motor position again if necessary.
4. Put on traction elements (V-belt, chain, etc.) and apply a preload by evenly adjusting the motor mounting platform. Do not stress the motor mounting platform and the columns against one another when doing this.
5. Tighten the threaded columns using the nuts which are not used for adjustment.

Only AD6/P and AD7/P:

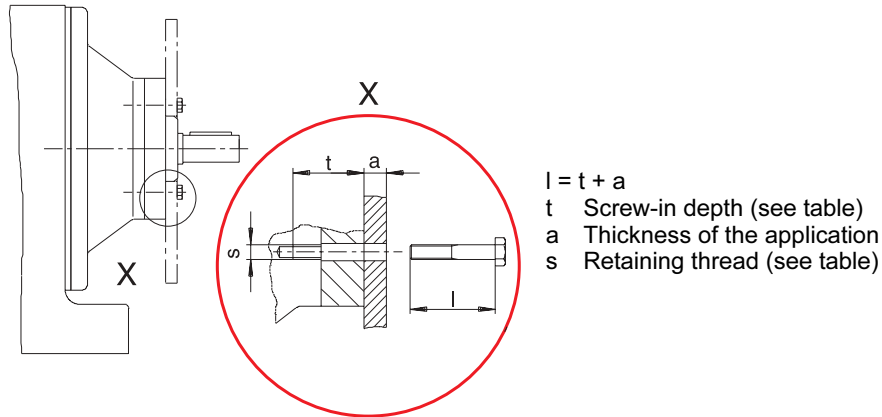
Unscrew the nuts on the stud bolts before adjustment to allow the stud bolts to move axially in the support without restriction. Do not tighten the nuts until the final adjustment position has been achieved. Do not adjust the motor mounting platform using the support.



Type with centering shoulder AD../ZR

Mounting applications on the input shaft assembly with centering shoulder

1. Provide retaining bolts of a suitable length for the application. The length l of the new bolts is calculated as follows:



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Round down the calculated bolt length to the next smaller standard length.

2. Remove the retaining bolts from the centering shoulder.
3. Clean the contact surface and the centering shoulder.
4. Clean the threads of the new bolts and apply a bolt locking compound (e.g. Loctite 243) to the first few threads.
5. Position the application against the centering shoulder and tighten the retaining bolts to the specified tightening torque T_A (see table).

Type	Screw-in depth t	Retaining thread s	Tightening torque T_A [Nm]
AD2/ZR	25.5	M8	25
AD3/ZR	31.5	M10	48
AD4/ZR	36	M12	86
AD5/ZR	44	M12	86
AD6/ZR	48.5	M16	210
AD7/ZR	49	M20	410
AD8/ZR	42	M12	86

Cover with backstop AD../RS

Check the direction of rotation of the drive before installation and startup. Please inform the SEW-EURODRIVE customer service if the direction of rotation is incorrect.

The backstop is maintenance-free in operation and does not require any further maintenance work.



5 Startup

5.1 Startup of helical-worm and Spiroplan® W gear units



Please note: The direction of rotation of the output shaft in series S..7 helical-worm gear units has been changed from CW to CCW; this is different from the S..2 series. Change direction of rotation: Swap over two motor feeder cables.

Running-in period

Spiroplan® and helical-worm gear units require a running-in period of at least 24 hours before reaching their maximum efficiency. A separate running-in period applies for each direction of rotation if the gear unit is operated in both directions of rotation. The table shows the average power reduction during the running-in period.

No. of starts	Worm		Spiroplan®	
	Power reduction	i range	Power reduction	i range
1 start	approx. 12 %	approx. 50 280	approx. 15 %	approx. 40 75
2 start	approx. 6 %	approx. 20 75	approx. 10 %	approx. 20 30
3 start	approx. 3 %	approx. 20 90	approx. 8 %	approx. 15
4 start	-	-	approx. 8 %	approx. 10
5 start	approx. 3 %	approx. 6 25	approx. 5 %	approx. 8
6 start	approx. 2 %	approx. 7 25	-	-

5.2 Startup of helical, parallel shaft helical and helical-bevel gear units

No special startup instructions are required for helical, parallel shaft helical and helical-bevel gear units provided the gear units have been installed in accordance with Sec. 'Mechanical Installation'.



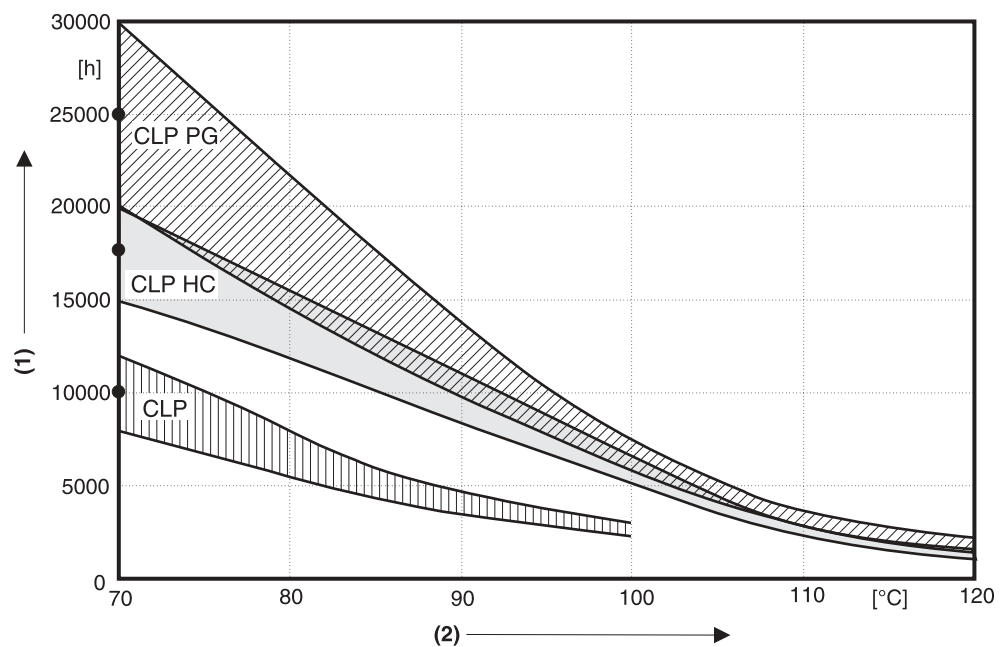
6 Inspection and Maintenance

6.1 Inspection and maintenance intervals

Frequency	What to do
• Every 3000 machine hours, at least every 6 months	• Check the oil
• Depending on the operating conditions (see illustration below), at the latest every 3 years	• Change mineral oil • Renew the anti-friction bearing grease
• Depending on the operating conditions (see illustration below), at the latest every 5 years	• Change synthetic oil • Renew the anti-friction bearing grease
• R07, R17, R27, F27 and Spiroplan® gear units have lubrication for life and are therefore maintenance-free	
• Varying (depending on external factors)	• Touch up or renew the surface/anticorrosion coating

6.2 Lubricant change intervals

Change the oil more frequently when using special designs subject to more severe/aggressive environmental conditions!



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Figure 12: Oil change intervals for standard gear units under normal environmental conditions.

- (1) Operating hours
(2) Sustained oil bath temperature
• Average value per oil type at 70 °C



6.3 Inspection and maintenance of the gear unit

Do not mix synthetic lubricants with each other and do not mix synthetic with mineral lubricants!

Mineral oil is the standard lubricant.

The position of the oil level plug and oil drain plug as well as the breather valve depends on the mounting position. Refer to the diagrams of the mounting positions.

Checking the oil level



1. **De-energize the geared motor and secure it to prevent it from being switched back on inadvertently!**

Wait until the gear unit has cooled down – Danger of burns!

2. Refer to Sec. 'Installing the gear unit' when changing the mounting position!
3. For gear units with oil level plug: Remove the oil level plug, check the fill level and correct if necessary. Reinstall the oil level plug.

Check the oil



1. **De-energize the geared motor and secure it to prevent it from being switched back on inadvertently!**

Wait until the gear unit has cooled down – Danger of burns!

2. Remove some oil from the oil drain plug
3. Check the oil consistency
 - Viscosity
 - If you see that the oil is heavily contaminated, we recommend you change the oil even if it is outside the service intervals specified in 'Inspection and maintenance periods'
4. For gear units with oil level plug: Remove the oil level plug, check the fill level and correct if necessary. Reinstall oil level plug.

Changing the oil



Only change the oil when the gear unit is at operating temperature.

1. **De-energize the geared motor and secure it to prevent it from being switched back on inadvertently!**

Wait until the gear unit has cooled down – Danger of burns!

Note: The gear unit must still be warm since the high viscosity of cold oil will make it harder to drain the oil completely.

2. Place a container underneath the oil drain plug.
3. Remove the oil level plug, breather plug/breather valve and oil drain plug.
4. Drain all the oil.
5. Reinstall the oil drain plug.
6. Pour in new oil of the same type through the vent hole (if changing the oil type, please contact our customer service first)
 - Fill in the amount of oil in accordance with the mounting position (see Sec. 'Lubricant fill quantities') or as stated on the nameplate.
 - Check the oil level plug
7. Reinstall the oil level plug
8. Reinstall the breather plug/breather valve



7 Malfunctions

7.1 Gear unit malfunctions

Fault	Possible cause	Remedy
Unusual, regular running noise	A Meshing/grinding noise: Bearing damage B Knocking noise: Irregularity in the gearing	A Check the oil (see Sec. 'Inspection and Maintenance'), change bearings B Contact customer service
Unusual, irregular running noise	Foreign bodies in the oil	- Check the oil (see Sec. 'Inspection and Maintenance') - Stop the drive, contact customer service
Oil leaking ¹⁾ - from the gear cover plate - from the motor flange - from the motor oil seal - from the gear unit flange - from the output end oil seal	A Rubber seal on the gear cover plate leaking B Seal defective C Gear unit not vented	A Tighten the bolts on the gear cover plate and observe the gear unit. Oil still leaking: Contact customer service B Contact customer service C Vent the gear unit (see Sec. 'Mounting Positions')
Oil emerging from breather valve	A Too much oil B Drive used with the wrong mounting position C Frequent cold starts (oil foams) and/or high oil level	A Correct the oil level (see Sec. 'Inspection and Maintenance') B Mount the breather valve correctly (see Sec. 'Mounting Positions') and correct the oil level (see Sec. 'Lubricants')
Output shaft does not turn although the motor is running or the input shaft is rotated	Connection between shaft and hub in gear unit interrupted	Send in the gear unit/geared motor for repair

1) It is normal for small amounts of oil/grease to emerge from the oil seal during the running-in phase - (24 hour running time, see also DIN 3761).

Customer service

Please provide the following information when contacting customer service:

- Nameplate data (complete)
- Nature and extent of the fault
- Time and peripheral circumstances of the fault
- Presumed cause

8 Mounting Positions

8.1 General information about mounting positions

Mounting position designation

SEW differentiates between six mounting positions M1...M6 for gear units. The following figure shows the spatial orientation of the gear unit in mounting positions M1...M6.

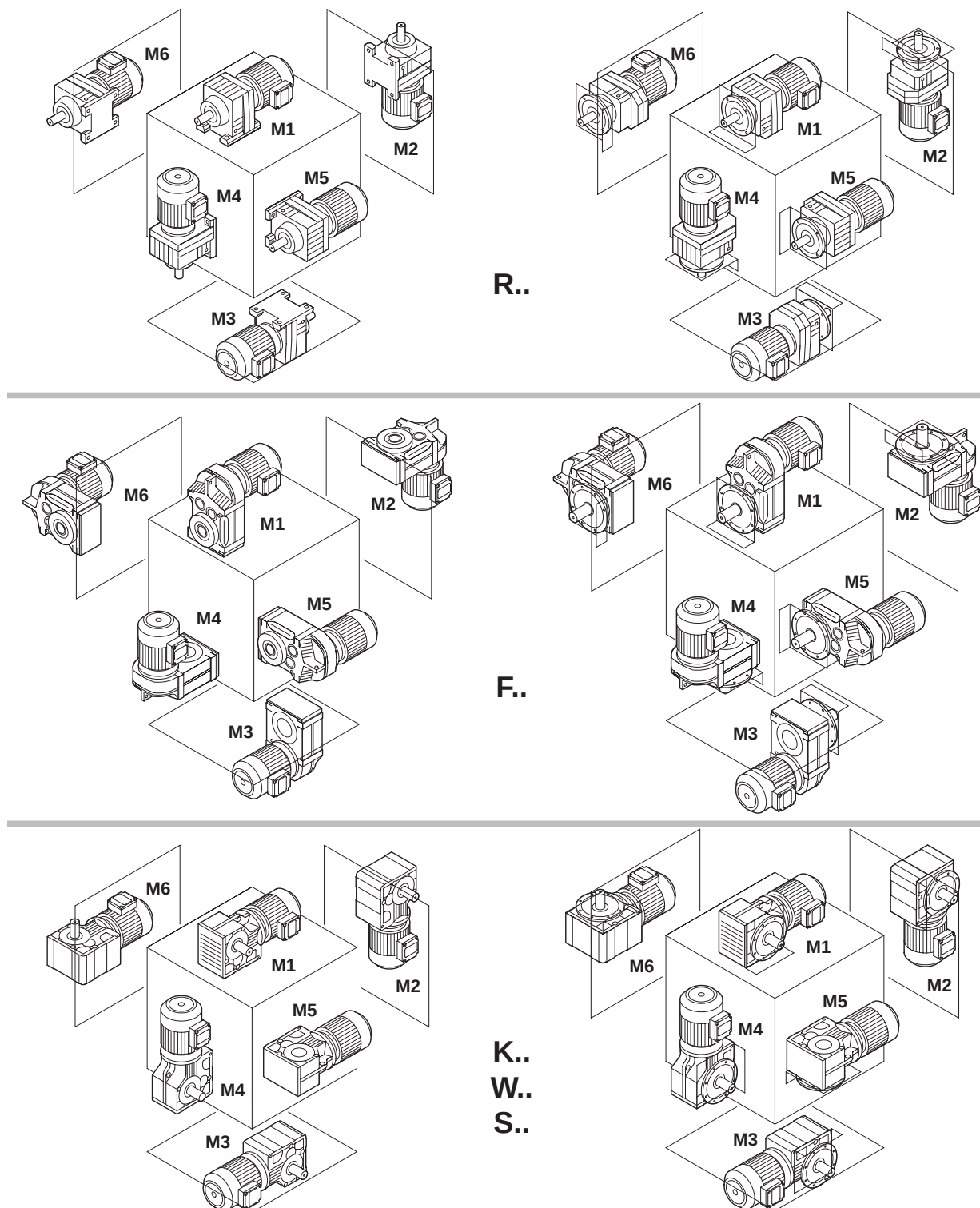


Figure 13: Depiction of mounting positions M1 M6

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8.2 Key to the mounting position sheets

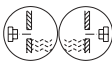


Spiroplan® geared motors do not change with their mounting position. However, mounting positions M1 to M6 are also shown for Spiroplan® geared motors to assist you in working with this documentation.

Important: No breather valves or oil level checking and drain plugs can be fitted on Spiroplan® geared motors.

Symbols used

The following table shows which symbols are used in the mounting position sheets and what they mean:

Symbol	Meaning
	Breather valve
	Oil level plug
	Oil drain plug

Churning losses



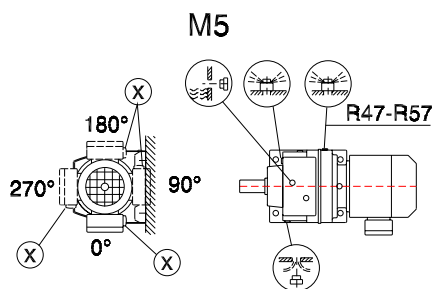
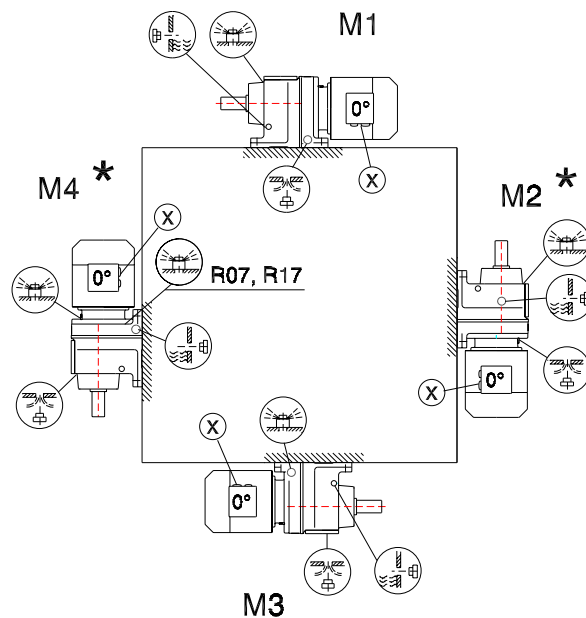
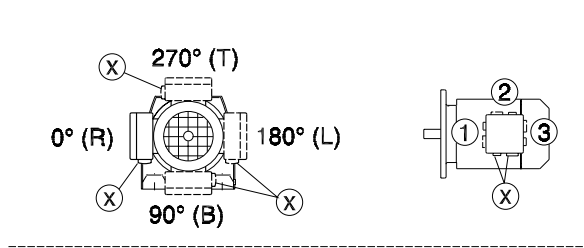
Increased churning losses may arise in some mounting positions. Please contact SEW-EURODRIVE in case of the following combinations:

Mounting position	Gear unit type	Gear unit size	Input speed [1/min]
M2, M4	R	97 ... 107	> 2500
		> 107	> 1500
M2, M3, M4, M5, M6	F	97 ... 107	> 2500
		> 107	> 1500
	K	77 ... 107	> 2500
		> 107	> 1500
	S	77 ... 97	> 2500

8.3 Mounting positions of R helical gear units

R07-R167

04 040 200

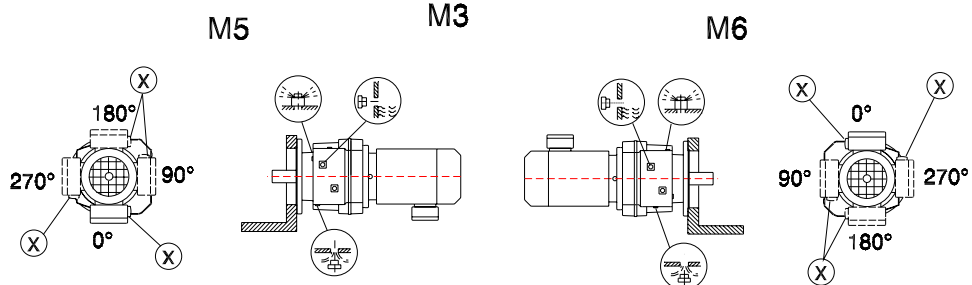
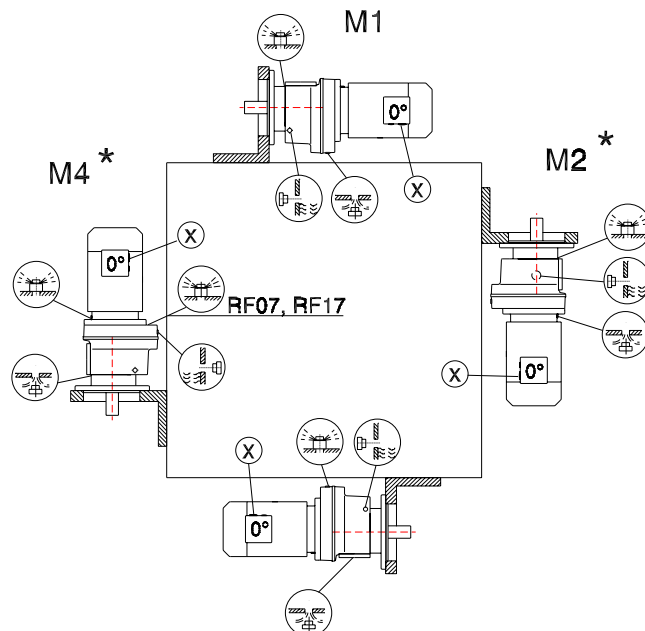
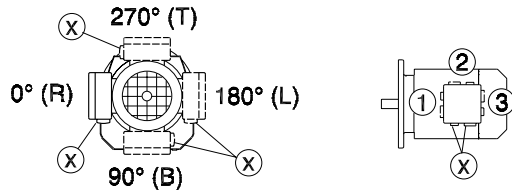


R07		M1, M2, M3, M5, M6
R17, R27		M1, M3, M5, M6
R07, R17, R27		
R47, R57		M5

* → page 35

RF07-RF167

04 041 200

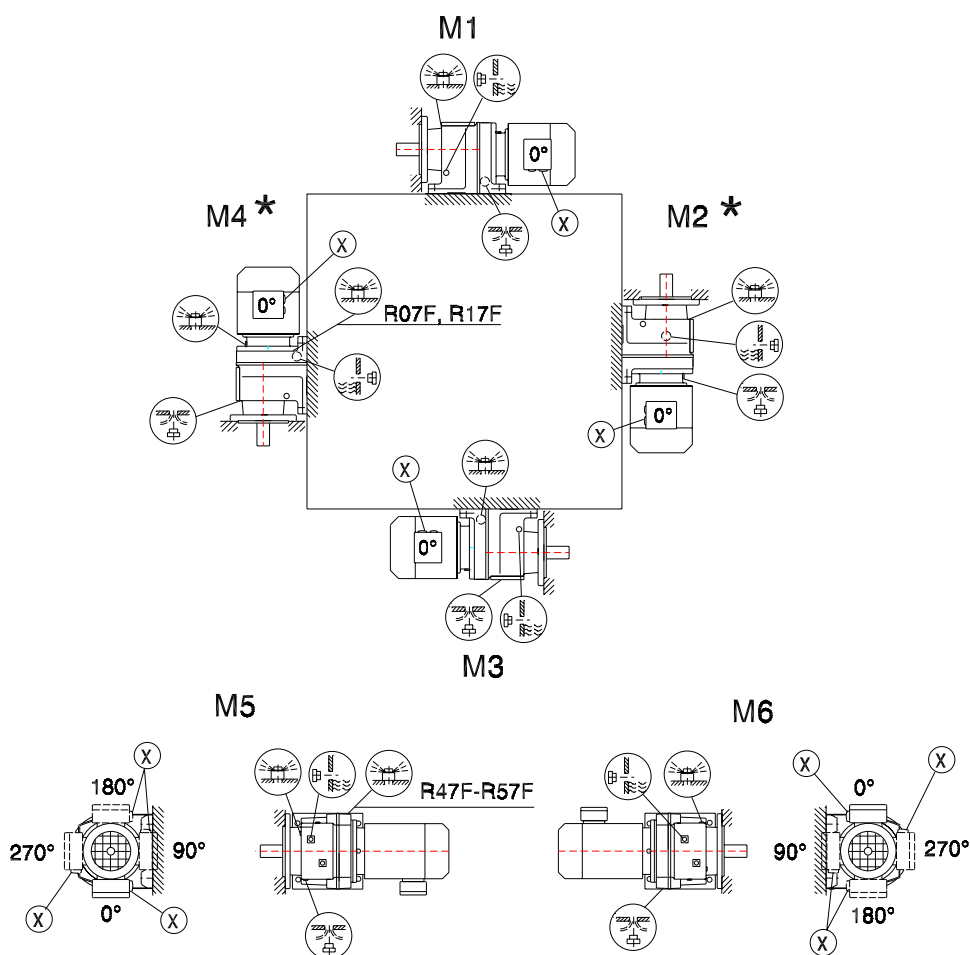
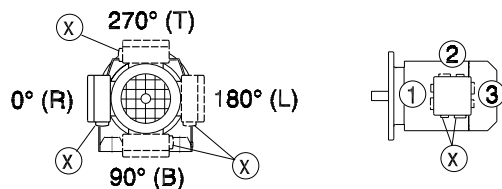


RF07		M1, M2, M3, M5, M6
RF17, RF27		M1, M3, M5, M6
RF07, RF17, RF27		
RF47, RF57		M5

* → page 35

R07F-R87F

04 042 200



R07F		M1, M2, M3, M5, M6
R17F, R27F		M1, M3, M5, M6
R07F, R17F, R27F		
R47F, R57F		M5

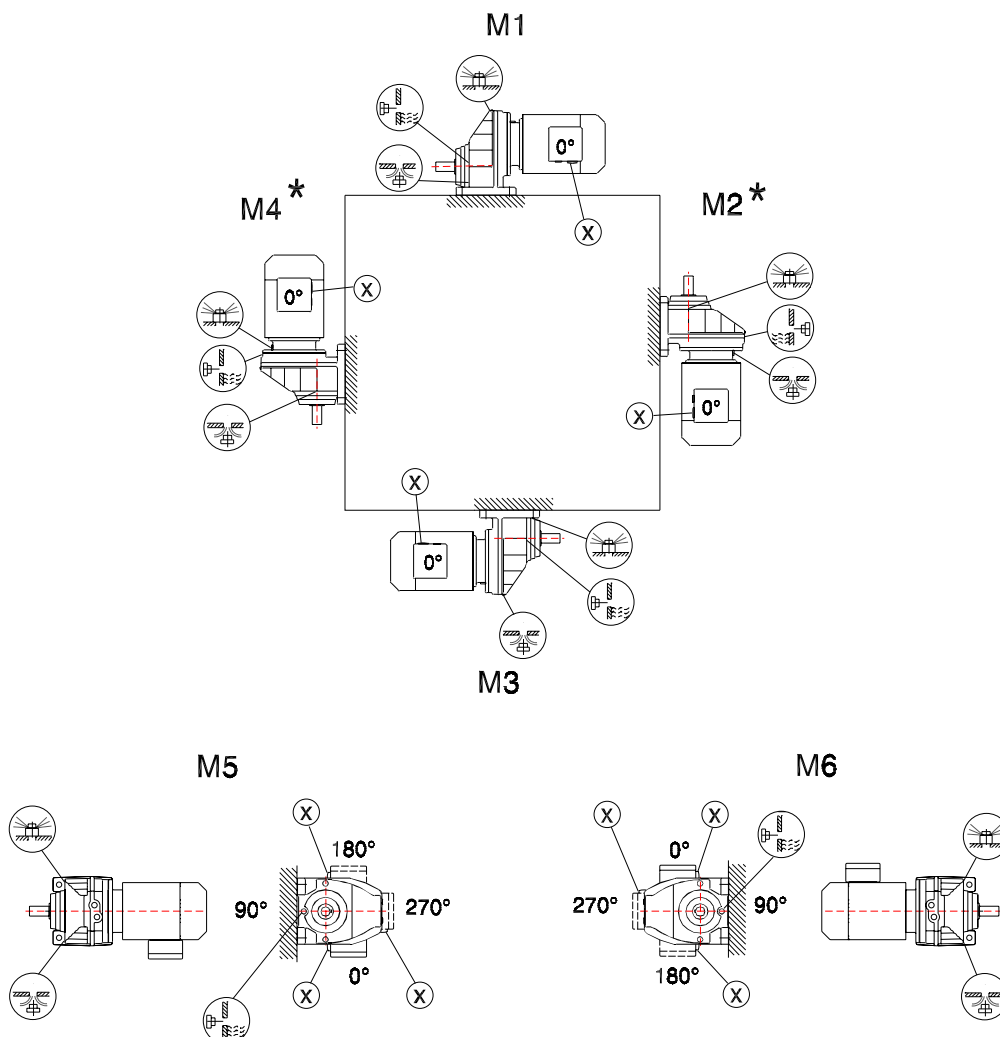
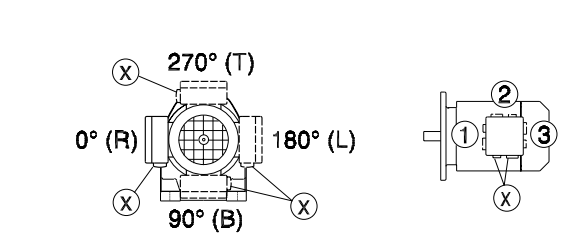
* → page 35

Important: Please refer to the information in the 'Geared Motors' catalog, Sec. 'Project Planning for Gear Units/Overhung and axial loads'.

8.4 Mounting positions of RX helical gear units

RX57-RX107

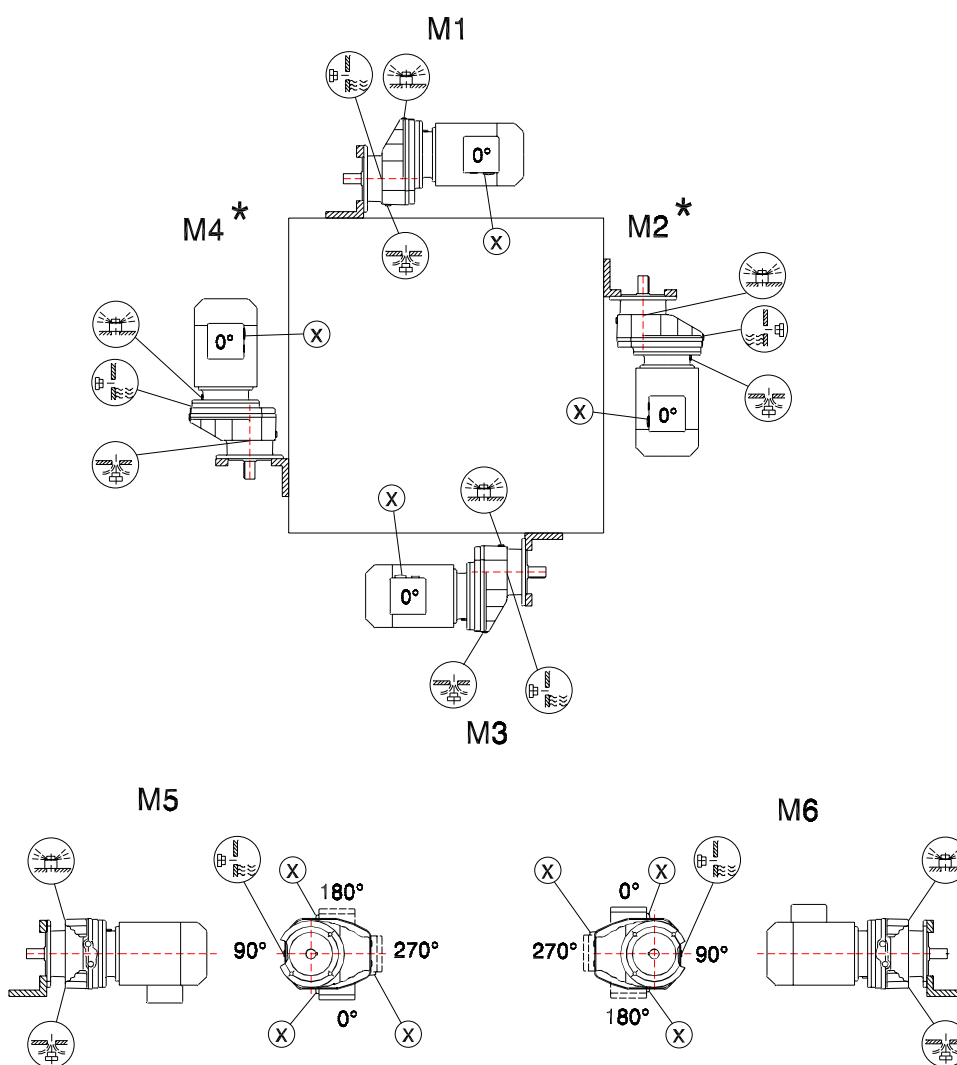
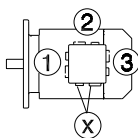
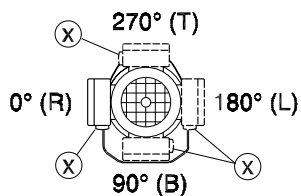
04 043 200



* → page 35

RXF57-RXF107

04 044 200

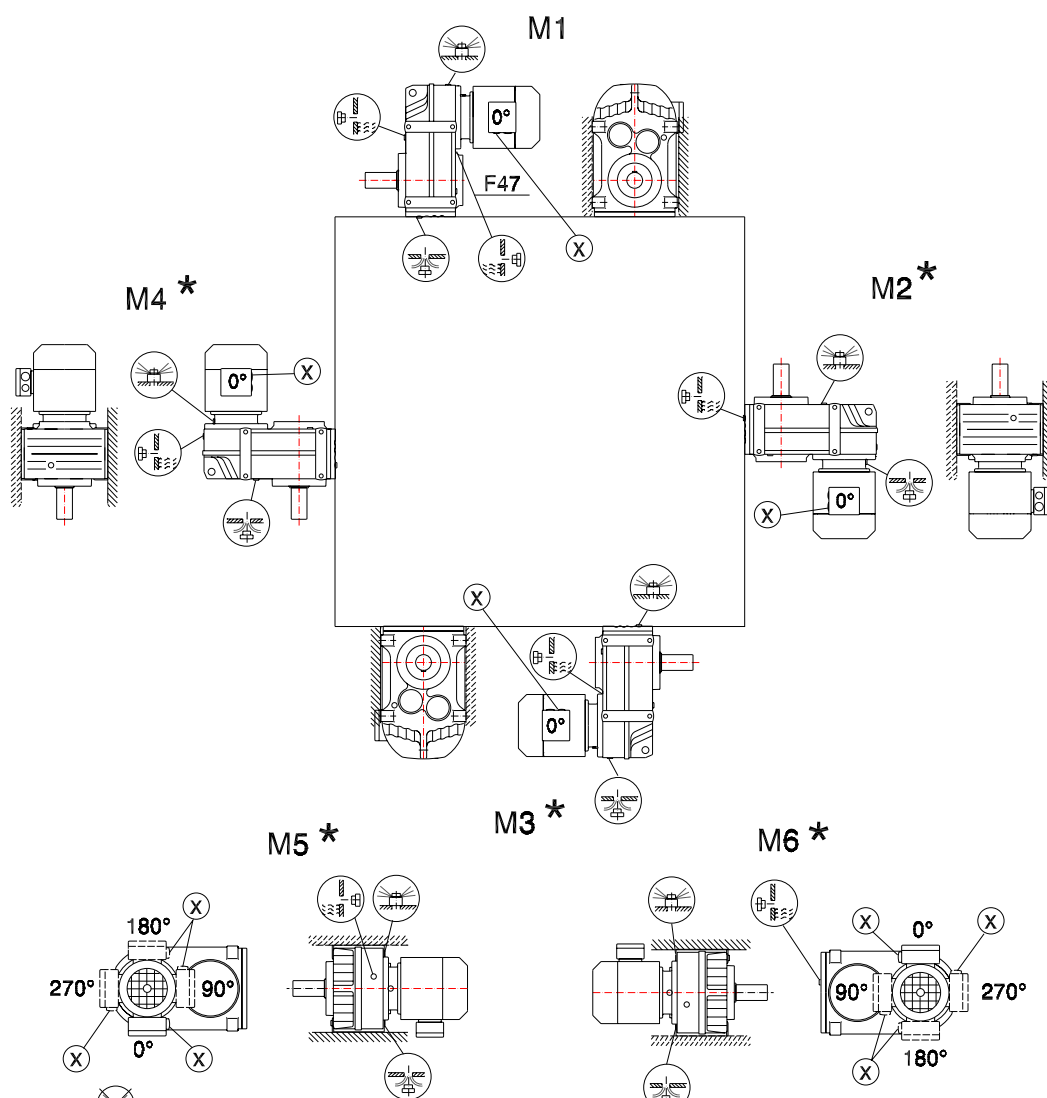
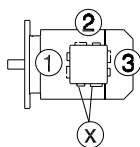
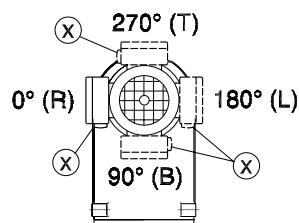


* → page 35

8.5 Mounting positions of parallel shaft helical gear units

F/FA..B/FH27B-157B, FV27B-107B

42 042 200



F..27 M1, M3, M5, M6

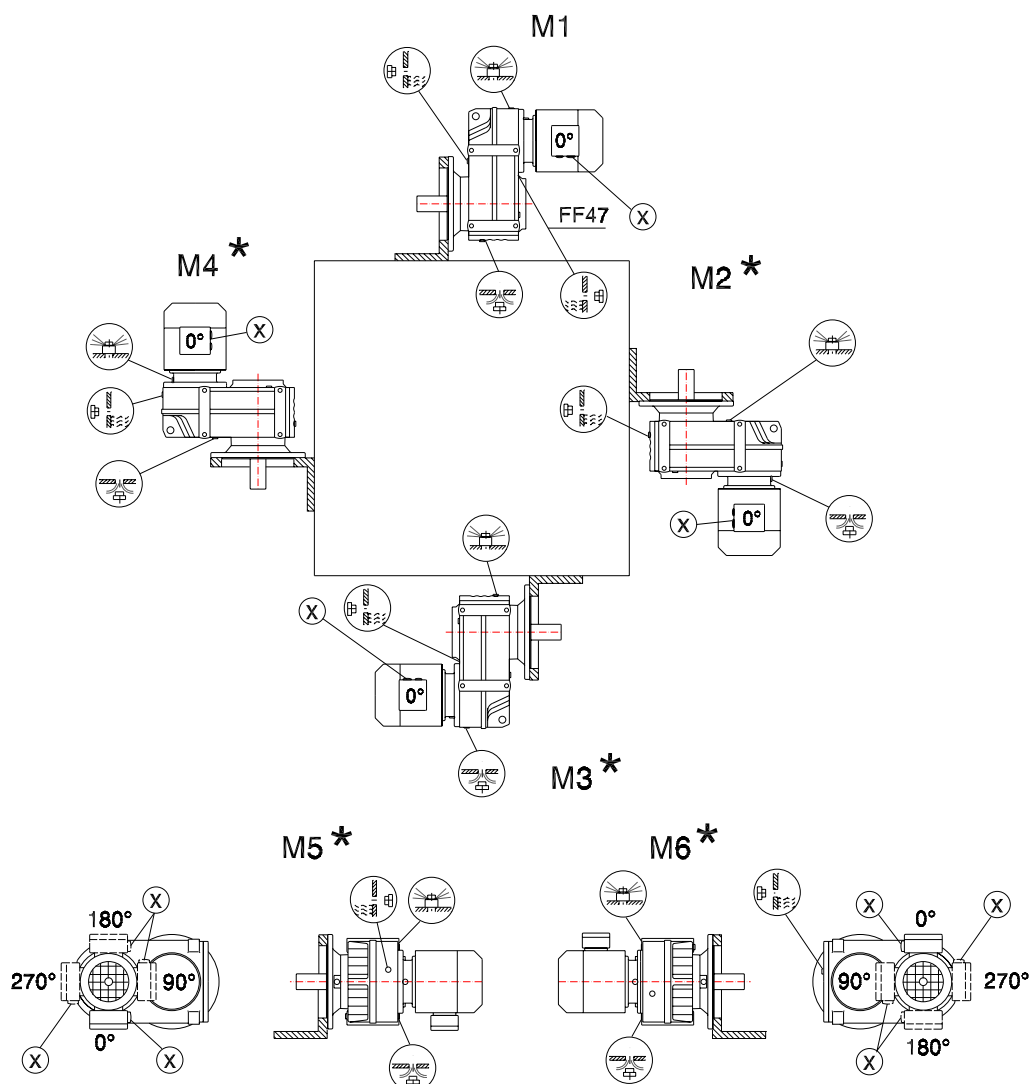
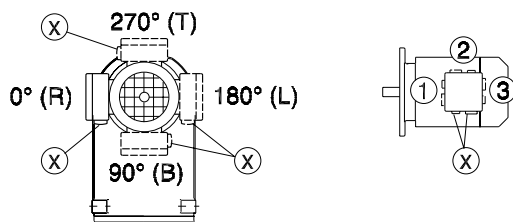
F..27 M1 - M6

F..27 M1, M3, M5, M6

* → page 35

FF/FAF/FHF/FAZ/FHZ27-157, FVF/FVZ27-107

42 043 200

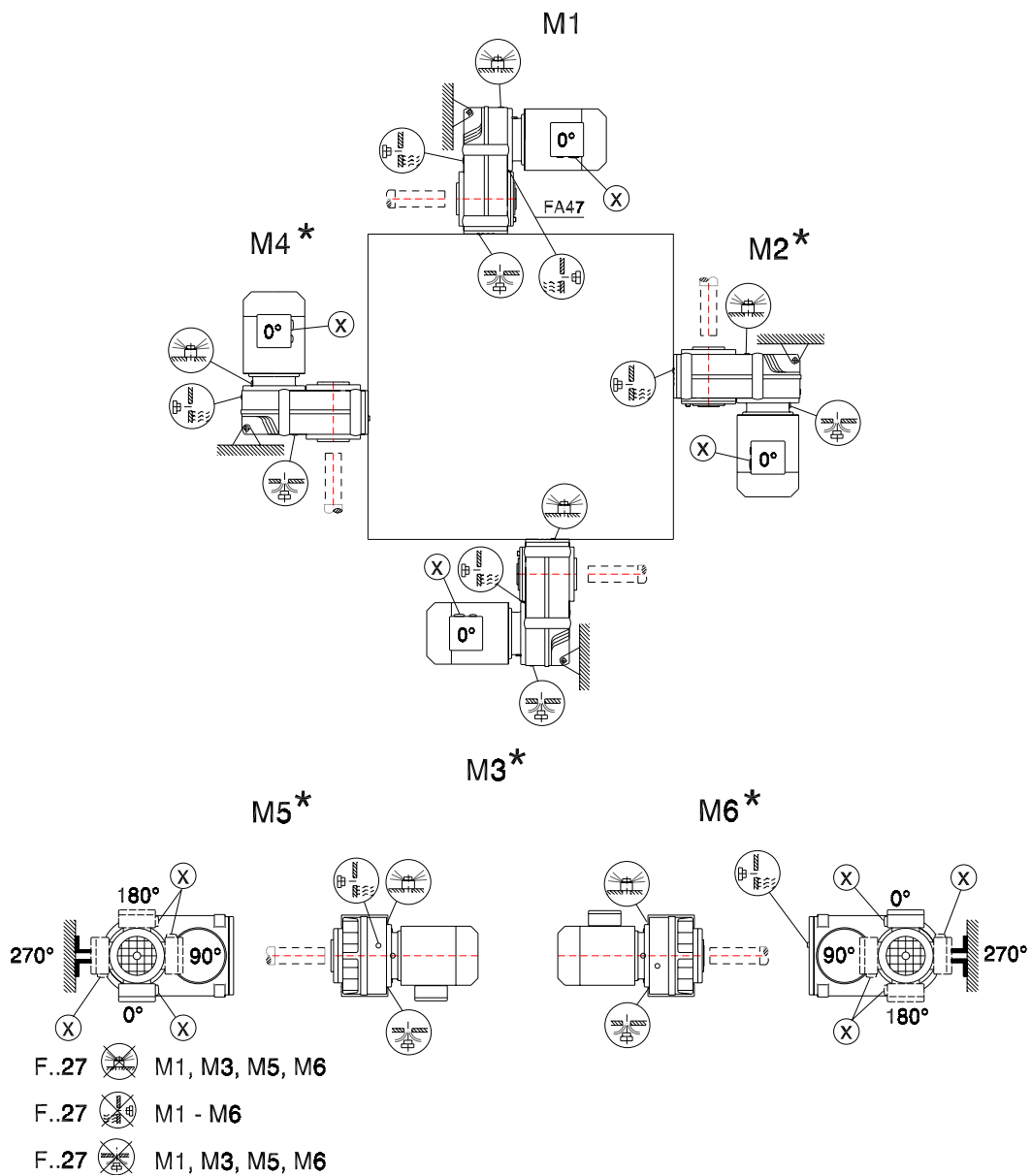
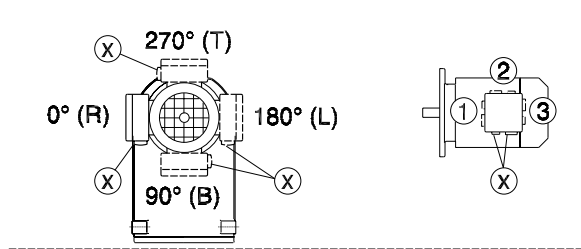


- F..27 M1, M3, M5, M6
- F..27 M1 - M6
- F..27 M1, M3, M5, M6

* → page 35

FA/FH27-157, FV27-107

42 044 200

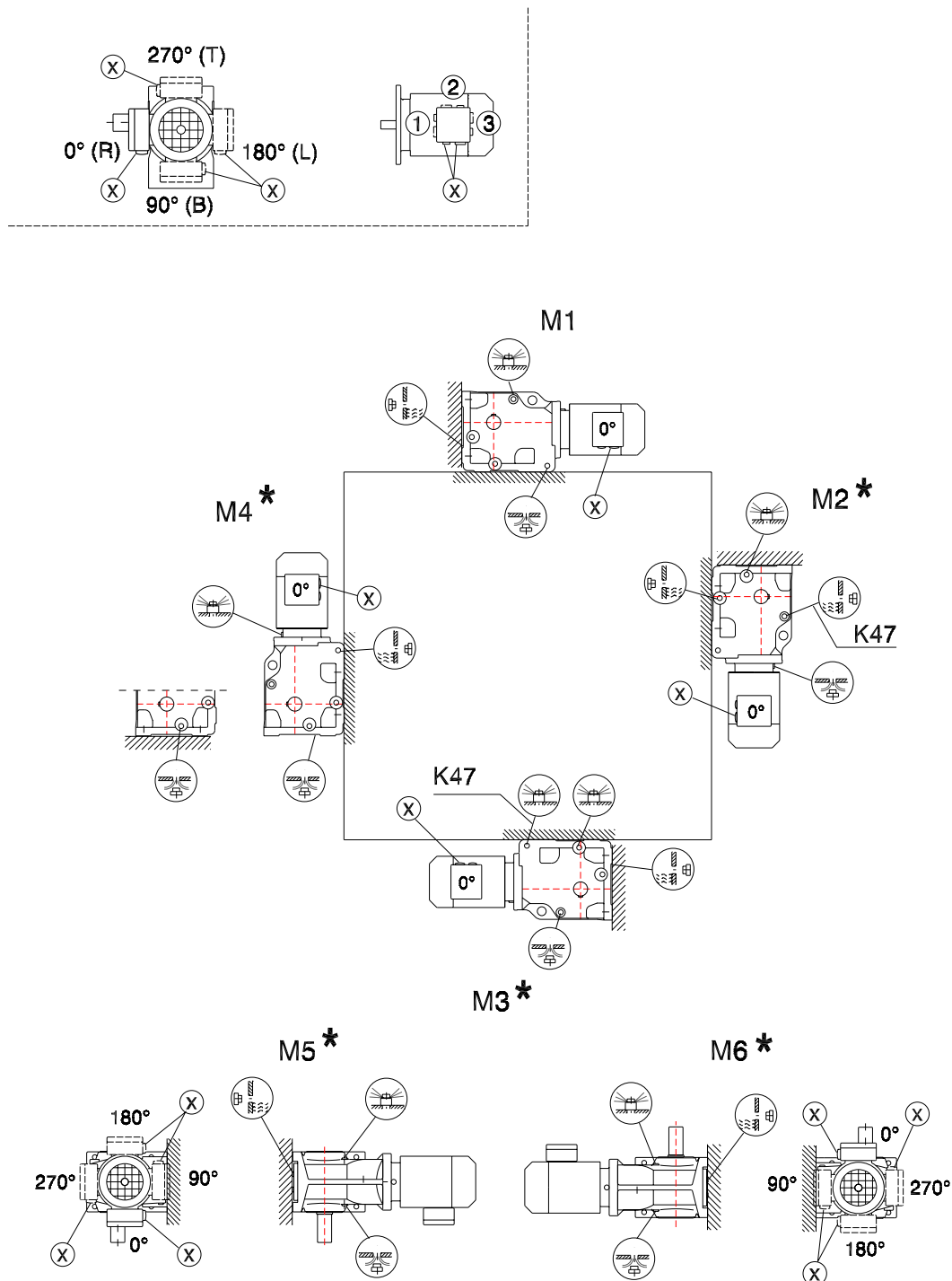


* → page 35

8.6 Mounting positions of helical-bevel gear units

K/KA..B/KH37B-157B, KV37B-107B

34 025 200

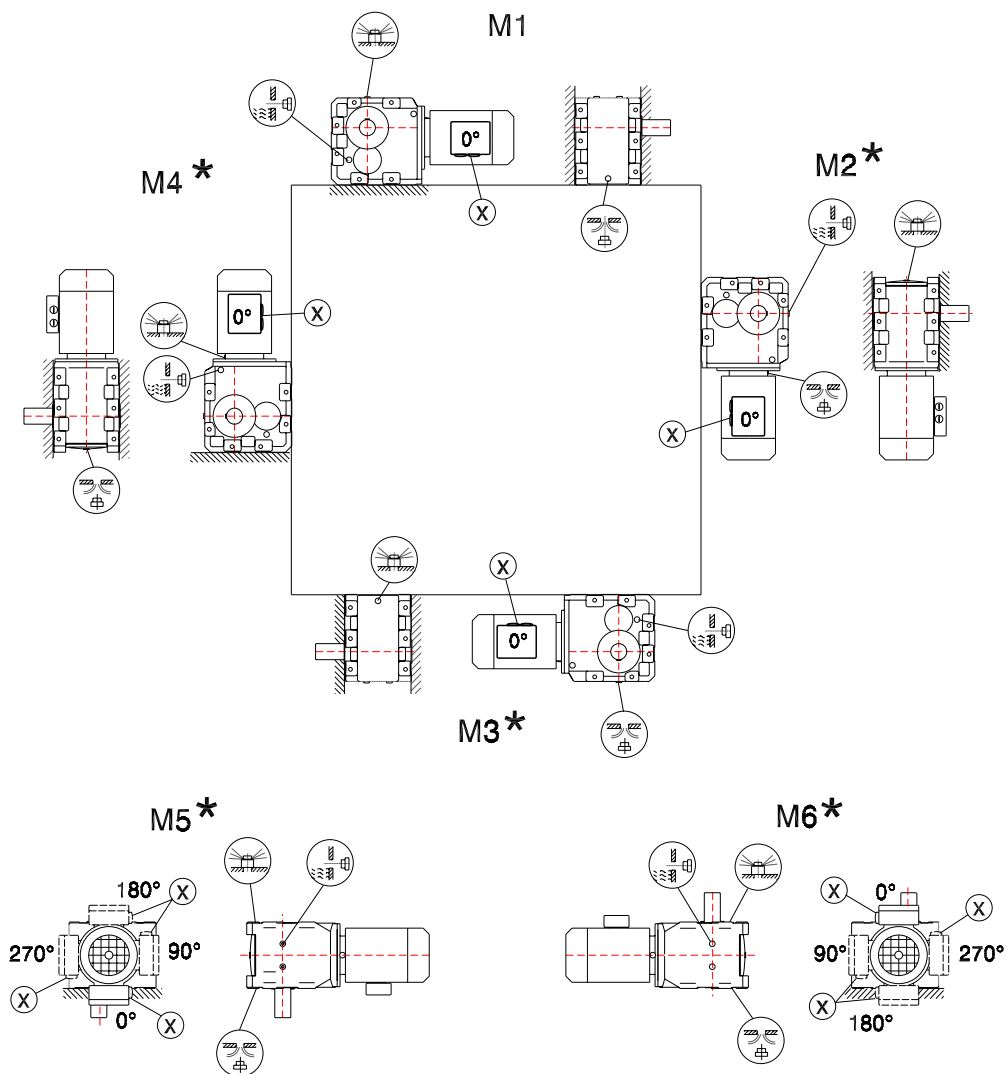
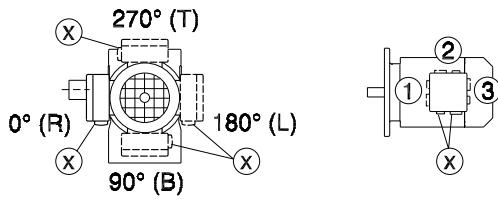


* → page 35

Important: Please refer to the  information in the 'Geared Motors' catalog, Sec. 'Project Planning for Gear Units/ Overhung and axial loads'.

K167-187, KH167B-187B

34 026 200

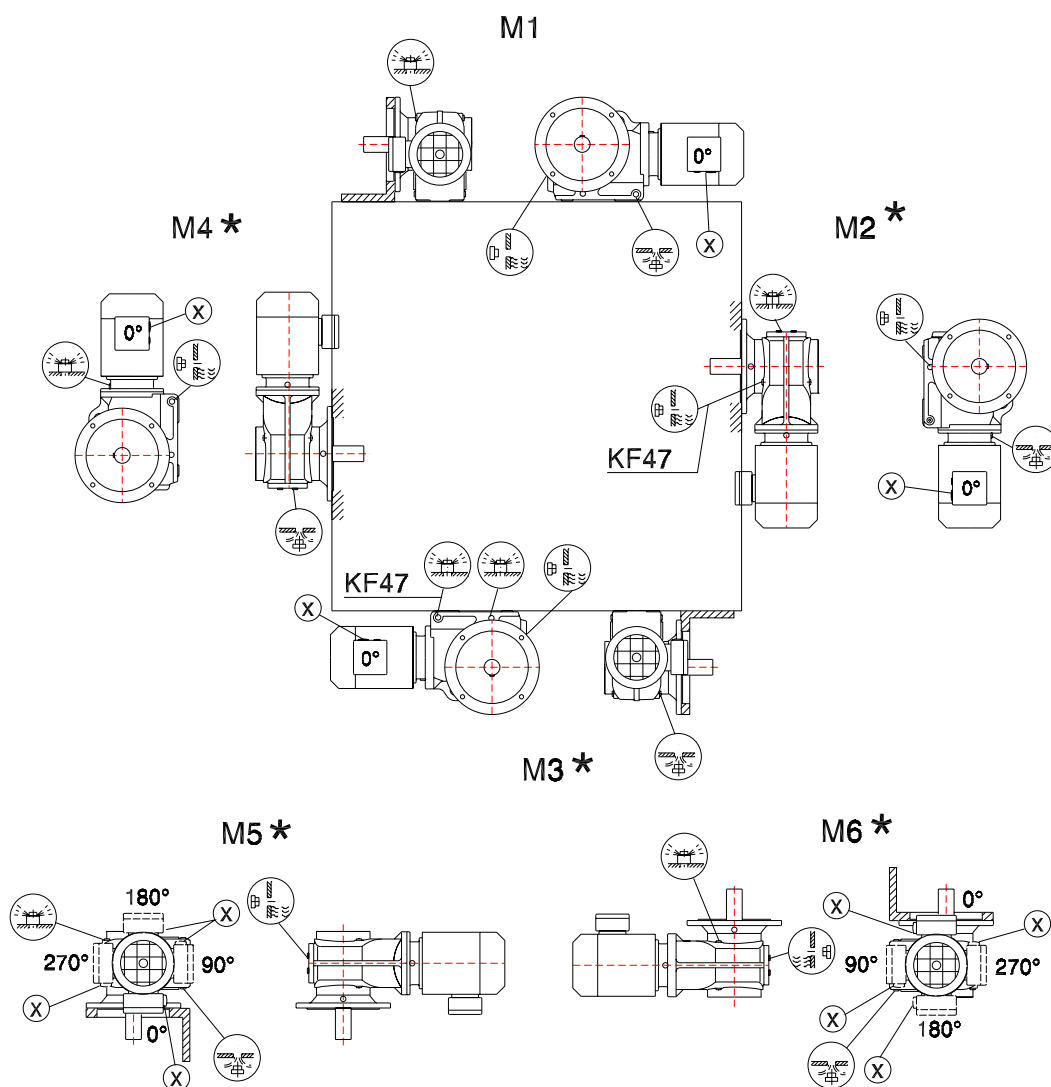
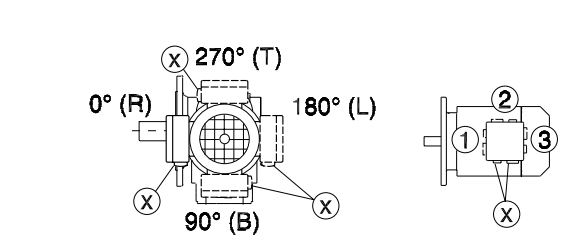


* → page 35

Important: Please refer to the  information in the 'Geared Motors' catalog, Sec. 'Project Planning for Gear Units/ Overhung and axial loads'.

KF/KAF/KHF/KAZ/KHZ37-157, KVF/KVZ37-107

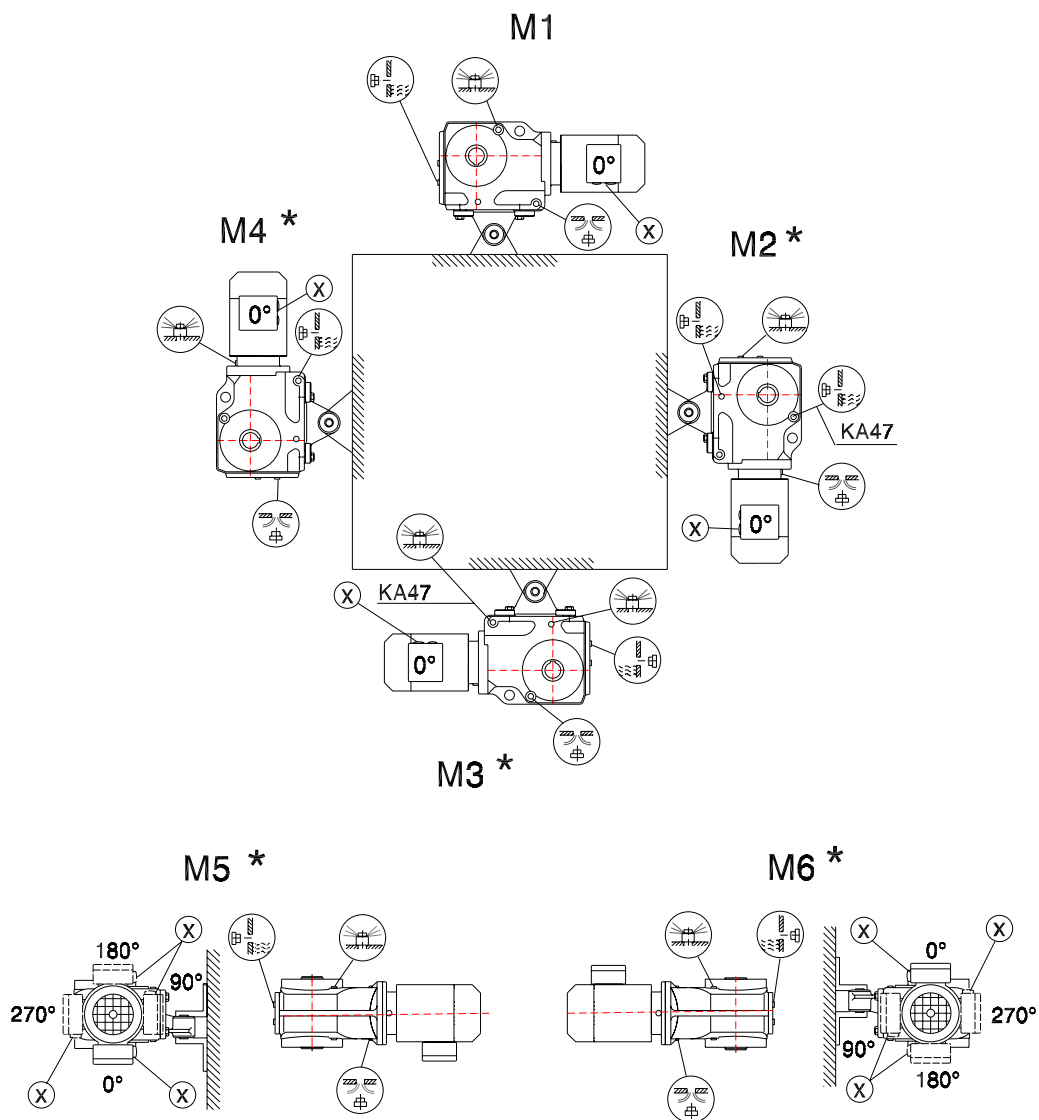
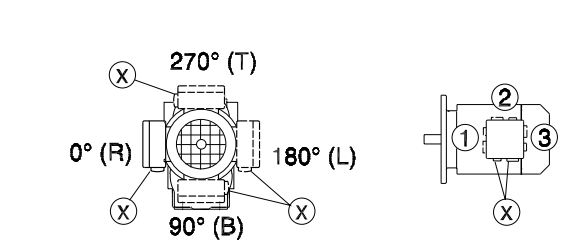
34 027 200



* → page 35

KA/KH37-157, KV37-107

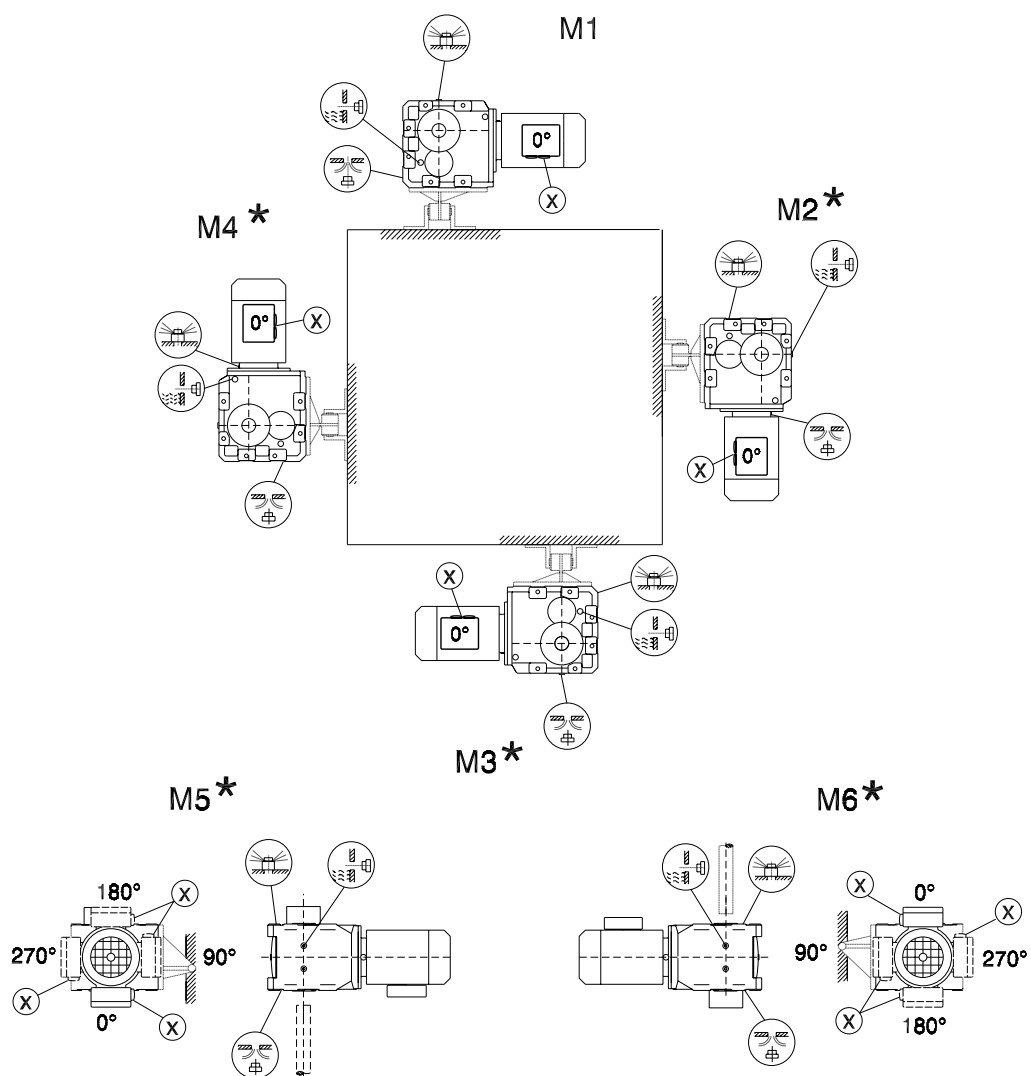
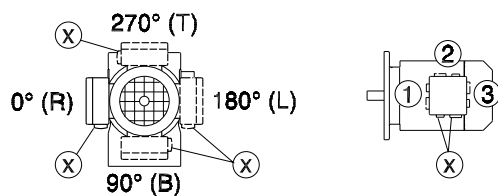
39 025 200



* → page 35

KH167-187

39 026 200

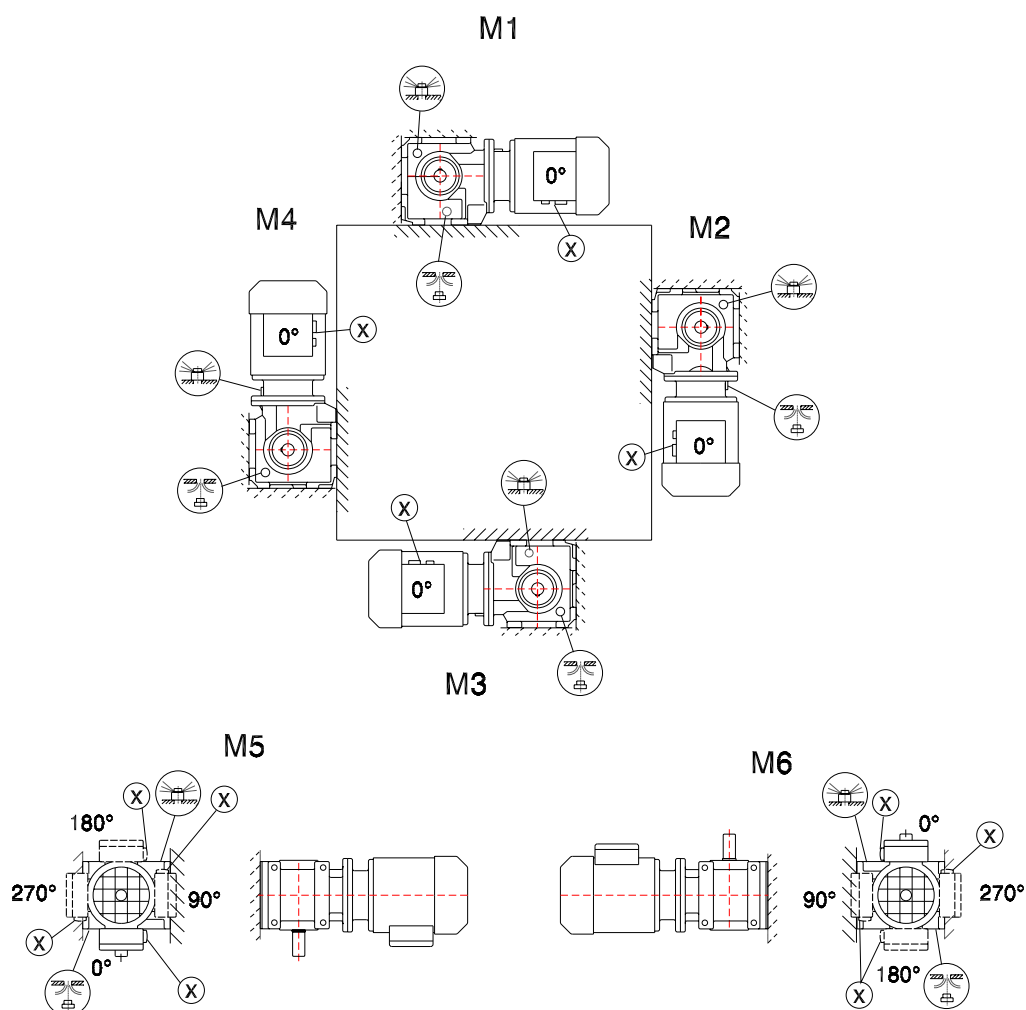
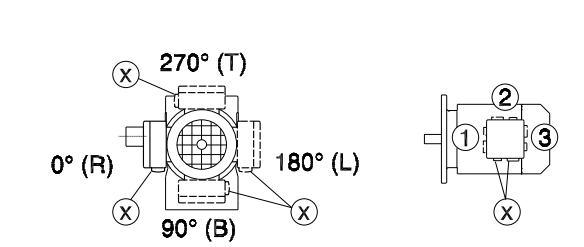


* → page 35

8.7 Mounting positions of helical-worm gear units

S37

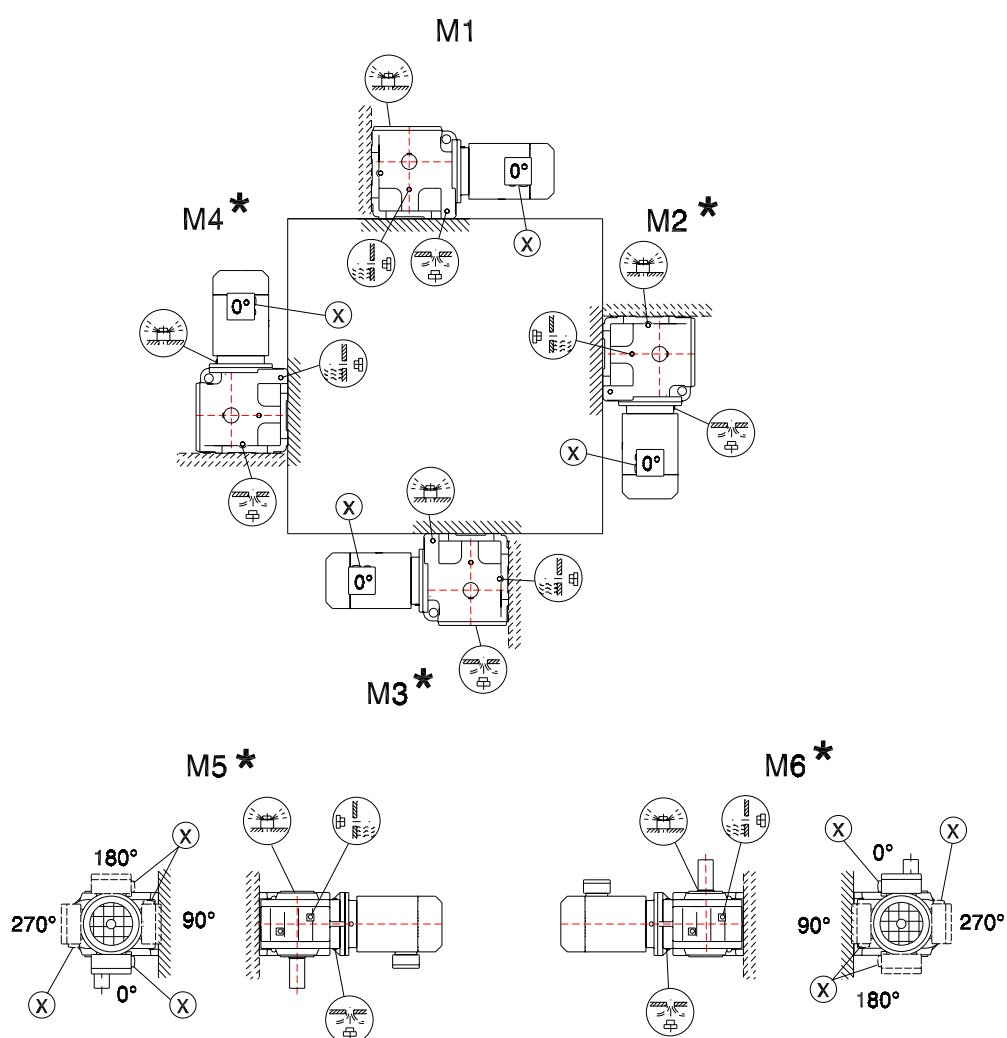
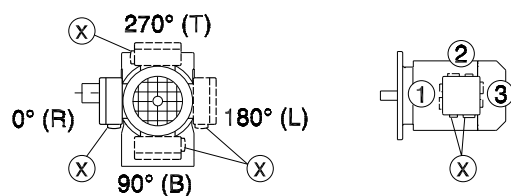
05 025 200



Important: Please refer to the  information in the 'Geared Motors' catalog, Sec. 'Project Planning for Gear Units/ Overhung and axial loads'.

S47-S97

05 026 200

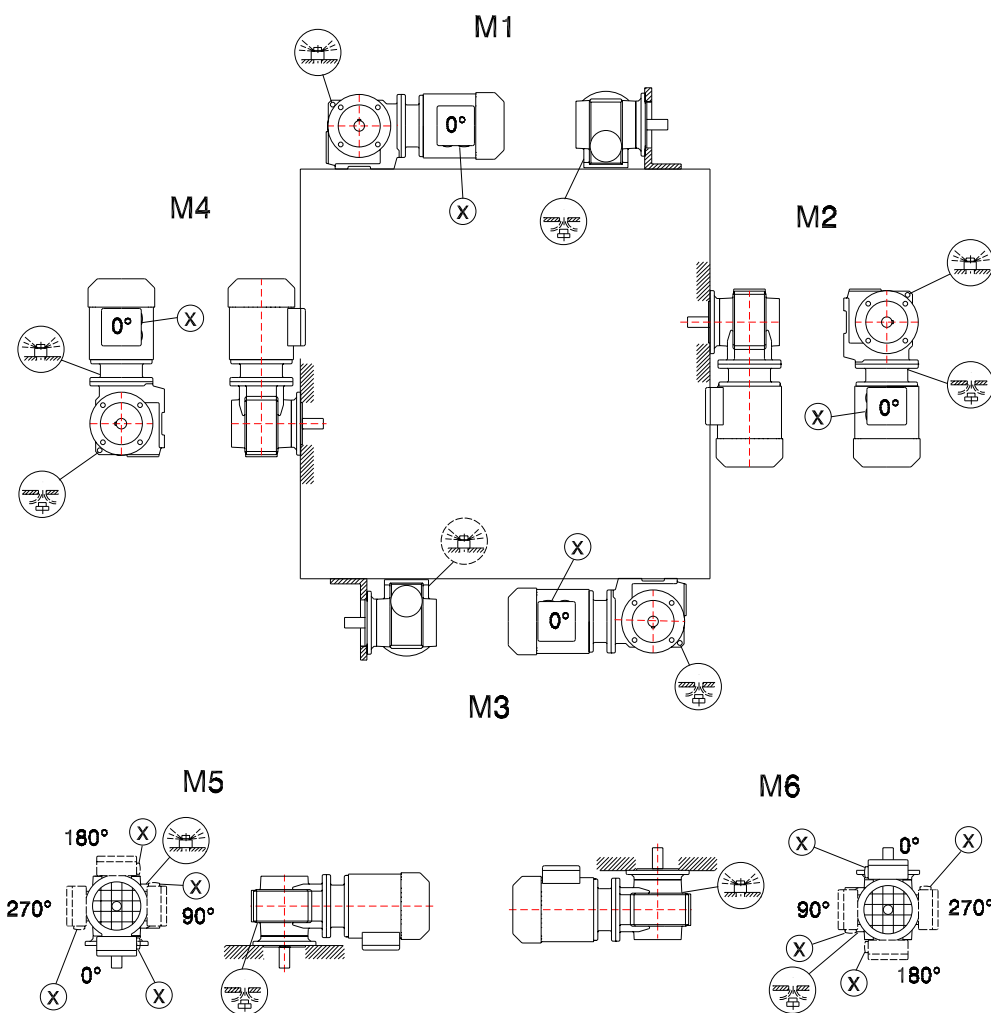
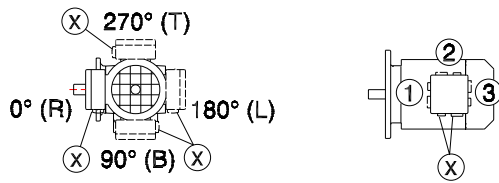


* → page 35

Important: Please refer to the  information in the 'Geared Motors' catalog, Sec. 'Project Planning for Gear Units/Overhung and axial loads'.

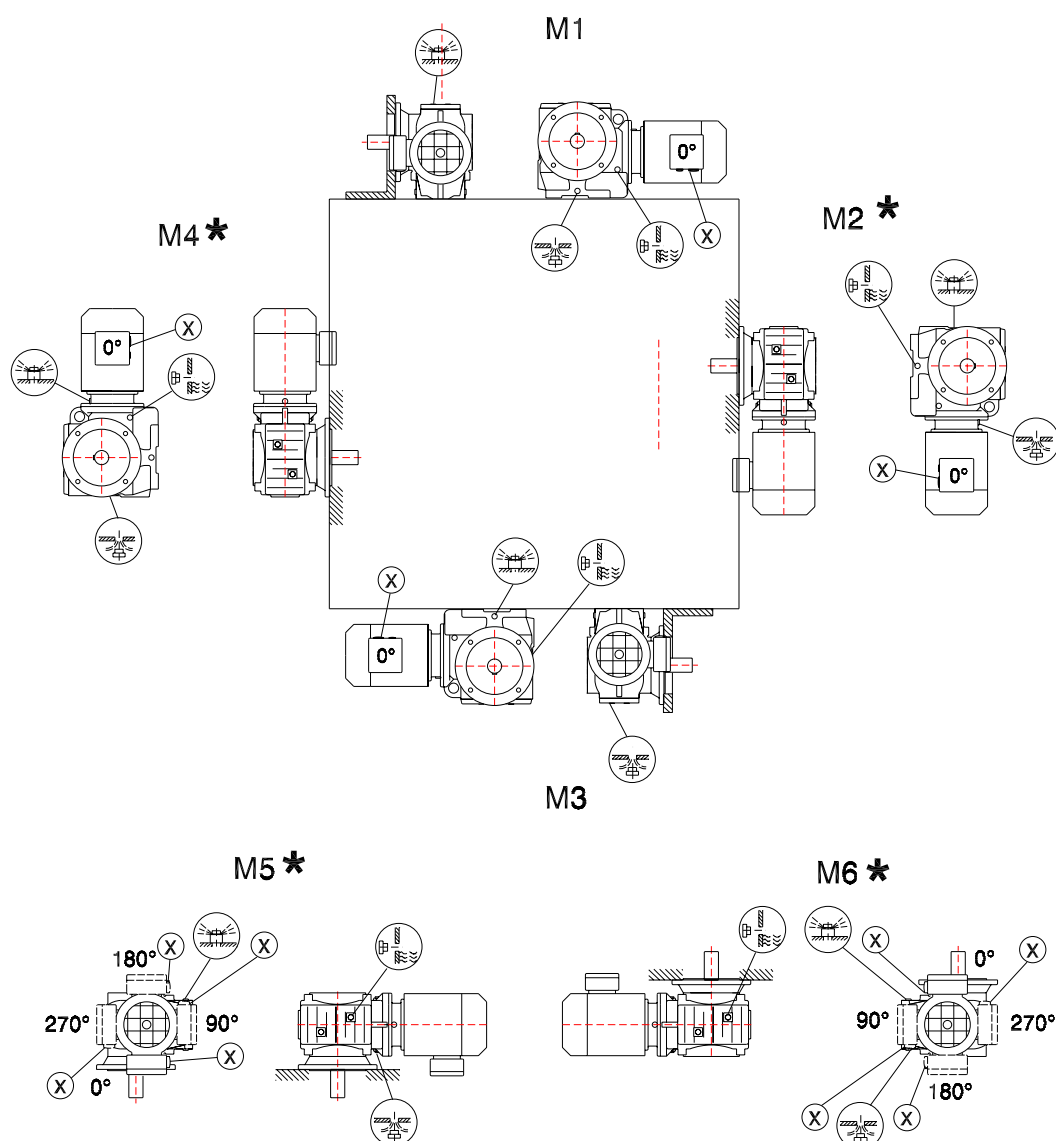
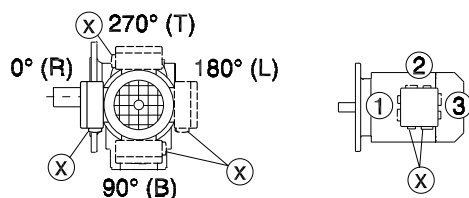
SF/SAF/SHF37

05 027 200



SF/SAF/SHF/SAZ/SHZ47-97

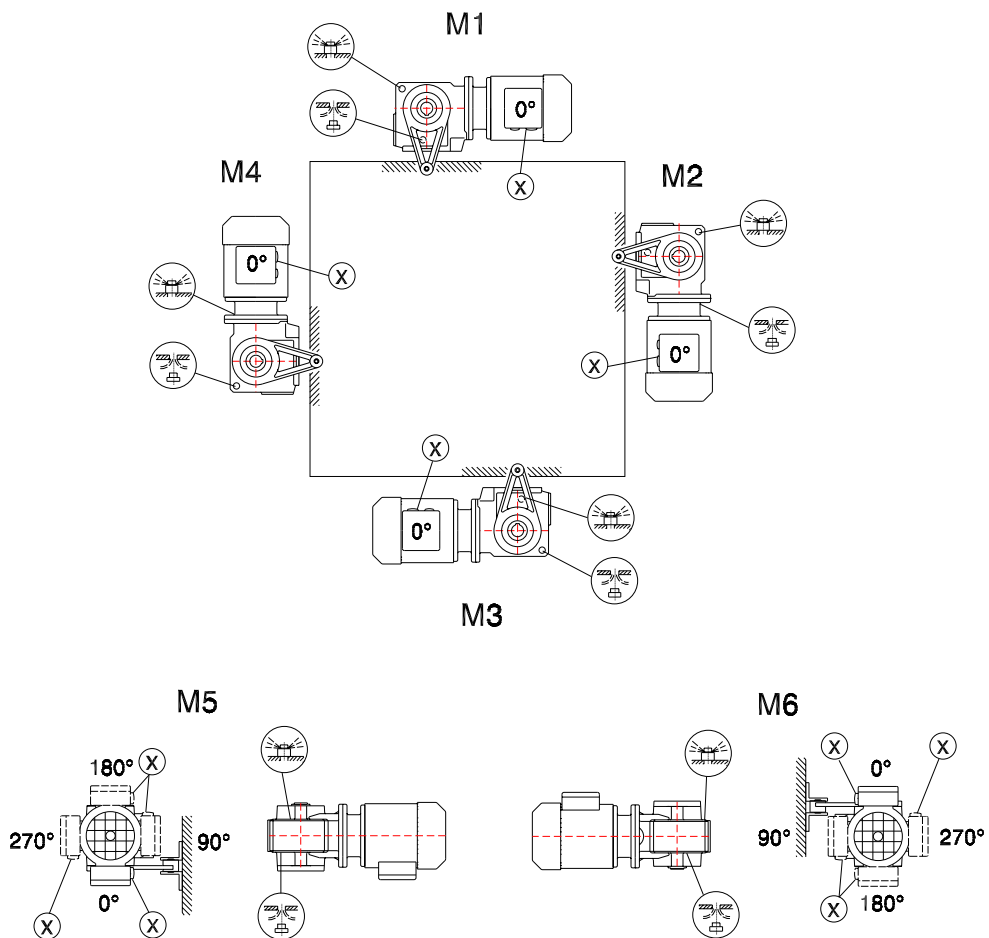
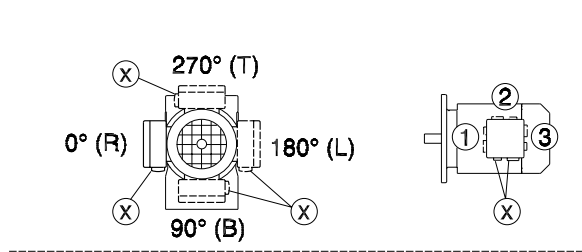
05 028 200



* → page 35

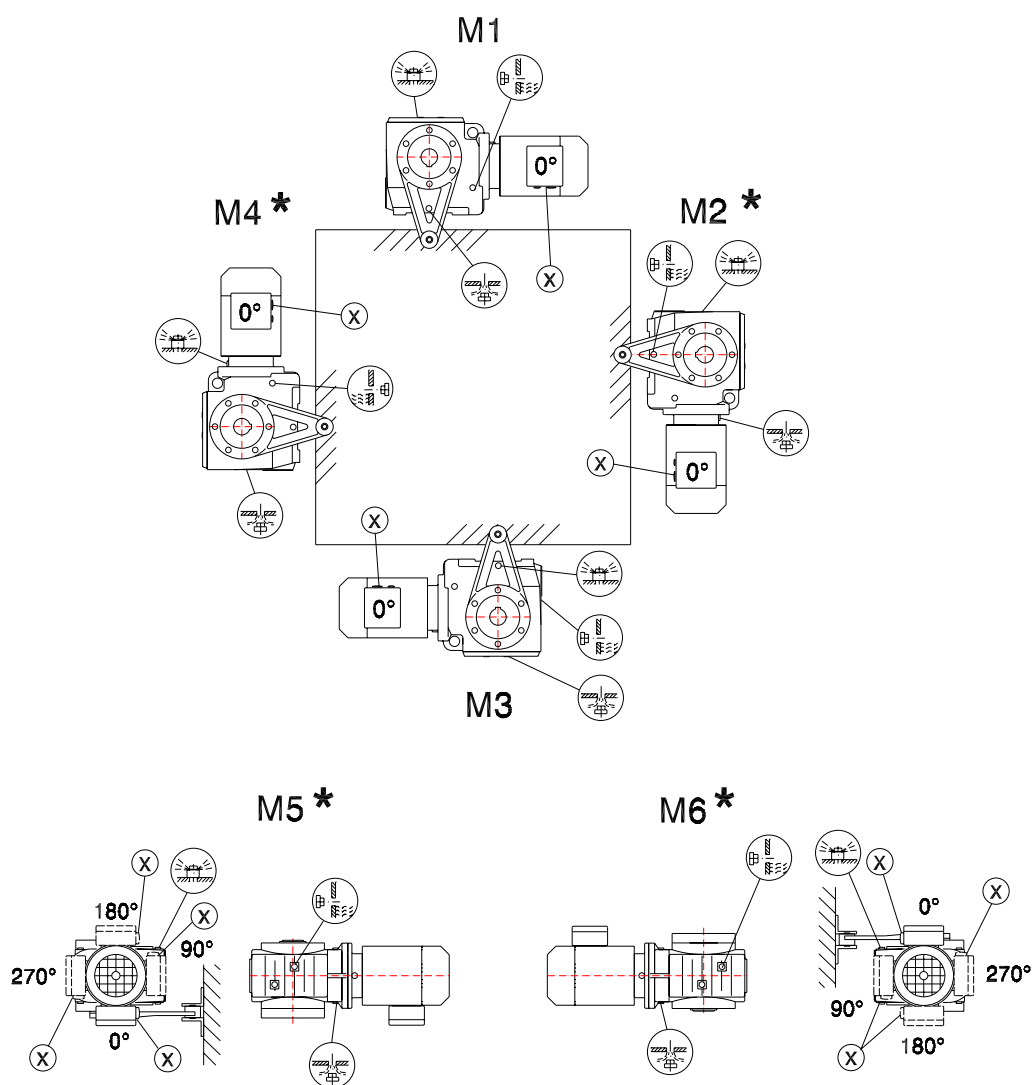
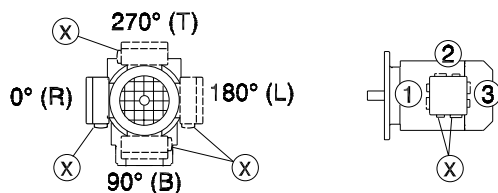
SA/SH37

28 020 200



SA/SH47-97

28 021 200

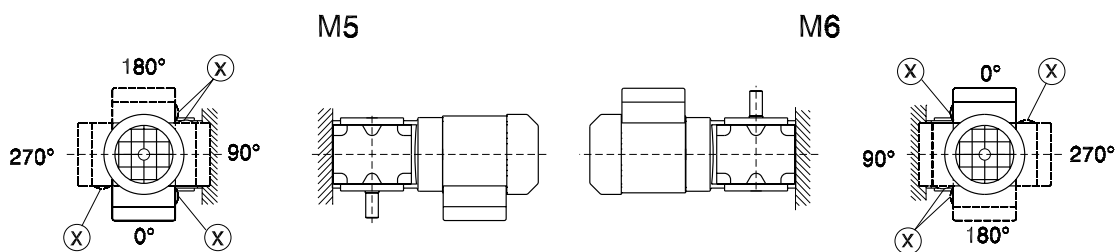
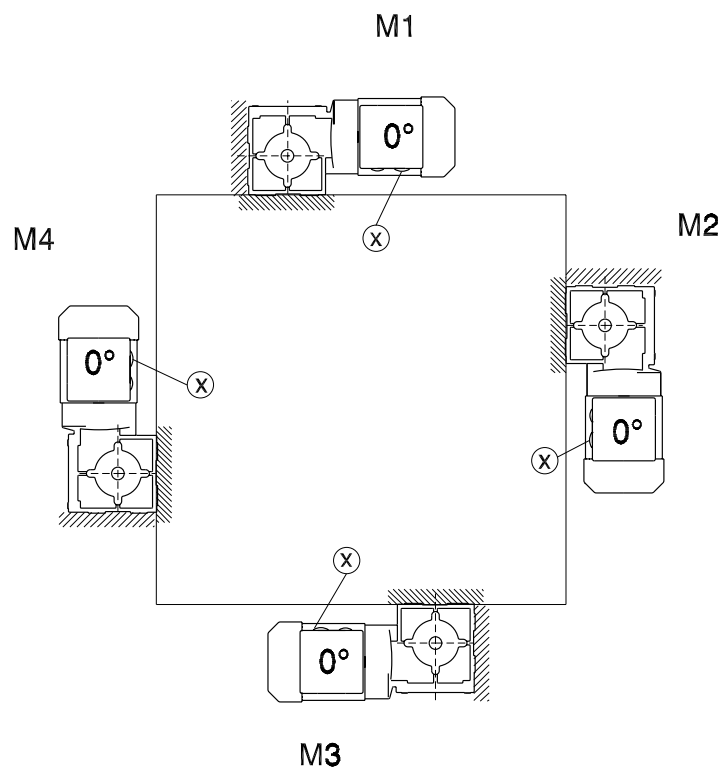
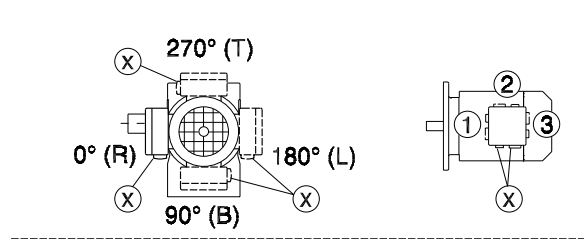


* → page 35

8.8 Mounting positions of Spiroplan® W gear units

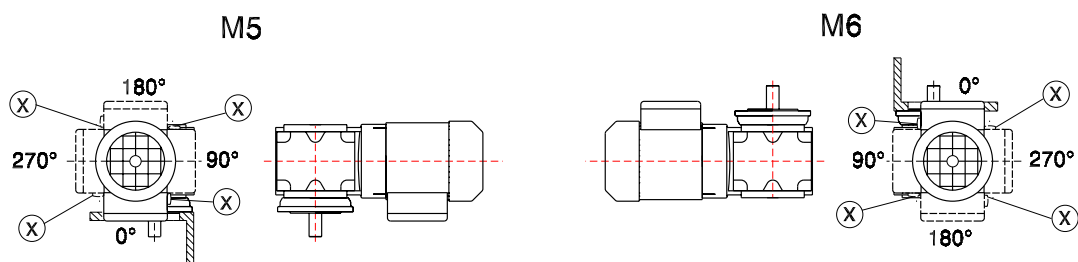
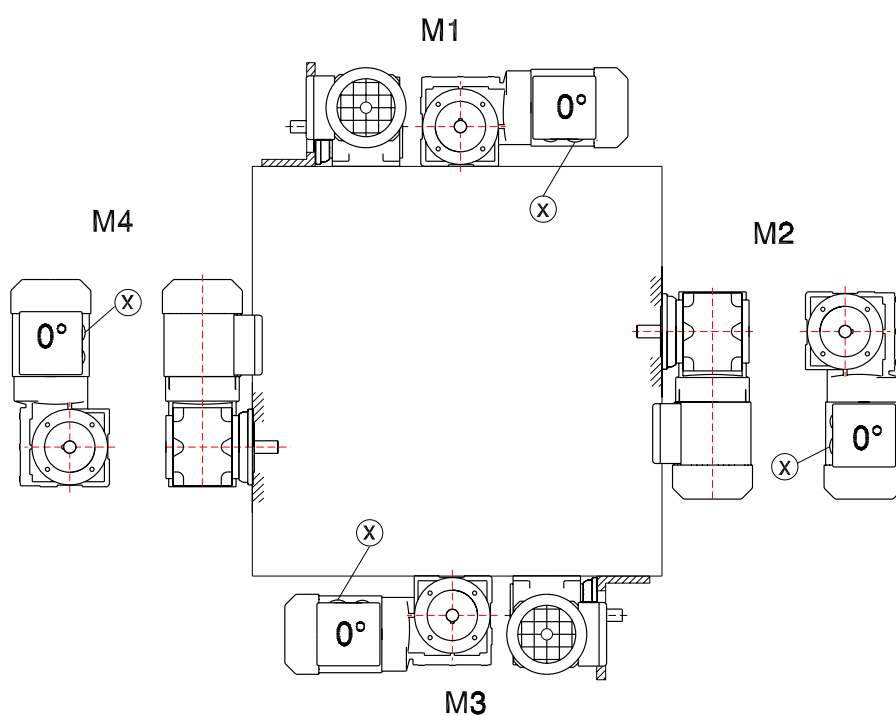
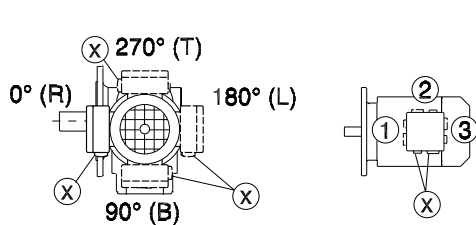
W10-30

20 001 002



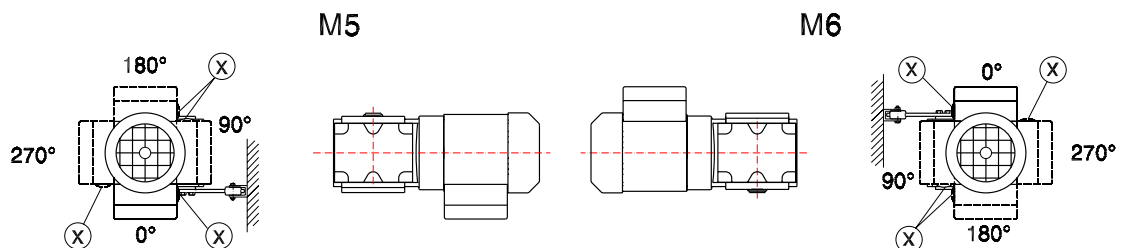
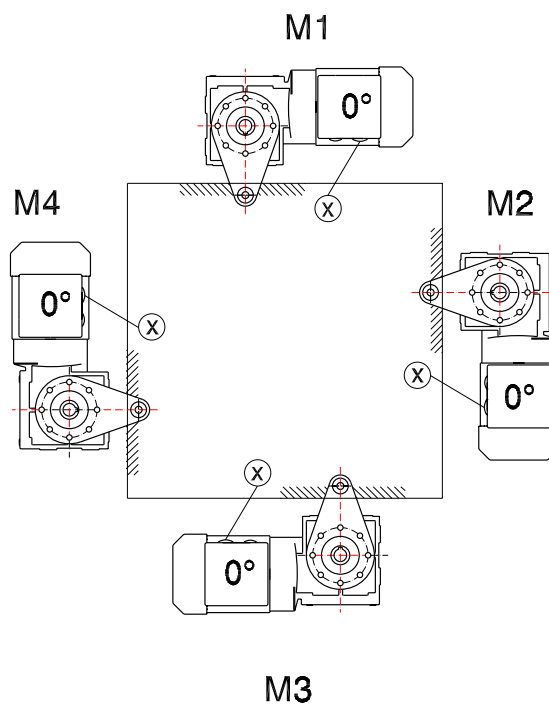
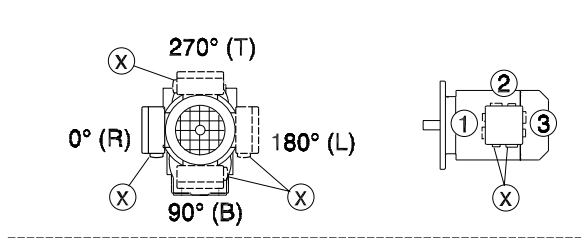
WF/WAF10-30

20 002 002



WA10-30

20 003 002





9 Lubricants

General information

Unless a special arrangement is made, SEW-EURODRIVE supplies the drives with a lubricant fill specifically for the gear unit and mounting position. The decisive factor is the mounting position (M1...M6, → Sec. 'Mounting Positions and Important Order Information') specified when ordering the drive. You must adapt the lubricant fill to any subsequent changes made to the mounting position (→ Lubricant fill quantities).

Lubricant table

The lubricant table on the following page shows the permitted lubricants for SEW-EURODRIVE gear units. Please note the following key to the lubricant table.

Key to the lubricant table

Abbreviations used, meaning of shading and notes:

CLP	= Mineral oil
CLP PG	= Polyglycol (W gear units, conforms to USDA-H1)
CLP HC	= Synthetic hydrocarbons
E	= Ester oil (water pollution danger category WGK 1)
HCE	= Synthetic hydrocarbons + ester oil (USDA - H1 certification)
HLP	= Hydraulic oil
	= Synthetic lubricant (= synthetic-based anti-friction bearing grease)
	= Mineral lubricant (= mineral-based anti-friction bearing grease)
1)	Helical-worm gear units with PG oil: Please contact SEW
2)	Special lubricant for Spiroplan® gear units only
3)	Recommendation: Select SEW $f_B \geq 1.2$
4)	Note critical starting behavior at low temperatures!
5)	Low-viscosity grease
6)	Ambient temperature
	Lubricant for the foodstuffs industry (food industry oil)
	Biodegradable oil (lubricant for use in agriculture, forestry and water resources)

Anti-friction bearing greases

The anti-friction bearings in gear units and motors are given a factory-fill with the greases listed below. SEW-EURODRIVE recommends regreasing anti-friction bearings with a grease fill at the same time as changing the oil.

	Ambient temperature	Manufacturer	Type
Anti-friction bearing in gear unit	-30 °C +60 °C	Mobil	Mobilux EP 2
	-40 °C +80 °C	Mobil	Mobiltemp SHC 100
Anti-friction bearing in motor	-25 °C +80 °C	Esso	Unirex N3
	-25 °C +60 °C	Shell	Alvania R3
	+80 °C +100 °C	Klüber	Barrierta L55/2
	-45 °C -25 °C	Shell	Aero Shell Grease 16
Special greases for anti-friction bearings in gear units:			
	-30 °C +40 °C	Aral	Aral Eural Grease EP 2
	-20 °C +40 °C	Klüber	Klüberbio M32-82



The following grease quantities are required:

- For fast-running bearings (motor and gear unit input end): Fill the cavities between the rolling elements one third full with grease.
- For slow-running bearings (in gear units and gear unit output end): Fill the cavities between the rolling elements two thirds full with grease.



Lubricant table

01 805 792

	6) °C -50 0 +50 +100 Standard -10 +40	DIN (ISO) Oil	ISO, NLGI	Mobil®	Shell	Klüber LUBROL	ARAL	BP	Tribol	TEXACO	Optimat	FUCHS
R...		CLP (CC)	VG 220	Mobilgear 630	Shell Omala 220	Klüberoil GEM 1-220	Aral Degol BG 220	BP Energol GR-XP 220	Tribol 1100/220	Meropa 220	Optigear BM 220	Renolin CLP 220
K... (HK...)	+80	CLP PG	VG 220	Mobil Glygoyle 30	Shell Tivela WB	Klüberoil GH 6-220	Aral Degol GS 220	BP Energol SG-XP 220	Tribol 800/220	Synlube CLP 220	Optiflex A 220	
F...	-40	CLP HC	VG 220	Mobil SHC 630	Shell Omala 220 HD	Klüberoil EG 4-220	Aral Degol PAS 220		Tribol 1510/220	Pinnacle EP 220	Optigear Synthetic A 220	Renolin Unisyn CLP 220
	-40		VG 150	Mobil SHC 629		Klüberoil EG 4-150				Pinnacle EP 150		
	-20	CLP (CC)	VG 150	Mobilgear 627	Shell Omala 100	Klüberoil GEM 1-150	Aral Degol BG 100	BP Energol GR-XP 100	Tribol 1100/100	Meropa 150	Optigear BM 100	Renolin CLP 150
	-30	HLP (HM)	VG 68-46	Mobil D.T.E. 13M	Shell Tellus T 32	Klüberoil GEM 1-68	Aral Degol BG 46		Tribol 1100/68	Rando EP Ashless 46	Optigear 32	Renolin B 46 HVI
	-40	CLP HC	VG 32	Mobil SHC 624		Klüber-Summit HySyn FG-32				Cetus PAO 46		
	-40	HLP (HM)	VG 22	Mobil D.T.E. 11M	Shell Tellus T 15	Isotex MT 30 ROT		BP Energol HLP-HM 10		Rando HDZ 15		
	Standard	CLP (CC)	VG 680	Mobilgear 636	Shell Omala 680	Klüberoil GEM 1-680	Aral Degol BG 680	BP Energol GR-XP 680	Tribol 1100/680	Meropa 680	Optigear BM 680	Renolin CLP 680
S... (HS...)	-20	CLP PG	VG 680 ¹⁾	Mobil Glygoyle HE 680		Klüberoil GH 6-680		BP Energol SG-XP 680	Tribol 800/680	Synlube CLP 680		
	-30	CLP HC	VG 460	Mobil SHC 634	Shell Omala 460 HD	Klüberoil EG 4-460				Pinnacle EP 460		
	-40		VG 150	Mobil SHC 629		Klüberoil EG 4-150				Pinnacle EP 150		
	-20	CLP (CC)	VG 150	Mobilgear 627	Shell Omala 100	Klüberoil GEM 1-150	Aral Degol BG 100	BP Energol GR-XP 100	Tribol 1100/100	Meropa 100	Optigear BM 100	Renolin CLP 150
	-25	CLP PG	VG 220 ¹⁾	Mobil Glygoyle 30		Klüberoil GH 6-220			Tribol 800/220	Synlube CLP 220	Optiflex A 220	
	-40	CLP HC	VG 32	Mobil SHC 624		Klüber-Summit HySyn FG-32				Cetus PAO 46		
R..., K... (HK...), F..., S... (HS...)	-30	HCE	VG 460		Shell Cassida Fluid GL 460	Klüberoil 4UH1-460	Aral Eural Gear 460				Optileb GT 460	
	-20	E	VG 460			Klüberoil CA2-460	Aral Degol BAB 460				Optisyn BS 460	
W... (HW...)	Standard	SEW PG	VG 460 ²⁾			Klüber SEW HT-460-5						
	-40	API GL5	SAE 75W90 (~VG 100)	Mobilube SHC 75 W90-LS								
	-20	CLP PG	VG 460 ³⁾			Klüberoil UH1 6-460						
R32 R302	-25	DIN 51 818 ⁵⁾	00	Glygoyle Grease 00	Shell Tivela Compound A	Klüberoil GE 46-1200				Multifak 6833 EP 00	Longtime PD 00	Renolin SF 7 - 041
	Standard		000 - 0	Mobilux EP 004	Shell Alvania GL 00		Aralub MFL 00	BP Energol LS-EP 00		Multifak EP 000		

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Lubricant fill quantities

The specified fill quantities are **recommended values**. The precise values vary depending on the stages and gear ratio. When filling, it is essential to check the **oil level plug since it indicates the precise oil capacity**.

The following tables show recommended values for lubricant fill quantities depending on the mounting position M1...M6.

Helical (R) gear units

Gear unit type R..., R..F	Fill quantity in liters					
	M1 ¹⁾	M2 ¹⁾	M3	M4	M5	M6
R07/R07F	0.12	0.20	0.20	0.20	0.20	0.20
R17/R17F	0.25	0.55	0.35	0.55	0.35	0.35
R27/R27F	0.25/0.40	0.70	0.50	0.70	0.50	0.50
R37/R37F	0.30/0.95	0.85	0.95	1.05	0.75	0.95
R47/R47F	0.70/1.50	1.60	1.50	1.65	1.50	1.50
R57/R57F	0.80/1.70	1.90	1.70	2.1	1.70	1.70
R67/R67F	1.10/2.3	2.6/3.5	2.8	3.2	1.80	2.0
R77/R77F	1.20/3.0	3.8/4.1	3.6	4.1	2.5	3.4
R87/R87F	2.3/6.0	6.7/8.2	7.2	7.7	6.3	6.5
R97	4.6/9.8	11.7/14.0	11.7	13.4	11.3	11.7
R107	6.0/13.7	16.3	16.9	19.2	13.2	15.9
R137	10.0/25.0	28.0	29.5	31.5	25.0	25.0
R147	15.4/40.0	46.5	48.0	52	39.5	41.0
R167	27.0/70	82	78	88	66	69
Gear unit type RF..	Fill quantity in liters					
	M1 ¹⁾	M2 ¹⁾	M3	M4	M5	M6
RF07	0.12	0.20	0.20	0.20	0.20	0.20
RF17	0.25	0.55	0.35	0.55	0.35	0.35
RF27	0.25/0.40	0.70	0.50	0.70	0.50	0.50
RF37	0.35/0.95	0.90	0.95	1.05	0.75	0.95
RF47	0.65/1.50	1.60	1.50	1.65	1.50	1.50
RF/RM57	0.80/1.70	1.80	1.70	2.0	1.70	1.70
RF/RM67	1.20/2.5	2.7/3.6	2.7	3.1	1.90	2.1
RF/RM77	1.20/2.6	3.8/4.1	3.3	4.1	2.4	3.0
RF/RM87	2.4/6.0	6.8/7.9	7.1	7.7	6.3	6.4
RF/RM97	5.1/10.2	11.9/14.0	11.2	14.0	11.2	11.8
RF/RM107	6.3/14.9	15.9	17.0	19.2	13.1	15.9
RF/RM137	9.5/25.0	27.0	29.0	32.5	25.0	25.0
RF/RM147	16.4/42.0	47.0	48.0	52	42.0	42.0
RF/RM167	26.0/70	82	78	88	65	71

1) The output end gear unit of multi-stage gear units must be filled with the larger oil volume.



Helical (RX) gear units

Gear unit type RX..	Fill quantity in liters					
	M1	M2	M3	M4	M5	M6
RX57	0.60	0.80	1.30	1.30	0.90	0.90
RX67	0.80	0.80	1.70	1.90	1.10	1.10
RX77	1.10	1.50	2.6	2.7	1.60	1.60
RX87	1.70	2.5	4.8	4.8	2.9	2.9
RX97	2.1	3.4	7.4	7.0	4.8	4.8
RX107	3.9	5.6	11.6	11.9	7.7	7.7
Gear unit type RXF..	Fill quantity in liters					
	M1	M2	M3	M4	M5	M6
RXF57	0.50	0.80	1.10	1.10	0.70	0.70
RXF67	0.70	0.80	1.50	1.40	1.00	1.00
RXF77	0.90	1.30	2.4	1.80	1.60	1.60
RXF87	1.60	2.0	4.9	4.0	2.9	2.9
RXF97	2.1	3.7	7.1	6.3	4.8	4.8
RXF107	3.1	5.7	11.2	9.3	7.2	7.2

Parallel shaft helical (F) gear units

F.., FA..B, FH..B, FV..B:

Gear unit type	Fill quantity in liters					
	M1	M2	M3	M4	M5	M6
F..27	0.60	0.80	0.70	0.70	0.60	0.60
F..37	0.95	1.25	0.70	1.25	1.00	1.10
F..47	1.50	1.80	1.10	1.90	1.50	1.70
F..57	2.6	3.5	2.1	3.5	2.8	2.9
F..67	2.7	3.8	1.9	3.8	2.9	3.2
F..77	5.9	7.3	4.3	8.0	6.0	6.3
F..87	10.8	13.0	7.7	13.8	10.8	11.0
F..97	18.5	22.5	12.6	25.2	18.5	20.0
F..107	24.5	32.0	19.5	37.5	27.0	27.0
F..127	40.5	55	34.0	61	46.5	47.0
F..157	69	104	63	105	86	78

FF..:

Gear unit type	Fill quantity in liters					
	M1	M2	M3	M4	M5	M6
FF27	0.60	0.80	0.70	0.70	0.60	0.60
FF37	1.00	1.25	0.70	1.30	1.00	1.10
FF47	1.60	1.85	1.10	1.90	1.50	1.70
FF57	2.8	3.5	2.1	3.7	2.9	3.0
FF67	2.7	3.8	1.90	3.8	2.9	3.2
FF77	5.9	7.3	4.3	8.1	6.0	6.3
FF87	10.8	13.2	7.8	14.1	11.0	11.2
FF97	19.0	22.5	12.6	25.5	18.9	20.5
FF107	25.5	32.0	19.5	38.5	27.5	28.0
FF127	41.5	56	34.0	63	46.5	49.0
FF157	72	105	64	106	87	79



FA.., FH.., FV.., FAF.., FHF.., FVF.., FAZ.., FHZ.., FVZ..:

Gear unit type	Fill quantity in liters					
	M1	M2	M3	M4	M5	M6
F..27	0.60	0.80	0.70	0.70	0.60	0.60
F..37	0.95	1.25	0.70	1.25	1.00	1.10
F..47	1.50	1.80	1.10	1.90	1.50	1.70
F..57	2.7	3.5	2.1	3.4	2.9	3.0
F..67	2.7	3.8	1.90	3.8	2.9	3.2
F..77	5.9	7.3	4.3	8.0	6.0	6.3
F..87	10.8	13.0	7.7	13.8	10.8	11.0
F..97	18.5	22.5	12.6	25.0	18.5	20.0
F..107	24.5	32.0	19.5	37.5	27.0	27.0
F..127	39.0	55	34.0	61	45.0	46.5
F..157	68	103	62	104	85	77

*Helical-bevel (K)
gear units*

K.., KA..B, KH..B, KV..B:

Gear unit type	Fill quantity in liters					
	M1	M2	M3	M4	M5	M6
K..37	0.50	1.00	1.00	1.30	0.95	0.95
K..47	0.80	1.30	1.50	2.0	1.60	1.60
K..57	1.20	2.3	2.5	3.0	2.6	2.4
K..67	1.10	2.4	2.6	3.4	2.6	2.6
K..77	2.2	4.1	4.4	5.9	4.2	4.4
K..87	3.7	8.0	8.7	10.9	8.0	8.0
K..97	7.0	14.0	15.7	20.0	15.7	15.5
K..107	10.0	21.0	25.5	33.5	24.0	24.0
K..127	21.0	41.5	44.0	54	40.0	41.0
K..157	31.0	62	65	90	58	62
K..167	33.0	95	105	123	85	84
K..187	53	152	167	200	143	143

KF..:

Gear unit type	Fill quantity in liters					
	M1	M2	M3	M4	M5	M6
KF37	0.50	1.10	1.10	1.50	1.00	1.00
KF47	0.80	1.30	1.70	2.2	1.60	1.60
KF57	1.30	2.3	2.7	3.2	2.9	2.7
KF67	1.10	2.4	2.8	3.6	2.7	2.7
KF77	2.1	4.1	4.4	6.0	4.5	4.5
KF87	3.7	8.2	9.0	11.9	8.4	8.4
KF97	7.0	14.7	17.3	21.5	15.7	16.5
KF107	10.0	22.0	26.0	35.0	25.0	25.0
KF127	21.0	41.5	46.0	55	41.0	41.0
KF157	31.0	66	69	92	62	62



KA.., KH.., KV.., KAF.., KHF.., KVF.., KAZ.., KHZ.., KVZ..:

Gear unit type	Fill quantity in liters					
	M1	M2	M3	M4	M5	M6
K..37	0.50	1.00	1.00	1.40	1.00	1.00
K..47	0.80	1.30	1.60	2.1	1.60	1.60
K..57	1.30	2.3	2.7	3.2	2.9	2.7
K..67	1.10	2.4	2.7	3.6	2.6	2.6
K..77	2.1	4.1	4.6	6.0	4.4	4.4
K..87	3.7	8.2	8.8	11.1	8.0	8.0
K..97	7.0	14.7	15.7	20.0	15.7	15.7
K..107	10.0	20.5	24.0	32.0	24.0	24.0
K..127	21.0	41.5	43.0	52	40.0	40.0
K..157	31.0	66	67	87	62	62
KH167	33.0	95	105	123	85	84
KH187	53	152	167	200	143	143

Spiroplan® (W)
gear units

The fill quantity of Spiroplan® gear units does not vary with their mounting position:

Gear unit type	Fill quantity in liters, regardless of mounting position
W..10	0.16
W..20	0.26
W..30	0.50

Helical-worm (S)
gear units

S..:

Gear unit type	Fill quantity in liters					
	M1	M2	M3 ¹⁾	M4	M5	M6
S37	0.25	0.40	0.50	0.55	0.40	0.40
S47	0.35	0.80	0.70/0.90	1.00	0.80	0.80
S57	0.50	1.20	1.00/1.20	1.45	1.30	1.30
S67	1.00	2.0	2.2/3.1	3.1	2.6	2.6
S77	1.90	4.2	3.7/5.4	5.9	4.4	4.4
S87	3.3	8.1	6.9/10.4	11.3	8.4	8.4
S97	6.8	15.0	13.4/18.0	21.8	17.0	17.0

1) The large gear unit of multi-stage gear units must be filled with the larger oil volume.

SF..:

Gear unit type	Fill quantity in liters					
	M1	M2	M3 ¹⁾	M4	M5	M6
SF37	0.25	0.40	0.50	0.55	0.40	0.40
SF47	0.40	0.90	0.90/1.10	1.05	1.00	1.00
SF57	0.50	1.20	1.00/1.50	1.55	1.40	1.40
SF67	1.00	2.2	2.3/3.0	3.2	2.7	2.7
SF77	1.90	4.1	3.9/5.8	6.5	4.9	4.9
SF87	3.8	8.0	7.1/10.1	12.0	9.1	9.1
SF97	7.4	15.0	13.8/18.8	22.6	18.0	18.0

1) The large gear unit of multi-stage gear units must be filled with the larger oil volume.



SA.., SH.., SAF.., SHF.., SAZ.., SHZ..:

Gear unit type	Fill quantity in liters					
	M1	M2	M3 ¹⁾	M4	M5	M6
S..37	0.25	0.40	0.50	0.50	0.40	0.40
S..47	0.40	0.80	0.70/0.90	1.00	0.80	0.80
S..57	0.50	1.10	1.00/1.50	1.50	1.20	1.20
S..67	1.00	2.0	1.80/2.6	2.9	2.5	2.5
S..77	1.80	3.9	3.6/5.0	5.8	4.5	4.5
S..87	3.8	7.4	6.0/8.7	10.8	8.0	8.0
S..97	7.0	14.0	11.4/16.0	20.5	15.7	15.7

1) The large gear unit of multi-stage gear units must be filled with the larger oil volume.

10 Index

10.1 Index of changes

The following additions and changes have been made to the previous edition of the Gear Units operating instructions (publication number: 1050 3005, edition 05/2001):

**General
information**

- The R07 series has been adopted into the operating instructions.

Sec. Safety Notes

- A summary table has been added dealing with the conditions for extended storage of gear units in the various climate zones.

**Sec. Mounting
Positions**

- All mounting position sheets have been revised. The mounting position sheets for Spiroplan® gear units have been added. The comparison between old and new mounting positions has been deleted.

Sec. Lubricants

- The lubricant table has been completely revised. The lubricant fill quantities have been updated and supplemented by the values for the R07 series.



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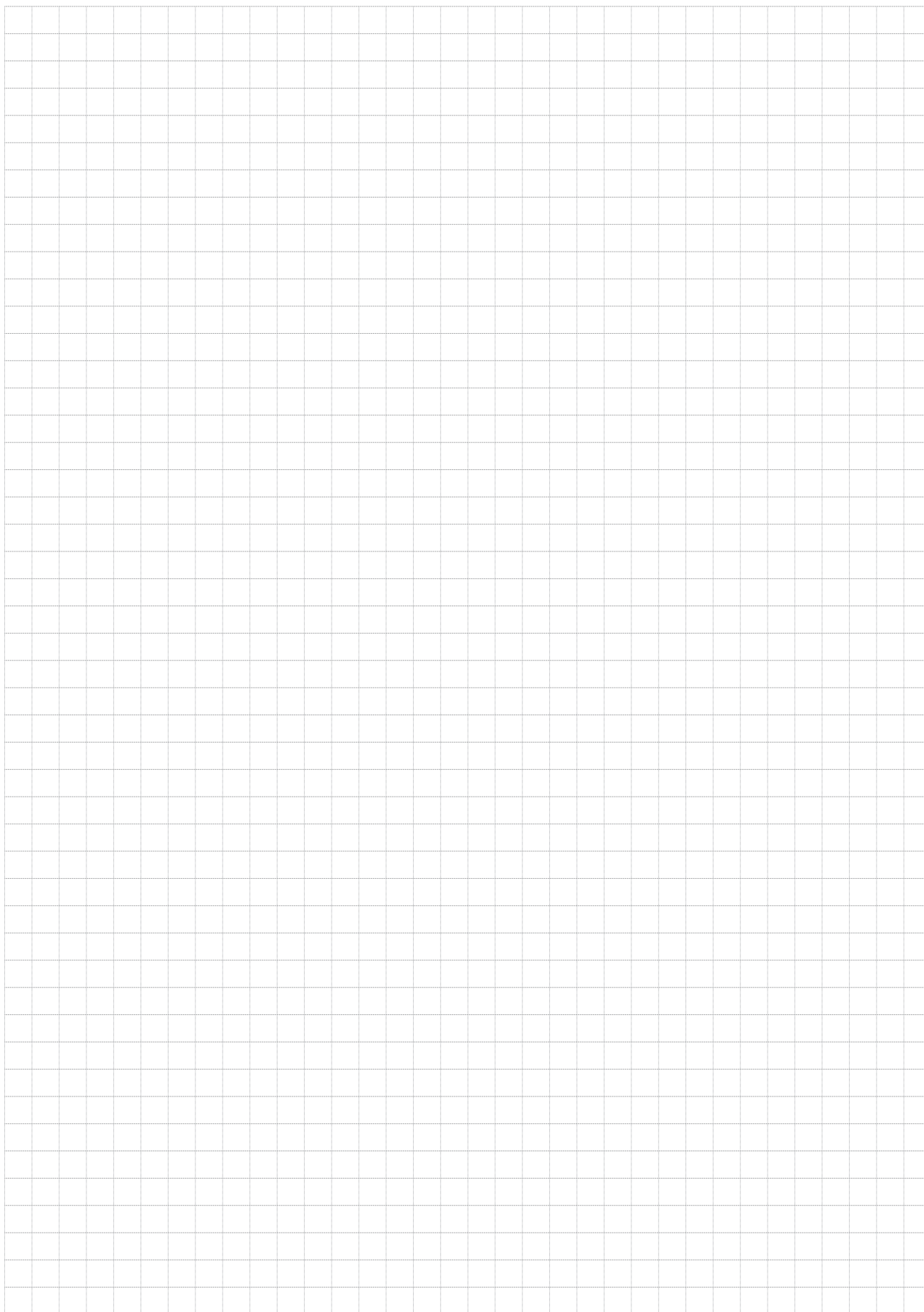


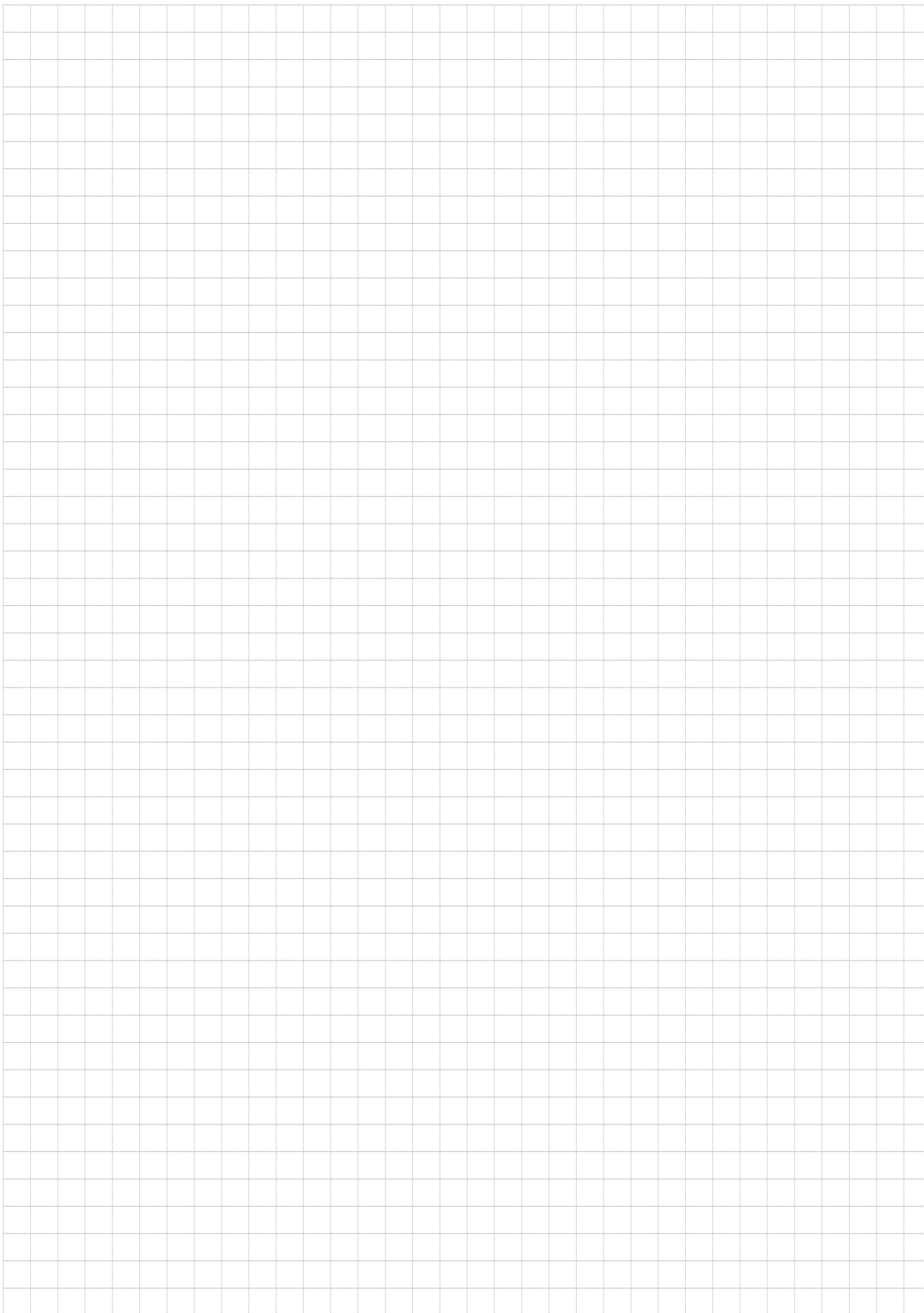
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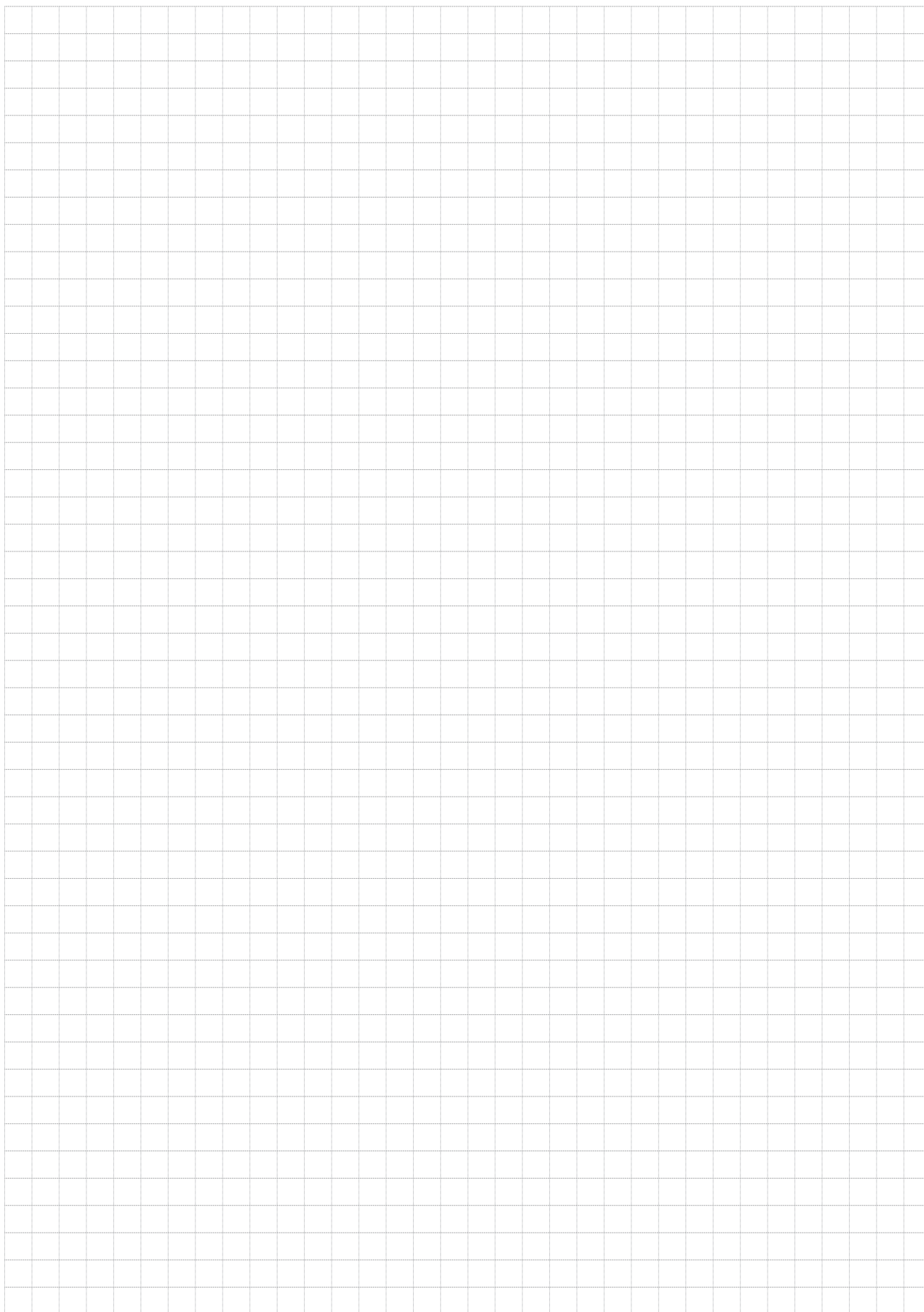


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SEW
EURODRIVE

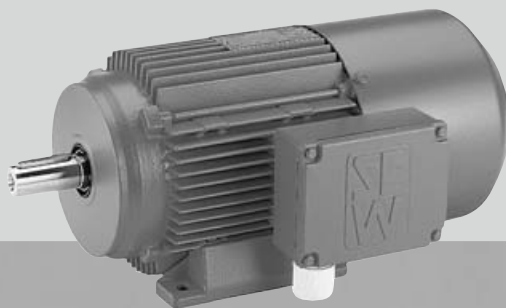




AC Motors DR/DV/DT/DTE/DVE
Asynchronous Servomotors CT/CV

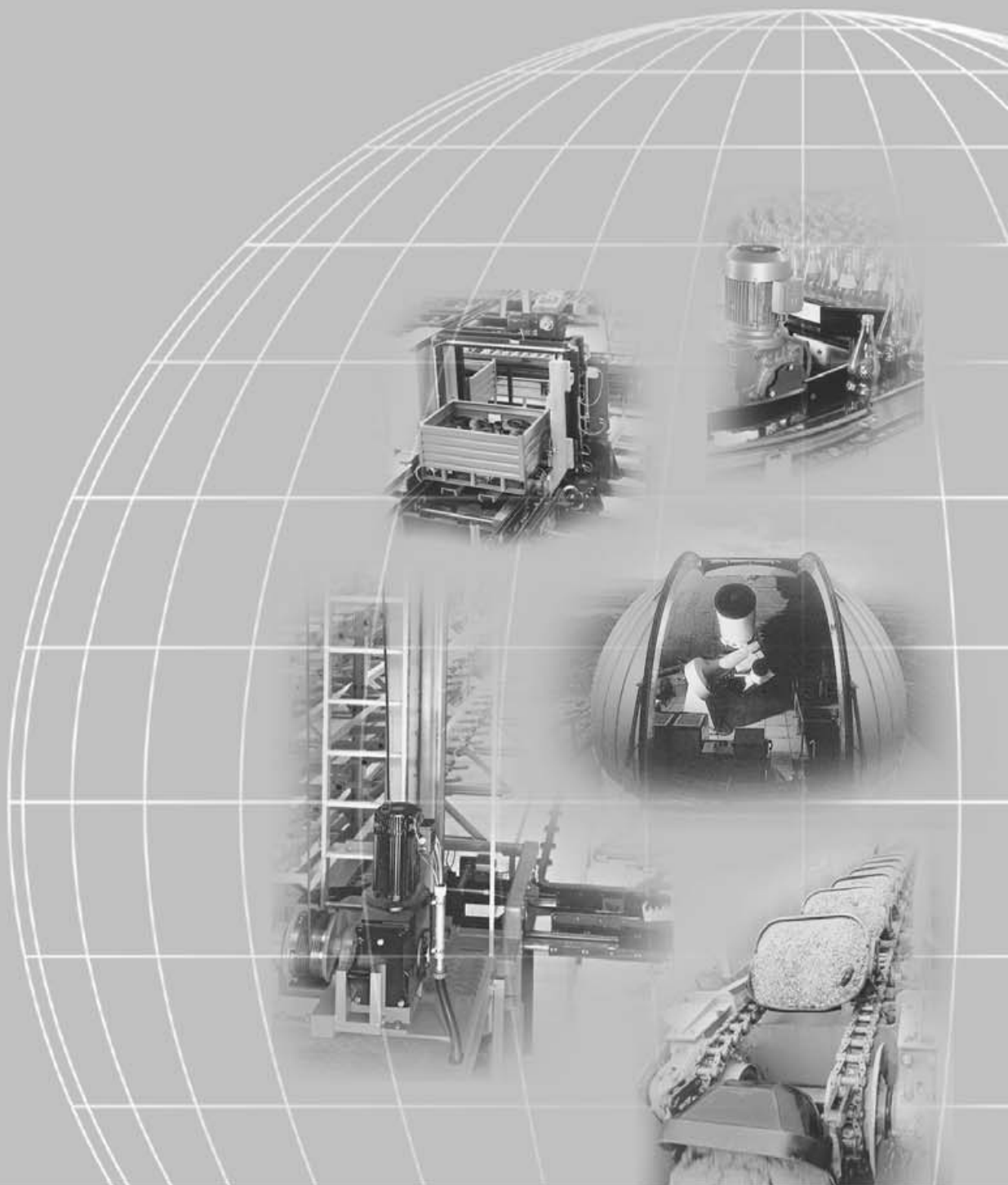
Edition

02/2003



Operating Instructions

1056 7917 / EN



SEW-EURODRIVE





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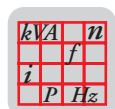
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1 Important Notes

Safety and warning instructions

Always follow the safety and warning instructions contained in this publication!



Electrical hazard

Possible consequences: Severe or fatal injuries.



Hazard

Possible consequences: Severe or fatal injuries.



Hazardous situation

Possible consequences: Slight or minor injuries.



Harmful situation

Possible consequences: Damage to the drive and the environment.



Tips and useful information.



A requirement of fault-free operation and fulfillment of any rights to claim under guarantee is that you adhere to the information in the operating instructions. Consequently, read the operating instructions before you start operating the drive!

The operating instructions contain important information about service; as a result, they should be kept in the vicinity of the unit.

Waste disposal



This product consists of:

- Iron
- Aluminum
- Copper
- Plastic
- Electronics components

Please dispose of the parts in accordance with the applicable regulations.



2 Safety Notes

Preliminary remarks

The following safety notes are principally concerned with the use of motors. When operating **geared motors**, please also refer to the safety notes for gear units in the corresponding operating instructions.

Please also pay attention to the supplementary safety notes in the individual sections of these operating instructions.

General information

During and after operation, motors and geared motors have live and moving parts and their surfaces may be hot.

All work related to transportation, putting into storage, setting up/mounting, connection, startup, maintenance and repair should only be performed by trained personnel observing

- the corresponding detailed operating instructions and wiring diagrams,
- the warning and safety signs on the motor/geared motor,
- the specific regulations and requirements for the system and
- national/regional regulations governing safety and the prevention of accidents.

Severe injuries and damage to property may result from

- incorrect use,
- incorrect installation or operation,
- removal of required protective covers or the housing when this is not permitted.

Designated use

These electric motors are intended for industrial systems. They comply with the applicable standards and regulations and meet the requirements of the Low Voltage Directive 73/23/EEC.

The technical data and the information about permitted conditions can be found on the nameplate and in the documentation.

It is essential to observe all specified information!

Transportation

Inspect the shipment for any damage in transit as soon as you receive the delivery. Inform the shipping company immediately. It may be necessary to preclude startup.

Tighten screwed in transport lugs firmly. They are only designed for the weight of the motor/geared motor; do not attach any additional loads.

The installed lifting eyebolts are in accordance with DIN 580. The loads and regulations specified in that document must always be observed. If the geared motor is equipped with two suspension eye lugs or lifting eyebolts, then both of the suspension eye lugs should be used for transportation. In this case, the tension force vector of the slings must not exceed a 45° angle in accordance with DIN 580.

Use suitable, sufficiently rated handling equipment if necessary. Remove any transportation fixtures prior to startup.

Installation/mounting

Follow the instructions in Sec. 'Mechanical Installation!'

Inspection and maintenance

Follow the instructions in Sec. 'Inspection and Maintenance!'

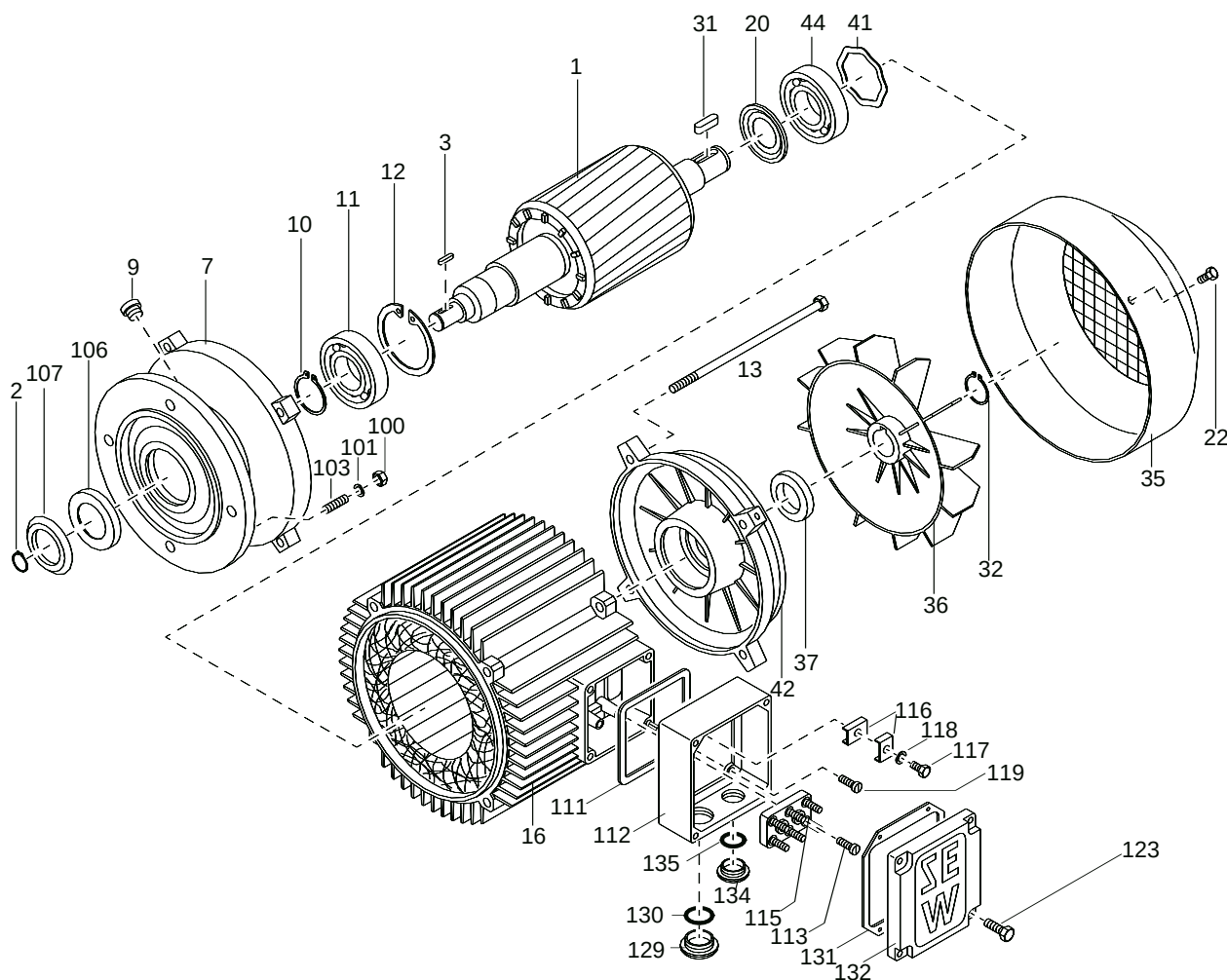


3 Motor Design



The following illustration is intended to explain the general structure. Its purpose is only to make it easier to assign components to the spare parts lists. Discrepancies are possible depending on the motor size and version!

3.1 Configuration principles of AC motors



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Key

1 Rotor, cpl.	31 Key	107 Oil flinger	131 Sealing washer
2 Circlip	32 Circlip	111 Seal	132 Terminal box cover
3 Key	35 Fan guard	112 Terminal box lower part	134 Screw plug
7 Flanged end shield	36 Fan	113 Machine screw	135 Sealing washer
9 Screw plug	37 V-ring	115 Terminal board	
10 Circlip	41 Equalizing ring	116 Terminal yoke	
11 Grooved ball bearing	42 B bearing end shield	117 Hex head screw	
12 Circlip	44 Grooved ball bearing	118 Lock washer	
13 Hex head screw (tie rod)	100 Hex nut	119 Machine screw	
16 Stator, cpl.	101 Lock washer	123 Hex head screw	
20 Nilos ring	103 Stud	129 Screw plug	
22 Hex head screw	106 Oil seal	130 Sealing washer	



3.2 Nameplate, unit designation

Nameplate

Example: Brake motor DFV 160 M4 /BM

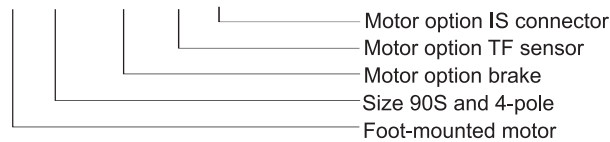
SEW-EURODRIVE		Bruchsal / Germany			
Typ	DFV 160 M 4 /BM				
Nr.	01.3001234568.0001.00		IM	B5	
kW	11 S1		cos φ	0.83	
50Hz V	220 - 240 Δ / 380 - 415 Y		A	39.0 / 22.5	
60Hz V	240 - 266 Δ / 415 - 460 Y		A	35.5 / 20.5	
r/min	1440 / 1740		IP	55	
Bremse V	230 AC		Nm	150	
Kg	109		Ma		
Schmierstoff			Made in Germany 184 103 3.16		

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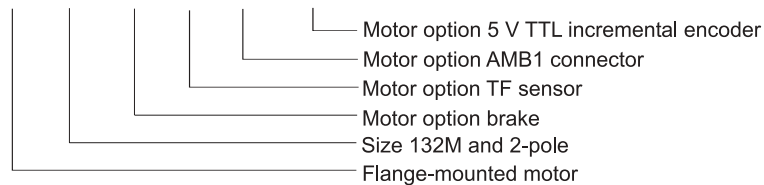
Unit designation

Examples: AC(brake)motors DR/DT/DV/DTE/DVE

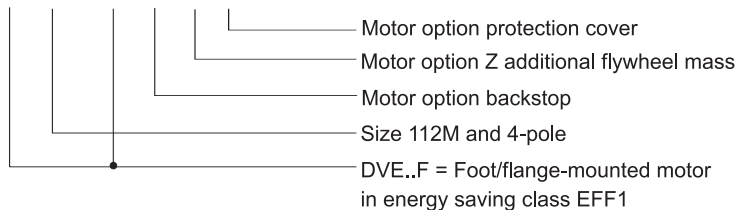
DT 90S4 / BMG / TF / IS



DFV 132M2 / BM / TF / AMA1 / EV1T



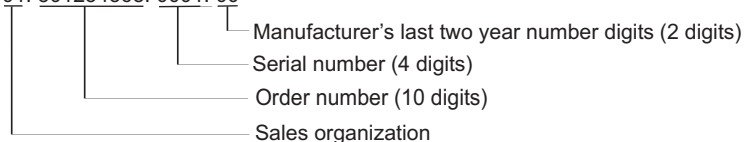
DVE 112 M4 F / RS / Z / C



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Example: Factory number

01. 301234568. 0001. 00



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**Nameplate**

Example: Servo brake motor CT90L4 / BMG / TF / ES1S

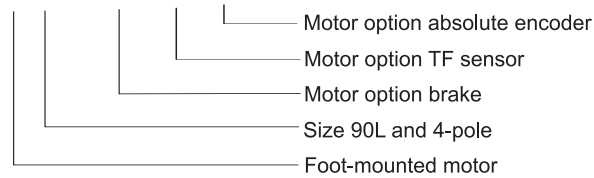
SEW-EURODRIVE		Bruchsal / Germany		CE	
Typ	CT90L4 BMG TF / ES1S			3 ~ IEC 34	
Nr.	01.3410069302.0001.00				
Nm	30.5	max. Motor	i	:1	
○	Nm 10.5	r/min	3000	○	
		Hz	103	V	345
		A	7.9		
IM	B5	kg	28	IP	54
		iso. Kl.	F		
Bremsen V		230~	Nm	20	Gleichrichter BGE 1.5
Schmierstoff		Made in Germany 186 475 0.13			

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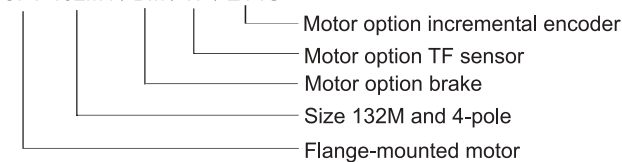
Unit designation

Examples: Servo (brake) motors CT/CV

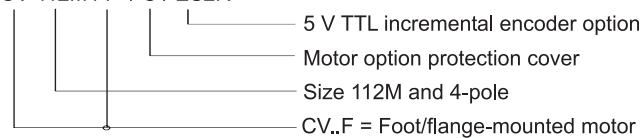
CT 90L4 / BMG / TF / AV1Y



CFV 132M4 / BM / TF / EV1S



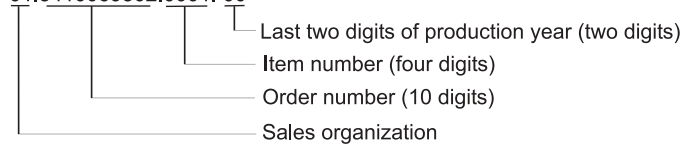
CV 112M4 F / C / ES2R



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Example: Factory number

01.3410069302.0001.00



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4 Mechanical Installation



It is essential to comply with the safety notes in Sec. 2 during installation!

4.1 Before you begin

The drive may only be installed if

- the entries on the nameplate of the drive and/or the output voltage of the frequency inverter match the voltage supply system,
- the drive is undamaged (no damage caused by transportation or storage) and
- it is certain that the following requirements have been fulfilled:
 - Ambient temperature between -25°C and $+40^{\circ}\text{C}^1$
 - No oil, acid, gas, vapors, radiation, etc.
 - Installation altitude max. 1000 m above sea level
 - Note the restrictions for encoders
 - Special designs: drive configured in accordance with the ambient conditions

4.2 Preliminary work

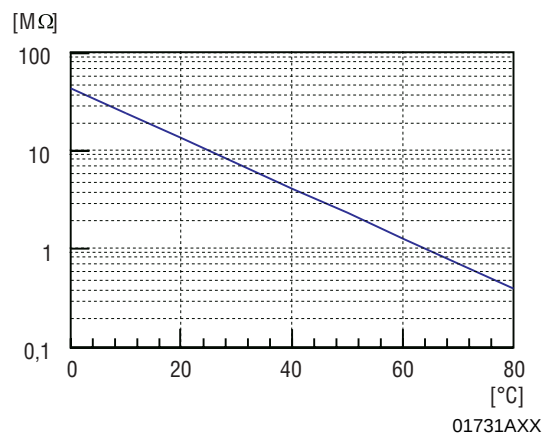
Motor shaft ends must be thoroughly cleaned of anti-corrosion agents, contamination or such like (use a commercially available solvent). Do not allow the solvent to penetrate the bearings or shaft seals – this could cause material damage!

Extended storage of motors

- Please note the reduced grease utilization period of the ball bearings after storage periods exceeding one year.
- Check whether the motor has absorbed moisture as a result of being stored for a long time. Measure the insulation resistance to do this (measuring voltage 500 V).



The insulation resistance (→ following figure) varies greatly depending on the temperature! The motor must be dried if the insulation resistance is not adequate.

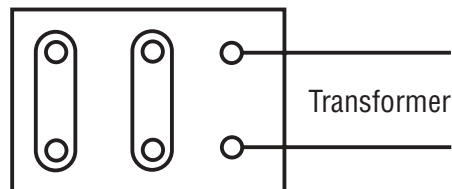


1. Minimum temperature for motors with backstop: -15°C , note that the temperature range of the gear unit may also be limited (→ gear unit operating instructions)

**Drying the motor**

Heat up the motor

- with hot air or
- using an isolation transformer
 - connect the windings in series (→ following figure)
 - auxiliary AC voltage supply max. 10% of the rated voltage with max. 20% of the rated current



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The drying process is finished when the minimum insulation resistance has been exceeded.

Check the terminal box to see whether

- the inside is clean and dry,
- the connections and fastening parts are free from corrosion,
- the joint seals are ok,
- the cable screw fittings are tight, otherwise clean or replace them.

4.3 Installing the motor

The motor or geared motor may only be mounted or installed in the specified mounting position on a level and torsionally rigid support structure which is not subject to shocks.

Carefully align the motor and the driven machine to avoid placing any unacceptable strain on the output shafts (observe permitted overhung load and axial thrust data!).

Do not butt or hammer the shaft extension.

Use an appropriate cover to protect motors in vertical mounting positions from objects or fluids entering (protection cowl C).

Ensure an unobstructed cooling air supply and that air heated by other apparatus cannot be drawn in or reused.

Balance components for subsequent mounting on the shaft with a half key (motor shafts are balanced with a half key).

Any condensation drain holes will be sealed by plastic plugs and should only be opened when necessary; open condensation drain holes are not permitted, since this invalidates higher classes of enclosure.

For brake motors with manual brake release, screw in either the hand lever (with self-reengaging manual brake release) or setscrew (with lockable manual brake release).

Note the following for encoder mounting:

Foot-mounted motors CT/DT71, CT/DT(E)90, CV/DV(E)132M, CV/DV(E)160L must be mounted on supports because the radius of the cover is greater than the shaft height.

For foot-mounted (brake)motors of sizes DTE90L and DVE132M, the shaft height is that of the next larger IEC standard motor (100 mm or 160 mm). The foot dimensions of the DTE90, DVE180 and DVE225 motors deviate from the IEC dimensions, see section "Notes on dimension sheets" in the Geared Motors catalog.



**Installation in
damp areas or in
the open**

If possible, arrange the terminal box so the cable entries are pointing downwards.

Coat the threads of cable screw fittings and pocket caps with sealant and tighten them well – then coat them again.

Seal the cable entry well.

Thoroughly clean the sealing surfaces of terminal boxes and terminal box covers prior to reassembly; gaskets must be glued in on one side. Fit new gaskets to replace brittle ones!

Restore the anticorrosive coating if necessary.

Check the enclosure.

4.4 Installation tolerances

Shaft end	Flanges
Diameter tolerance in accordance with DIN 748 • ISO k6 at $\varnothing \leq 50$ mm • ISO m6 at $\varnothing > 50$ mm • Center bore in accordance with DIN 332, shape DR..	Centering shoulder tolerance in accordance with DIN 42948 • ISO j6 at $\varnothing \leq 230$ mm • ISO h6 at $\varnothing > 230$ mm



5 Electrical Installation



It is essential to comply with the safety notes in Sec. 2 during installation!

Switch contacts in utilization category AC-3 to EN 60947-4-1 must be used for switching the motor and the brake.

5.1 Wiring notes

Comply with the safety notes during installation.

Protection against interfer- ence from brake control systems

Do not route brake cables alongside switched-mode power cables, since otherwise there is a risk of disrupting brake controllers.

Switched-mode power cables include, in particular:

- Output cables from frequency and servo controllers, converters, soft start units and brake units
- Connecting harnesses to braking resistors, etc.

Protection against interfer- ence from motor protection devices

To provide protection against interference from SEW motor protection devices (temperature sensors TF, winding thermostats TH):

- Route separately shielded feeder cables together with switched-mode power lines in one cable
- Do not route unshielded feeder cables together with switched-mode power lines in one cable

5.2 Special aspects for operation with a frequency inverter

When motors are powered by inverters, you must adhere to the wiring instructions issued by the inverter manufacturer. It is essential to pay close attention to the operating instructions for the frequency inverter.

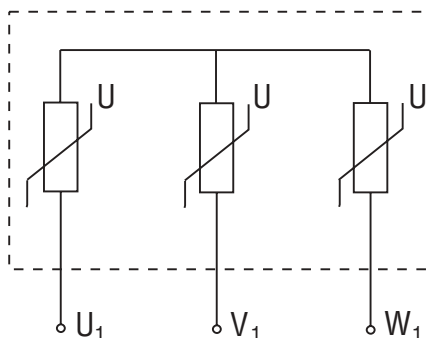
5.3 Special aspects of single-phase motors

Bear in mind that SEW single-phase motors are supplied without accessory equipment such as capacitors, starting relays or centrifugal switches (exception: ET56L4 → Sec. 'Single-phase version ET56'). Any parts you need must be obtained from your dealer and connected according to the corresponding instructions and wiring diagrams.



5.4 *Special aspects of torque motors and low-speed motors*

Due to the design of torque motors and low-speed motors, very high induction voltages may be generated when they are switched off. Consequently, we recommend protection using the varistor circuit shown below. The size of the varistors depends, among other factors, on the starting frequency – note for project planning!



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5.5 *Special aspects in switching operation*

When the motors are used in switching operation, any possible malfunctions of the switchgear must be excluded by appropriate wiring. According to EN 60204 (electrical equipment of machines), motor windings must have interference suppression in order to protect the numerical or programmable logic controllers. Since it is primarily switching operations which lead to the disruptions, we recommend installing protective circuitry on the switching devices.



5.6 Connecting the motor



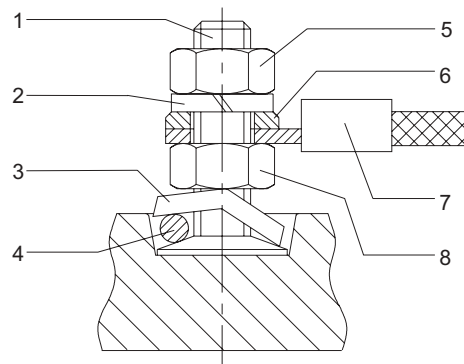
Connecting the motor via terminal boxes

In case of operation with electronic control units, it is essential to adhere to the corresponding operating instructions/wiring diagrams!

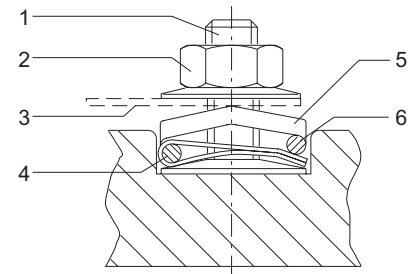
- According to the wiring diagram provided
- Check the cross sections of cables
- Arrange terminal links correctly
- Tighten screw connections and protective earth conductors firmly
- In terminal boxes: Check winding connections and tighten them if necessary

Small connection parts

Please note: In the case of motor sizes DR63 - DV132S, the small connection parts (connection nuts for feeder cables, terminal links, lock washer and washers) are supplied in a bag. Depending on the type of terminal board, install the parts in accordance with the figure below. In the connection on the right in the figure below, the second retaining nut, the lock washer and the washer are not used. The external connection (6) can be installed directly or as a lug (4) below the terminal washer (5).



- 1 Terminal stud
- 2 Lock washer
- 3 Terminal washer
- 4 Motor terminal lead
- 5 Top nut
- 6 Washer
- 7 External connection
- 8 Bottom nut



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- 1 Terminal stud
- 2 Hex nut with flange
- 3 Terminal link
- 4 Motor with Stock connection terminal
- 5 Terminal washer
- 6 External connection



The asynchronous servomotors of the CT/CV series will be shipped with switched terminal links according to the nameplate.

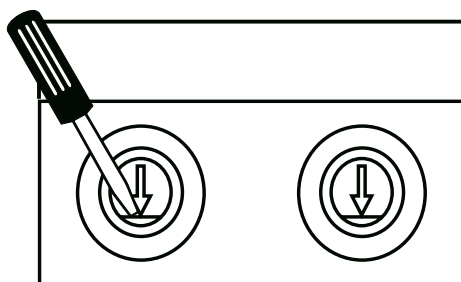


5.7 Preparing motor sizes 56 and 63



Important: Wear safety glasses – danger of injury from fragments!

- Put on the terminal box cover and tighten screws
- Define which cable entries to open
- Open the cable entries
 - with a chisel or similar (hold at an angle)
 - by tapping lightly with a hammer



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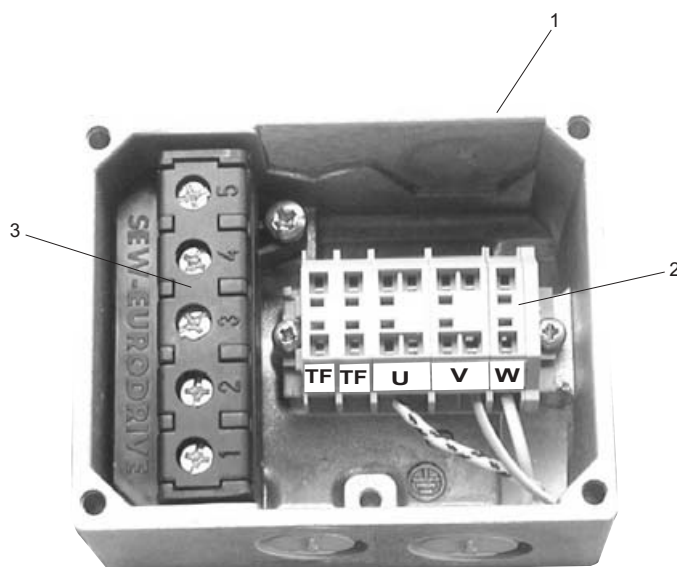


Caution – Do not punch through into the inside of the terminal box!

- Open the terminal box, remove any pieces that may have broken off
- Secure the cable screw fittings with the supplied lock nuts

5.8 Connecting motor DT56...+/BMG

The motor has a star point with three fixed connection points in the winding overhang. The power supply leads (L1, L2, L3) are connected to a spring cage terminal block (2) in the terminal box (1). The brake BMG02 is controlled using the brake rectifier BG1.2 (3). As an alternative, the brake can be controlled from the switch cabinet using BM series rectifiers.



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5.9 Single-phase version ET56

The single-phase motor ET56 is supplied with a mounted and connected running capacitor :

1~230 V, 50 Hz $C_B = 4 \mu\text{F}$

1~230 V, 60 Hz $C_B = 4 \mu\text{F}$

1~110 V, 60 Hz $C_B = 20 \mu\text{F}$



No full-load startup is possible with the running capacitor alone! The single-phase motor cannot be combined with a TF.

5.10 Connecting the motor using the IS plug connector



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The IS plug connector is supplied with its base fully wired-up, including additional features such as a brake rectifier. The upper section of the IS connector is included in the scope of delivery and must be connected as shown in the wiring diagram.

The IS plug connection has CSA approval up to 600 V. (Note for application according to CSA regulations: Tighten the M3 terminal screws to a torque of 0.5 Nm! See the following table for American Wire Gauge (AWG) line cross sections!)

Line cross section

Make sure the type of line corresponds to the applicable regulations. The rated currents are specified on the motor nameplate. The line cross sections that can be used are listed in the following table.

Without variable terminal link	With variable terminal link	Link cable	Double assignment (motor and brake/SR)
0.25 - 4.0 mm ²	0.25 - 2.5 mm ²	max. 1.5 mm ²	max. 1 x 2.5 and 1 x 1.5 mm ²
23 - 12 # AWG	23 - 14 # AWG	max. 16 # AWG	max. 1 x 14 # and 1 x 16 # AWG



Wiring the upper section of the plug connector

- Loosen the housing cover screws
 - Remove the housing cover
- Remove the screws from the upper section of the plug connector
 - Remove the upper section of the plug connector from the cover
- Strip the insulation off the connection lead
 - Strip about 9 mm insulation off the connecting leads
- Pass the cable through the cable screw fitting

Wiring as shown in wiring diagram DT82, DT83

- Connect the lines as shown in the wiring diagram
 - Tighten the clamping screws carefully!
- Install the plug connector (→ Sec. 'Installing the plug connector')

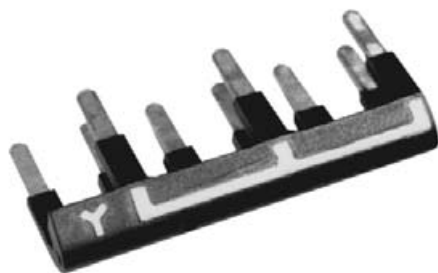
Wiring as shown in wiring diagram DT81

For Δ / Δ startup:

- Connect with six lines
 - Tighten the clamping screws carefully!
 - Motor contactors in the switch cabinet
- Install the plug connector (→ Sec. 'Installing the plug connector')

For Δ or Δ operation:

- Connect as shown in the wiring diagram
- Install the variable terminal link as shown in the following figures according to desired motor operation (m or W)
- Install the plug connector (→ Sec. 'Installing the plug connector')



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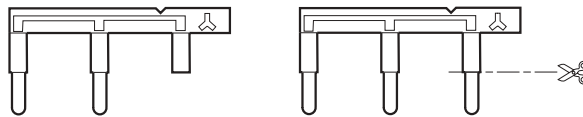


Connecting the motor using the IS plug connector

Brake control system BSR – preparing the variable terminal link

For Δ operation:

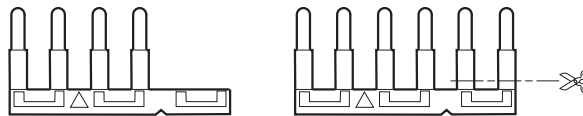
On the W side of the variable terminal link as shown in the following figure: Remove only the bare metal pin of the marked prong horizontally – touch guard!



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For Δ operation:

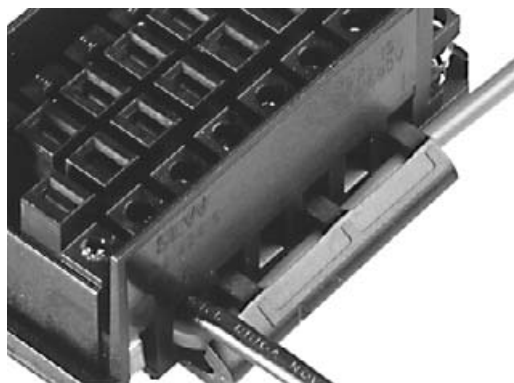
On the m side of the variable terminal link as shown in the following figure: Completely remove two prongs horizontally.



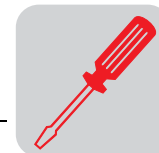
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Wiring according to the DT81 wiring diagram for W or m operation with double terminal assignment

- At terminal point for double assignment:
 - Connect the link cable
- When operation is as required:
 - Insert the link cable in the variable terminal link
- Install the variable terminal link
- At terminal point for double assignment:
 - Connect the motor lead above the variable terminal link
- Connect the other lines as shown in the wiring diagram
- Install the plug connector (→ Sec. 'Installing the plug connector')



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Installing the plug connector

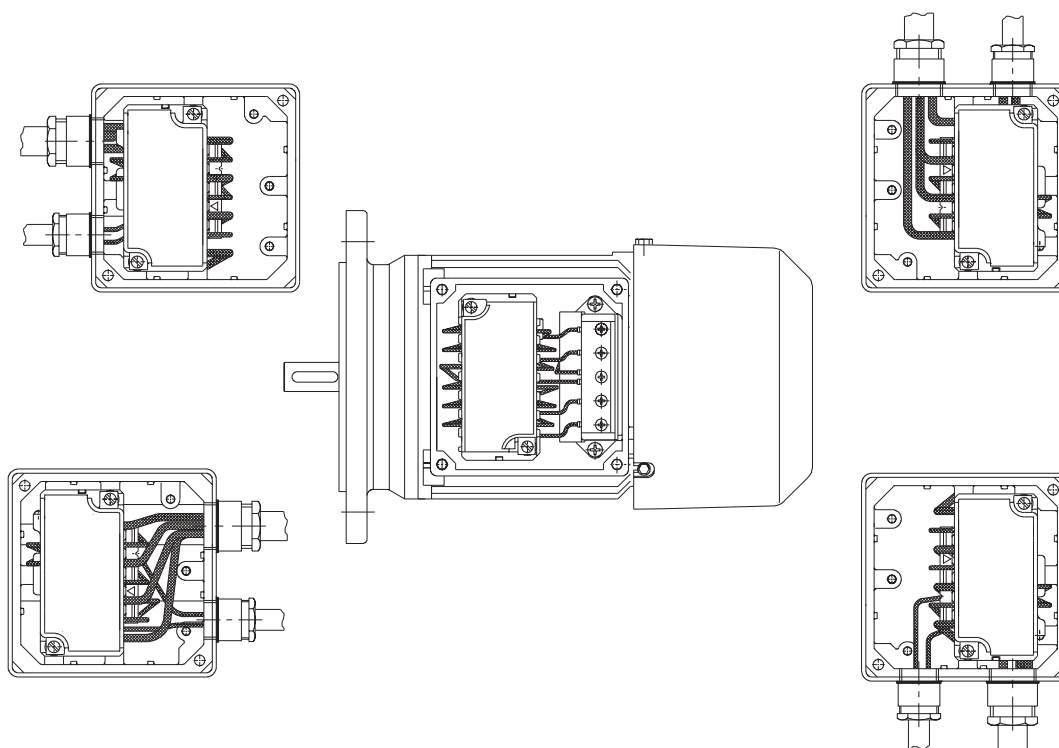
The housing cover of the IS plug connector can be screwed onto the lower section of the plug connector depending on the required position of the cable lead. The upper section of the plug connector shown in the following figure must first be installed in the housing cover so it will match the position of the lower section of the plug connector:

- Define the required mounting position
- Install the upper section of the plug connector into the housing cover in accordance with the mounting position
- Close the plug connector
- Tighten the cable screw fitting



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Mounting position of the upper section of the plug connector in the housing cover



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5.11 Connecting the motor using plug connectors AS.. and AM..



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Plug connectors AS.. or AM.. are mounted on the side of the terminal box. They are fastened onto the terminal box either with two clips as in the AMA1, AMB1 and ASB1 or with one clip as in the AMD1, AME1, ASD1 and ASE1.

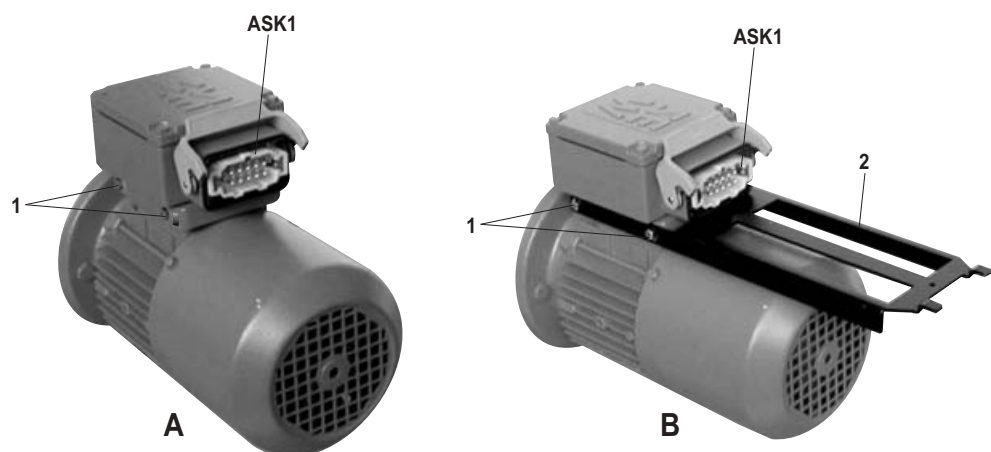


The ASD1 and ASE1 types with single clip closure correspond to the DESINA regulation issued by the Association of German Machine Tool Manufacturers (VDW).

The ASA1 / ASD1 or ASB1 / ASE1 and the AMA1 / AMD1 are supplied with their bases fully wired up, including additional features such as a brake rectifier.

The customer is responsible for obtaining the upper sections of the connectors from the dealer and connecting them in accordance with the wiring diagrams (supplied with the motor).

5.12 Connecting the motor using ASK1 plug connector



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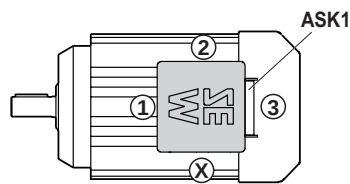
Drives with ASK1 plug connector are certified according to the ECOFAST specification (version 1.1). Switchgear or control units which also have to be certified can be connected to SEW-EURODRIVE motors using a pre-fabricated system cable or a carrier plate (installation integrated in the motor → Fig. B). The ASK1 plug connector with single-clip locking is mounted on the side of the terminal box and fully wired, including additional features such as a brake rectifier.



- The system cable pre-fabricated according to the ECOFAST specification must be obtained from a specialist retailer by the customer.
- For installation integrated in the motor according to the ECOFAST specification, the customer must obtain the carrier plate from SEW-EURODRIVE by quoting part number 0187 390 3. Carrier plates from other manufacturers do not fit on SEW-EURODRIVE motors.

Position of the plug connector

Possible positions of the ASK1 plug connector are "X" (= standard position), "1", "2" or "3." Unless specified otherwise, the unit is supplied with the plug connector in position "3." For installation integrated into the motor (using the carrier plate), units are supplied with the plug connector in position "3" only.



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Installing the carrier plate

- Loosen and remove four retaining screws (1) below the terminal box (→ Fig. A)
- Place the carrier plate (2) against the bores for the retaining screws and install it by tightening the four retaining screws (1) (→ Fig. B)



5.13 Connecting the brake

The brake is released electrically. The brake is applied mechanically when the voltage is switched off.



Comply with the applicable regulations issued by the relevant employer's liability insurance association regarding phase failure protection and the associated circuit/circuit modification!

- Connect the brake according to the wiring diagram supplied with the brake.
- **Note:** In view of the DC voltage to be switched and the high level of current load, it is essential to use either special brake contactors or AC contactors with contacts in utilization category AC-3 to EN 60947-4-1.
- If necessary, for version with manual brake release you can install
 - hand lever (for manually disengaging brake)
 - or manual brake release screw (for fixing brake in the disengaged position)
- After a new brake disk has been fitted, the maximum braking torque will be achieved after a few cycles.

Connecting the brake control system

The DC disk brake is powered from a brake control system with protection circuit. This control is accommodated in the terminal box / IS lower part or must be installed in the switch cabinet (→ Sec. 'Wiring notes').



- **Check the line cross sections - braking currents (→ Sec. 'Technical Data')**
- Connect the brake control system according to the wiring diagram supplied with the brake
- For motors in thermal classification H, install the brake rectifier in the switch cabinet!



5.14 Accessory equipment



Connect supplied accessory equipment according to the wiring diagrams included.

Temperature sensor TF



Do not apply any voltage!

The positive temperature coefficient (PTC) thermistors comply with DIN 44082.

Resistance measurement (measuring instrument with $V \leq 2.5 \text{ V}$ or $I < 1 \text{ mA}$):

- Standard measured values: 20...500 Ω , thermal resistance $> 4000 \text{ } \Omega$
- Measured values pole-changing with separate winding: 40...1000 Ω , thermal resistance $> 4000 \text{ } \Omega$

Winding thermostats TH

The thermostats are connected in series as standard and open when the permitted winding temperature is exceeded. They can be connected in the drive monitoring loop.

	V_{AC}		V_{DC}	
Voltage V [V]	250	400	60	24
Current ($\cos \varphi = 1.0$) [A]	2.5	0.75	1.0	1.6
Current ($\cos \varphi = 0.6$) [A]	1.6	0.5		
Contact resistance max. 1 ohm at 5 V = / 1 mA				

Forced cooling fan

Motor size 71 - 132S

VS System

- 1 x 230 V_{AC} , 50 Hz
- Connection in separate terminal box
- Max. connection cross section 3 x 1.5 mm^2
- Cable screw fitting M16x1.5



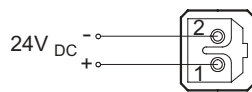
Please refer to the VS wiring diagram for information about connecting the VS system (order number: 0975 8385).



VR System

- $24\text{ V}_{\text{DC}} \pm 20\%$
- Connection via plug connector
- Max. connection cross section $3 \times 1\text{ mm}^2$
- Cable screw fitting Pg7 with inside diameter 7 mm

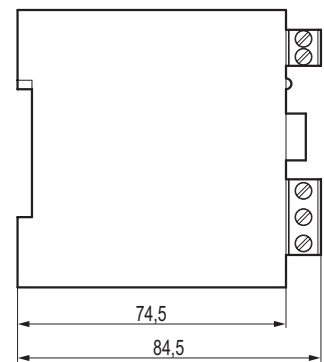
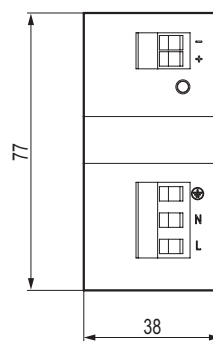
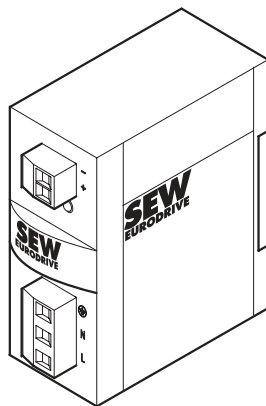
The **VR forced cooling fan** is available for 24 V DC voltage and for 100 ... 240 V AC voltage.



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The AC voltage type includes a VR forced cooling fan and the UWU51A switch-mode power supply (→ following figure).

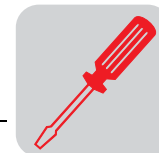
- Input: 100 ... 240 V_{AC} -6% / +10%, 50/60 Hz
- Output: 24 V_{DC} -1% / +2%, 1.25 A
- Connection: Screw terminals 0.2 ... 2.5 mm², separable
- Enclosure: IP20; attachment to EN 50022 DIN rail in the switch cabinet



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Please refer to the VR wiring diagram for information about connecting the VR forced cooling fan (order number: 0880 3198)



Motor size 132M - 280

V system

- 3 x 400 V_{AC}, 50 Hz
- Connection in separate terminal box
- Max. connection cross section 4 x 1.5 mm²
- Cable screw fitting M16x1.5



Please refer to the VS or V wiring diagram for information about connecting the VS and V systems (order no: 0975 8385).

A transformer may be present in the VS system in order to adapt to a voltage other than the standard. VS and V systems are also available for 60 Hz

Encoder overview

Encoder	for SEW motor	Type of encoder	Shaft	Specifica- tion	Supply	Signal
EH1T ¹⁾	DR63...	Encoder	Hollow shaft	-	5 V _{DC} regulated	5 V _{DC} TTL/RS-422
EH1S ²⁾					24 V _{DC}	1 V _{SS} sin/cos
EH1R						5 V _{DC} TTL/RS-422
EH1C						24 V _{DC} HTL
ES1T ¹⁾	CT/DT/CV/DV71...100 DTE/DVE90...100	Encoder	Spread shaft	-	5 V _{DC} regulated	5 V _{DC} TTL/RS-422
ES1S ²⁾					24 V _{DC}	1 V _{SS} sin/cos
ES1R						5 V _{DC} TTL/RS-422
ES1C						24 V _{DC} HTL
ES2T ¹⁾	CV/DV(E)112...132S				5 V _{DC} regulated	5 V _{DC} TTL/RS-422
ES2S ²⁾					24 V _{DC}	1 V _{SS} sin/cos
ES2R						5 V _{DC} TTL/RS-422
ES2C						24 V _{DC} HTL
EV1T ¹⁾	CT/CV71...200 DT/DV71...280 DTE/DVE90...225		Solid shaft		5 V _{DC} regulated	5 V _{DC} TTL/RS-422
EV1S ²⁾					24 V _{DC}	1 V _{SS} sin/cos
EV1R						5 V _{DC} TTL/RS-422
EV1C						24 V _{DC} HTL
NV11	DT/DV71...132 DTE/DVE90...132S	Proximity sensor	Solid shaft	A track	24 V _{DC}	1 pulse/revolution, NO contact
NV21				A+B track		2 pulses/revolution, NO contact
NV12				A track		
NV22				A+B track		
NV16				A track		6 pulses/revolution, NO contact
NV26				A+B track		
AV1Y	CT/CV71...200 DT/DV71...280 DTE/DVE90...225	Absolute encoder	Solid shaft	-	15/24 V _{DC}	MSSI interface and 1 V _{SS} sin/cos
AV1H ³⁾		HIPERFACE® encoder			12 V _{DC}	RS485 interface and 1 V _{SS} sin/cos

1) recommended encoder for operation with MOVITRAC[®] 31C

2) recommended encoder for operation with MOVIDRIVE[®]

3) recommended encoder for operation with MOVIDRIVE[®] compact



- Refer to the following wiring diagrams for information about connecting ES1./ES2./EV1. /EH1. encoders and AV1Y / AV1H absolute encoders:
 - Wiring diagram ES1./ES2./EV1./EH1. encoder: Order number 0918 6832
 - Wiring diagram AV1Y absolute encoder: Order number 0918 6808
 - Wiring diagram AV1H absolute encoder: Order number 1052 9705



Maximum vibration load for encoders $\leq 10g \approx 100 \text{ m/s}^2$

Encoder connection

When connecting the encoders to the inverters, always follow the operating instructions for the relevant inverter!

- Maximum line length (inverter - encoder):
 - 100 m with a capacitance per unit length $\leq 120 \text{ nF/km}$
- Core cross section: $0.20 \dots 0.5 \text{ mm}^2$
- Use a shielded cable with twisted pairs of insulated conductors (exception: cable for HTL sensor) and connect the shield over a large surface area at both ends:
 - to the encoder in the cable screw fitting or in the encoder plug
 - to the inverter on the electronics shield clamp or to the housing of the sub D plug
- Route the encoder cables separately from the power cables, maintaining a gap of at least 200 mm.



6 Startup

6.1 Prerequisites for startup



It is essential to comply with the safety notes in Sec. 2 during startup!

***Before startup,
make sure that***

- the drive is undamaged and not blocked,
- the measures stipulated in the 'Preliminary work' section are performed after extended storage,
- all connections have been made properly,
- the direction of rotation of the motor/geared motor is correct
 - (motor rotating clockwise: U, V, W to L1, L2, L3),
- all protective covers have been fitted correctly,
- all motor protection equipment is active and set for the rated motor current,
- in the case of hoist drives, the self-reengaging manual brake release is used,
- there are no other sources of danger present.

***During startup,
make sure that***

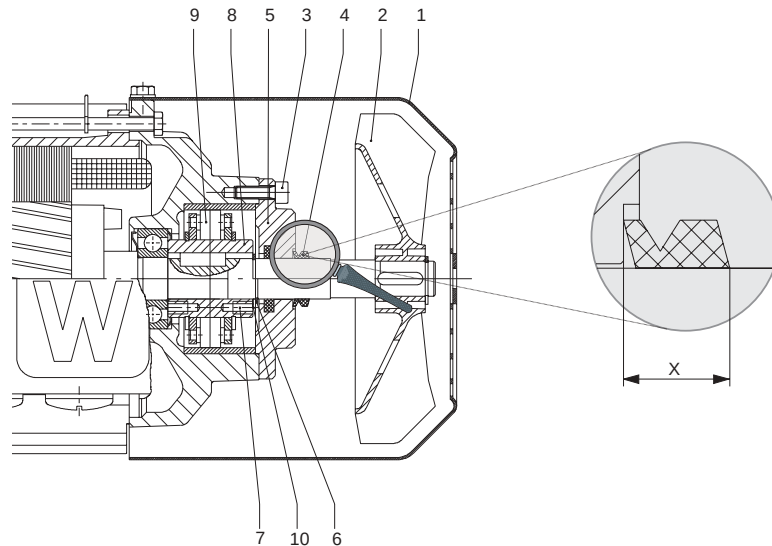
- the motor is running correctly (no overload, no speed fluctuation, no loud noises, etc.),
- the correct braking torque is set according to the specific application (→ Sec. 'Technical data').
- in case of problems (→ Sec. 'Malfunctions').



In brake motors with self-reengaging manual brake release, the manual brake release lever must be removed after startup. A bracket is provided for storing it on the outside of the motor.



6.2 Altering the blocking direction on motors with backstop



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Dimension 'x'
after installation

Motor	Dimension 'x' after installation
DT71/DT80	6.7 mm
DT(E)90/DV(E)100	9.0 mm
DV(E)112/132S	9.0 mm
DV(E)132M - 160M	11.0 mm
DV(E)160L - 225	11.0 mm
DV250 - 280	13.5 mm



Do not start up the motor in the blocking direction (note the phase angle when connecting). Note the direction of rotation of the output shaft and the number of stages when mounting the motor on a gear unit. The backstop can be operated once in the blocking direction at half the motor voltage for checking purposes.

1. **Isolate the motor from supply, safeguarding it against unintentional power-up.**
2. Remove the fan guard (1) and the fan (2); remove the machine screws (3).
3. Remove the V-ring (4) and sealing flange with felt ring (5). (Collect the grease for subsequent use.)
4. Remove the circlip (6) (not with DT71/80); in addition for DV132M-160M: remove the equalizing rings (10).
5. Pull the carrier (8) and wedge element train (9) completely off via the tapped bores (7), turn them by 180° and press them back on.
6. Refill the grease.
7. **Important: Do not exert pressure on or hit the wedge element train – danger of damaging the material!**
8. During the press-in operation – shortly before the wedge element penetrates the locking collar – slowly turn the rotor shaft by hand in the direction of rotation. This allows the wedge element to slide into the locking collar more easily.
9. Install the remaining parts of the backstop by following steps 4. to 2. in reverse order. Note the installation dimension 'x' for the V-ring (4).



7 Malfunctions

7.1 Motor faults

Fault	Possible cause	Remedy
Motor does not start up	Interruption in connecting harness	Check connections, correct if necessary
	Brake does not release	→ Sec. 'Brake faults'
	Fuse blown	Replace fuse
	Motor protection has tripped	Check motor protection is set correctly, rectify any fault
	Motor contactor does not switch, fault in control	Check motor contactor control, rectify any fault
Motor does not start or only with difficulty	Motor designed for delta connection but used in star connection	Correct circuit
	Voltage and frequency deviate markedly from setpoint, at least during switch-on	Provide better power supply system; check cross section of connecting harness
Motor does not start in star connection, only in delta connection	Torque not sufficient in star connection	Switch on directly if delta inrush current is not too great; otherwise use a larger motor or a special version (contact SEW)
	Contact fault on star delta switch	Rectify fault
Incorrect direction of rotation	Motor connected incorrectly	Swap over two phases
Motor hums and has high current consumption	Brake does not release	→ Sec. 'Brake faults'
	Winding defective	Send motor to specialist workshop for repair
	Rotor rubbing	
Fuses blow or motor protection trips immediately	Short circuit in line	Rectify short circuit
	Short circuit in motor	Send motor to specialist workshop for repair
	Lines connected incorrectly	Correct circuit
	Ground fault on motor	Send motor to specialist workshop for repair
Severe speed loss under load	Overload	Perform power measurement, use larger motor or reduce load if necessary
	Voltage drops	Increase cross section of connecting harness
Motor heats up excessively (measure temperature)	Overload	Perform power measurement, use larger motor or reduce load if necessary
	Inadequate cooling	Correct cooling air supply or clear cooling air passages, retrofit forced cooling fan if necessary
	Ambient temperature too high	Adhere to permitted temperature range
	Use delta connection for motor rather than star connection as provided for	Correct circuit
	Loose contact in connecting harness (one phase missing)	Rectify loose contact
	Fuse blown	Look for and rectify cause (see above); replace fuse
	Supply voltage deviates from rated motor voltage by more than 5%. A higher voltage has a particularly unfavorable effect in motors with low-speed winding since in these, the no-load current is already close to the rated current even when the voltage is normal.	Adapt motor to supply voltage
	Rated operating mode (S1 to S10, DIN 57530) exceeded, e.g. due to excessive starting frequency	Adapt rated operating mode of motor to required operating conditions; if necessary call in a specialist to determine what is the correct drive
Excessively loud	Ball bearing compressed, contaminated or damaged	Re-align motor, inspect ball bearing (→ Sec. 'Permitted ball bearing types') and replace if necessary (→ Sec. 'Lubricant table for anti-friction bearings of SEW motors')
	Vibration of rotating parts	Rectify cause, possibly imbalance
	Foreign bodies in cooling air passages	Clean the cooling air passages



7.2 Brake faults

Fault	Possible cause	Remedy
Brake does not release	Incorrect voltage on brake control unit	Apply correct voltage
	Brake control unit failed	Fit a new brake control system, check internal resistance and insulation of brake coil, check switchgear
	Max. permitted working air gap exceeded because brake lining worn down	Measure and set working air gap
	Voltage drop along connecting harness > 10%	Provide for correct connection voltage; check cable cross section
	Inadequate cooling, brake overheats	Replace type BG brake rectifier with type BGE
	Brake coil has interturn fault or short circuit to exposed conductive part	Replace complete brake and brake control system (specialist workshop), check switchgear
Motor does not brake	Working air gap not correct	Measure and set working air gap
	Brake lining worn down	Replace entire brake disk
	Incorrect braking torque	Change the braking torque (→ Sec. 'Technical data') - by the type and number of brake springs - Brake BMG 05: by installing the same brake coil body design as in the BMG 1 brake - Brake BMG 2: by installing the same brake coil body design as in the BMG 4 brake
	BM(G) only: Working air gap so large that setting nuts come into contact	Set the working air gap
	BR03, BM(G) only: Manual brake release device not set correctly	Adjust setting nuts
Brake is applied with time lag	Brake is switched on AC voltage side	Switch on DC and AC voltage sides (e.g. BSR); please refer to wiring diagram
Noises in vicinity of brake	Gearing wear caused by jolting startup	Check project planning
	Pulsating torques due to incorrectly set frequency inverter	Check/correct setting of frequency inverter according to operating instructions

7.3 Faults when operating with a frequency inverter



The symptoms described in Sec. 'Motor faults' may also occur when the motor is operated with a frequency inverter. Please refer to the frequency inverter operating instructions for the significance of the problems which occur and to find information about rectifying the problems.

Customer service

Please provide the following information if you require the assistance of our customer service department:

- Nameplate data (complete)
- Nature and extent of the fault
- Time and peripheral circumstances of the fault
- Presumed cause



8 Inspection and Maintenance



- Only use genuine spare parts in accordance with the valid parts list!
- Always replace the brake control system when you replace the brake coil!
- Motors can become very hot during operation – danger of burns!
- Secure hoist drives or lower them (danger of dropping).
- Isolate the motor and brake from the power supply before starting work, safeguarding them against unintentional power-up!

8.1 Inspection and maintenance periods

Equipment/components	Frequency	What to do
Brake • BMG02 • BR03 • BMG05-8 • BM15-62 • BMG61/122 without encoder mounting	• If used as a working brake: At least every 3000 hours of operation¹⁾	Inspecting the brake • Measure the brake disk thickness • Brake disk, lining • Measure and set working air gap • Pressure plate • Carrier / gearing • Pressure rings
Brake • BMG02 • BR03 • BMG05-8 • BM15-62 • BMG61/122 with encoder mounting	• If used as a holding brake: Every 2 to 4 years, depending on operating conditions ¹⁾	Inspecting the brake • Extract the abraded matter • Except for BMG61/122 with encoder mounting: Inspect the switch elements and change if necessary (e.g. in case of burn-out) • Measure the working air gap • BMG61/122 with encoder mounting: Maintenance by SEW-EURODRIVE only!
Motor	• Every 10,000 hours of operation	Inspect the motor: • Check ball bearings and replace if necessary • Replace the oil seal • Clean the cooling air passages
Motor with backstop		• Change the low-viscosity grease in the backstop
Tacho-generator		• Inspection / maintenance according to corresponding operating instructions
Drive	• Varies (depending on external factors)	• Touch up or renew the surface/anticorrosion coating

1) The periods of wear are affected by many factors and may be short. The system manufacturer must calculate the required inspection/maintenance intervals individually in accordance with the project planning documents (e.g. Drive Engineering - Practical Implementation, Vol. 4).

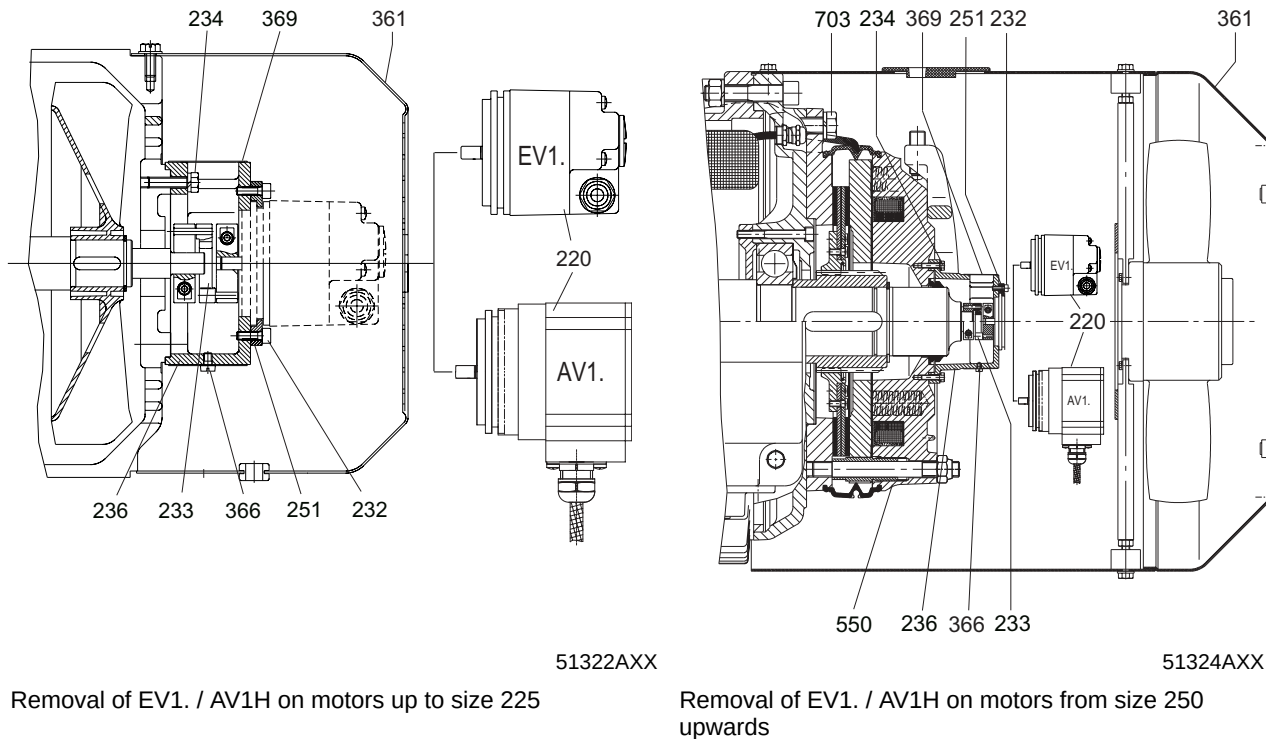


8.2 Preliminary work for motor and brake maintenance



Isolate the motor and brake from the power supply before starting work, safeguarding them against unintentional power-up!

Removing the incremental encoder EV1. / absolute encoder AV1H



Key

220 Encoder	236 Intermediate flange	369 Cover plate
232 Machine screw	251 Conical spring washer	550 Brake
233 Coupling	361 Hood cover / fan guard	703 Hex head screw
234 Hex head screw	366 Machine screw	

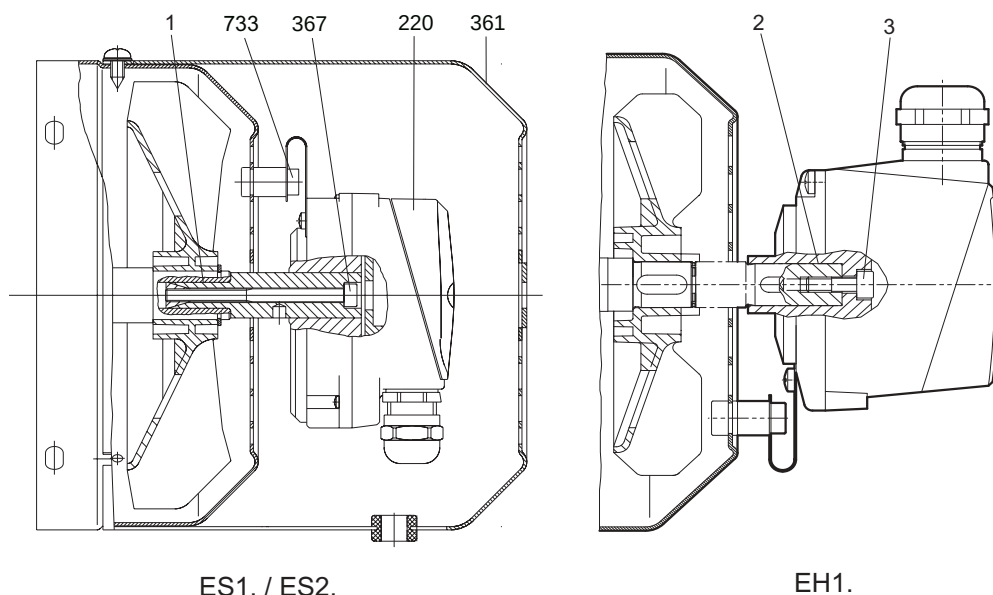
- Remove the hood cover or fan guard (361). If a forced cooling fan is fitted, remove it first.
- Remove the machine screw (366) from the intermediate flange and remove the cover plate (369).
- Unscrew the clamping hub connection of the coupling.
- Loosen machine screws (232) and turn the conical spring washers (251) outwards.
- Remove the encoder (220) together with the coupling (233).
- Lever off the intermediate flange (236) after removing the screws (234).

Note:

During re-assembly, make sure the runout of the shaft end is ≤ 0.05 mm.



Removing the incremental encoder ES1. / ES2.



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Key

220	Encoder	361	Hood cover
367	Retaining screw	733	Retaining screw for the torque arm

ES1. / ES2.

- Remove the hood cover (361).
- Loosen the retaining screws (733) for the torque arm.
- Open the screw cover on the back wall of the encoder (220).
- Unscrew the central retaining screw (367) by about 2 – 3 turns and loosen the cone by tapping lightly on the head of the screw. Then unscrew the retaining screw and pull off the encoder.

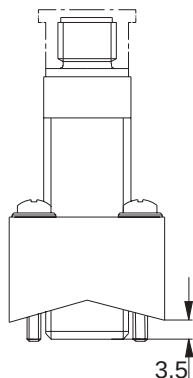
EH1.

- Unscrew the central retaining screws [3] and remove encoder.

Note:

During re-assembly:

- Apply NOCO[®] fluid to the encoder spigot [1] or shaft end [2].
- Tighten the central retaining screw (367) to 2.9 Nm.

***Removing the proximity sensor NV1. / NV2.***

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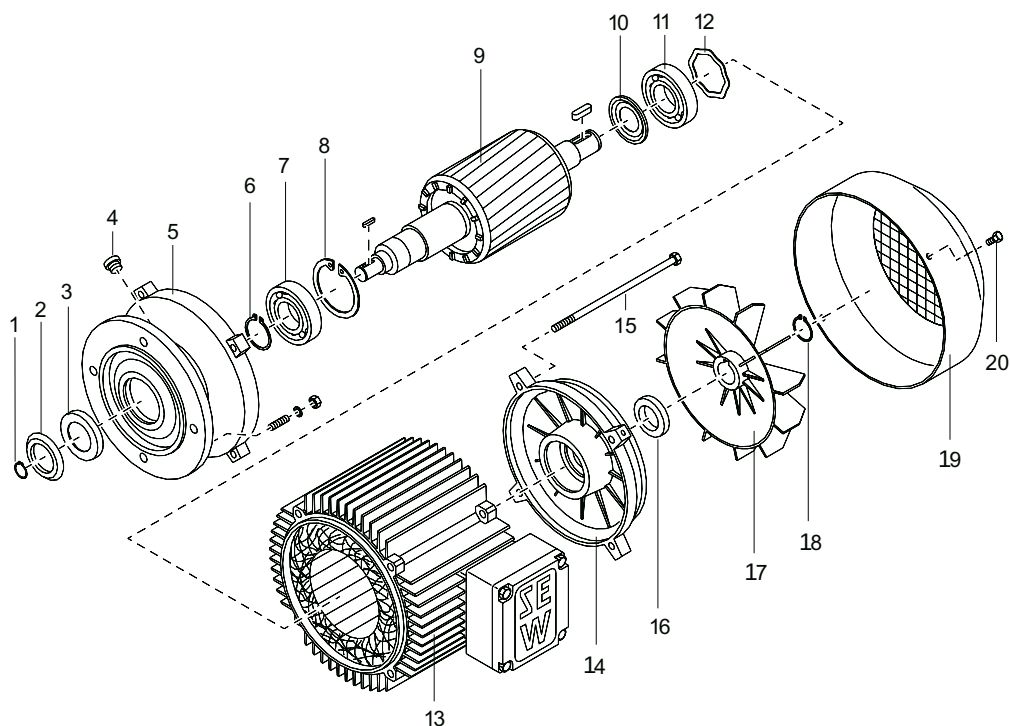
**Important! It is essential for the fan wheel to be at standstill!**

- Disconnect plug
- Pull off the fan guard including NV1. / NV2. Do not tilt it to avoid damaging the proximity switch.
- If the mounting block has been removed from the fan guard or has come loose, it is essential to ensure the following during re-assembly:
The switching surface of the proximity switch must be calibrated to a distance of 3.5 mm from the edge of the prismatic block (→ figure above).



8.3 Inspection and maintenance work on the motor

Example: Motor DFT90



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Key

1 Circlip	8 Circlip	15 Hex head screw
2 Oil flinger	9 Rotor	16 V-ring
3 Oil seal	10 Nilos ring	17 Fan
4 Screw plug	11 Ball bearing	18 Circlip
5 Drive end bearing bracket	12 Equalizing ring	19 Fan guard
6 Circlip	13 Stator	20 Housing screw
7 Ball bearing	14 Non drive end bearing bracket	



Sequence



Isolate the motor and brake from the power supply, safeguarding them against unintentional power-up!

1. Remove the forced cooling fan and encoder, if fitted (→ Sec. 'Preliminary work for motor and brake maintenance').
2. Remove the flange cover or fan guard (19) and the fan (17).
3. Remove the hexagon head cap screws (15) from the drive end bearing bracket (5) and the non-drive end bearing bracket (14), release the stator (13) from the drive end bearing bracket.
4. **Motors with brake BM/BMG:**
 - Open the terminal box cover, remove brake cable from the rectifier.
 - Push the non-drive end bearing bracket and the brake off the stator and carefully lift them off (if necessary, run the brake cable along with trailing wire).
 - Pull the stator back by approx. 3 to 4 cm.
5. **Motors with brake BMG02, BR03:**
 - Remove the complete brake with the releasing lever (on units with manual brake release).
6. Visual check: Are there traces of gear oil or condensation inside the stator?
 - If not, continue with 9.
 - If there is condensation, continue with 7.
 - If there is gear oil, have the motor repaired by a specialist workshop.
7. If there is moisture inside the stator:
 - With geared motors: Remove the motor from the gear unit.
 - With motors without a gear unit: Remove the drive end flange.
 - Remove the rotor (9).
8. Clean the winding, dry it and check it electrically (→ Sec. 'Preliminary work').
9. Replace new ball bearings (7, 11) (use authorized ball bearings only → Sec. 'Permitted ball bearing types').
10. Replace oil seal (3) in the drive end bearing bracket.
11. Reseal the stator seat and grease the V-ring or labyrinth seal (DR63).
12. Install the motor, brake etc.
13. Then check the gear unit (→ gear unit operating instructions).

Lubrication of the backstop

The backstop is supplied with Mobil LBZ low-viscosity grease as lubricant and anticorrosion protection. If you want to use a different grease, make sure it complies with NLGI class 00/000, with a base oil viscosity of 42 mm²/s at 40 °C on a lithium saponified and mineral oil base. The application temperature range is from -50 °C to +90 °C. See the following table for the amount of grease required.

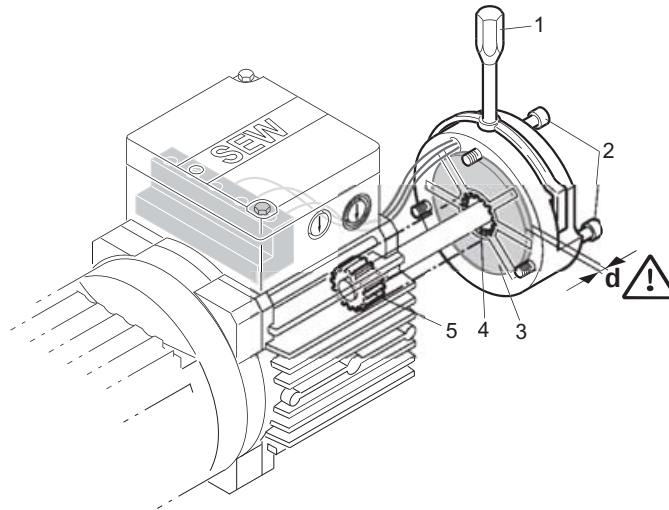
Motor type	71/80	90/100	112/132S	132M-160M	160L...225	250/280
Grease [g]	9	15	15	20	45	80



8.4 Inspection and maintenance of the brake BMG02

Measure the brake disk thickness, replace brake BMG02

The status of the brake disk is determined by measuring the brake disk thickness. Replace the brake BMG02 when the brake disk thickness reaches the minimum value (→ figure below). It is not possible to adjust the working air gap.



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1. **Isolate the motor and brake from the power supply, safeguarding them against unintentional power-up!**
2. Remove the hand lever (1, on units with manual brake release). Remove the fan guard and the fan.
3. Loosen the screws (2) and remove the complete brake with the releasing lever (on units with manual brake release)
4. Measure the thickness 'd' of the brake disk (3):

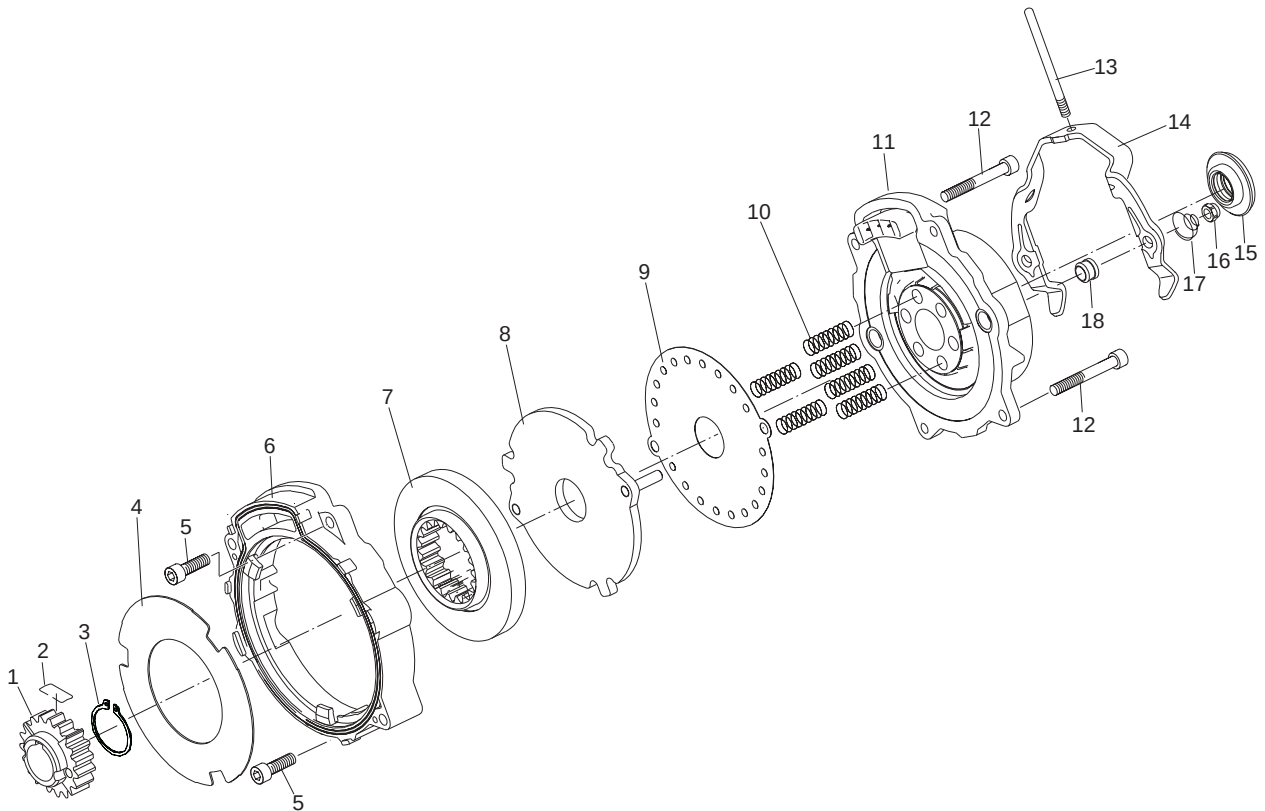
Brake Type	Thickness 'd' of the brake disk [mm]		Max. braking torque [Nm]
	max.	min.	
BMG02	6	5.4	0.8
		5.6	1.2



5. **Replace the complete brake when the brake disk thickness has reached the minimum value.**
6. Mount the complete brake on the motor:
 - Make sure that the gearing of the brake disk (4) engages in the gearing of the carrier (5).
 - Route the electric brake leads through the non drive-end bearing bracket and the inside of the motor into the terminal box.
7. Use bolts (2) to fit the brake back onto the non drive-end bearing bracket.
8. Re-install the fan, fan guard and hand lever (1) (on units with manual brake release).



8.5 Inspection and maintenance of the brake BR03



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Key

1 Carrier	7 Brake disk	13 Hand lever
2 Clip	8 Pressure plate with stud	14 Releasing lever
3 Circlip	9 Damping plate	15 Sealing washer
4 Friction plate	10 Brake springs	16 Lock nut
5 Bolt	11 Brake coil body	17 Conical coil spring
6 Guide ring	12 Bolt	18 Sealing element

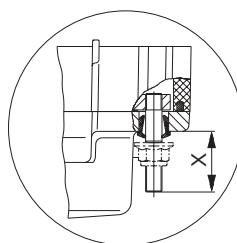


Inspecting brake BR03, measure the working air gap



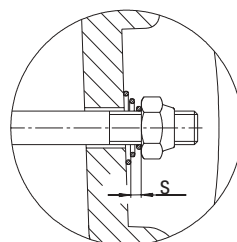
The working air gap cannot be adjusted and can only be measured by means of the stroke of the pressure plate when the brake is released.

1. **Isolate the motor and brake from the power supply, safeguarding them against unintentional power-up!**
2. Remove the hand lever (13, on units with manual brake release). Remove the fan guard and the fan.
3. Remove the lock nuts (16) and, if manual brake release is fitted, remove the conical coil springs (17) and the releasing lever (14).
4. Measure clearance x (→ following figure) with the brake at rest:



50066AXX

- From the end of the stud on the pressure plate (8) to the brake coil body (11).
5. Release the brake electrically.
 6. Measure clearance x with the brake released:
 - From the end of the stud on the pressure plate (8) to the brake coil body (11).
 7. The differential corresponds to the working air gap, i.e. the stroke of the pressure plate (18):
 - If the working air gap ≤ 0.8 mm, reinstall the conical coil springs (17), releasing lever (14) and lock nuts (16).
 - If the working air gap ≥ 0.8 mm, replace the entire brake.
 - Use setting nuts to set the floating clearance 's' between the conical coil springs (pressed flat) and the setting nuts (→ following figure).



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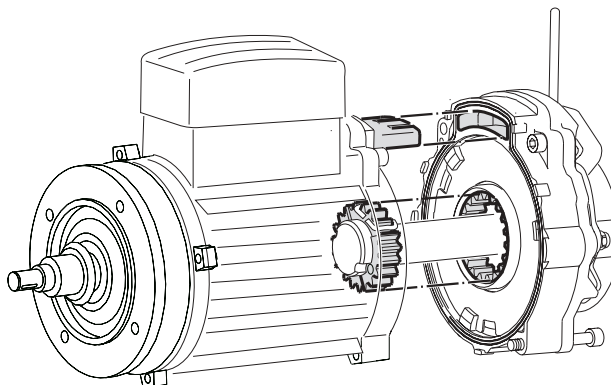
Brake	Floating clearance s [mm]
BR03	2



Important: The floating clearance 's' is necessary so that the pressure plate can move up as the brake lining wears. Otherwise, reliable braking is not guaranteed.



8. Reassemble the removed parts. Connect the new brake (replaced if the working air gap ≥ 0.8 mm) to the motor (→ following figure):
 - Make sure the gearing of the brake disk engages in the gearing of the carrier and that the plug on the motor end fits into the socket on the brake end.



50175AXX

Altering the braking torque BR03

The braking torque can be changed in steps (→ Sec. 'Work done, working air gap, braking torques of brake BR03, BMG05-8')

- by installing different brake springs.
- by changing the number of brake springs.



1. **Isolate the motor and brake from the power supply, safeguarding them against unintentional power-up!**
2. Remove the hand lever (13, on units with manual brake release). Remove the fan guard and the fan.
3. Loosen the bolts (12) and remove the complete brake with the releasing lever (on units with manual brake release).
4. Loosen the screws (5) and remove the guide ring (6) with friction plate (4), brake disk (7), pressure plate (8) and damping plate (9).
5. Remove the brake springs (10) from the brake coil body (11) and replace them by new ones.
6. Position the new brake springs symmetrically.
7. Slide the damping plate (9) over two studs attached to the pressure plate (8) so the embossing pattern is located with the projecting side facing the pressure plate.
8. Pressure plate (8):
 - Place on the brake springs (10) together with the damping plate (9).
 - Guide the studs attached to the pressure plate (8) through the holes in the brake coil body (6) and make sure the pressure plate is in the correct position.
9. Place the flat side of the brake disk (7) on the pressure plate (8).

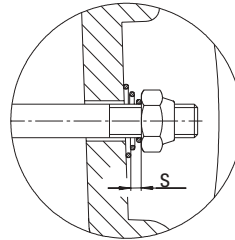
Note: Do not bring the disk into contact with grease or oil!
10. Place the guide ring (6) and friction disk (4) onto the brake disk (7), press down and install the screws (5).





11. Units with manual brake release:

- Put on the conical coil springs (17) and releasing lever (14), install the lock nuts (16).
- With manual brake release: Use setting nuts to set the floating clearance 's' between the conical coil springs (pressed flat) and the setting nuts (→ following figure).



01111BXX

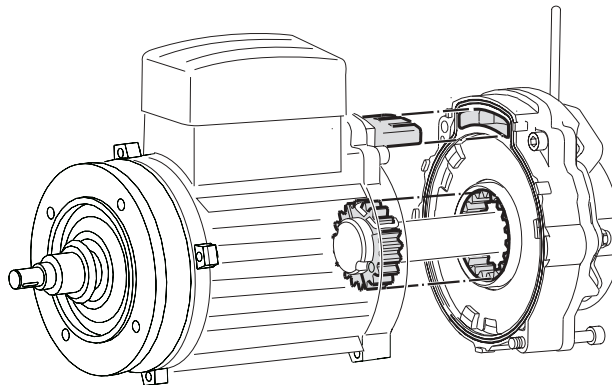
Brake	Floating clearance s [mm]
BR03	2



Important: The floating clearance 's' is necessary so that the pressure plate can move up as the brake lining wears. Otherwise, reliable braking is not guaranteed.

12. Connect the complete brake back onto the motor (→ following figure):

- Make sure the gearing of the brake disk engages in the gearing of the carrier and that the plug on the motor end fits into the socket on the brake end.



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13. Re-install the fan, fan guard and hand lever (10) (on units with manual brake release).

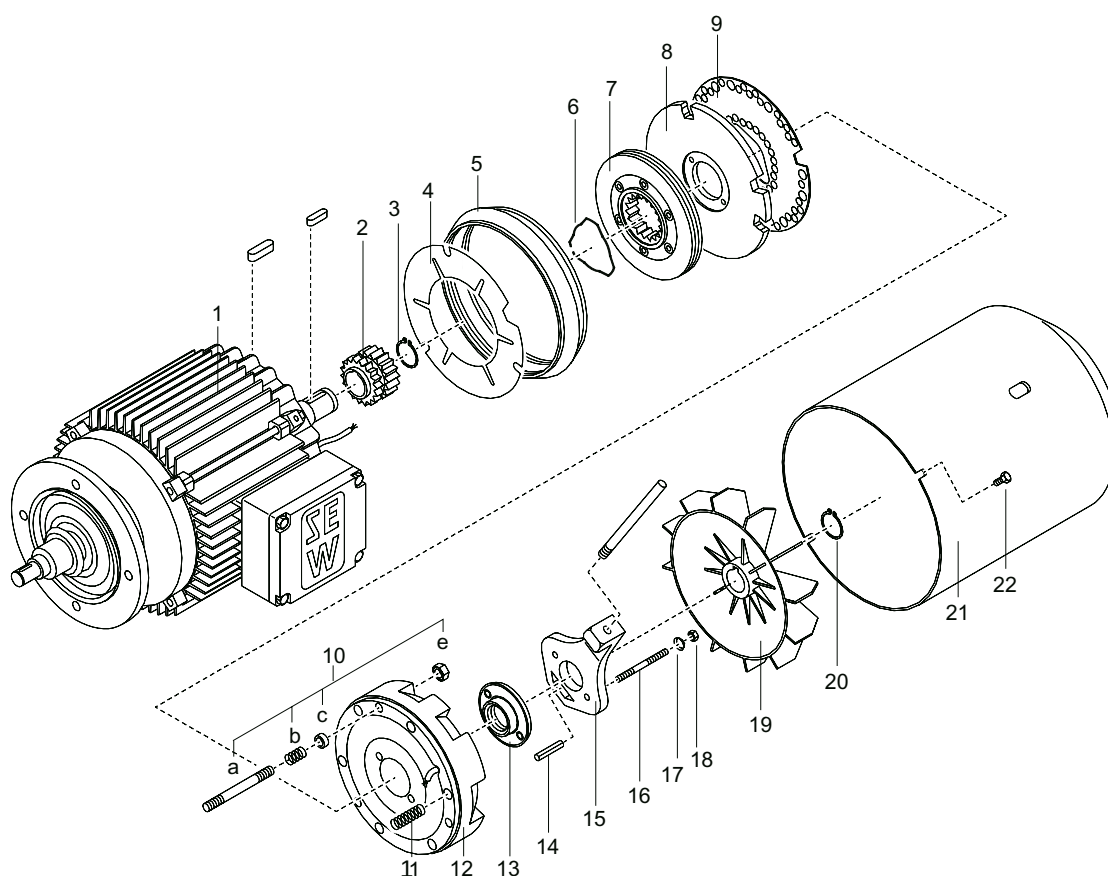


8.6 Inspection and maintenance of the brakes BM15-62, BMG05-122



Brakes BMG61 / BMG122 with encoder mounting are only used as holding brakes.
Maintenance work by SEW-EURODRIVE only.

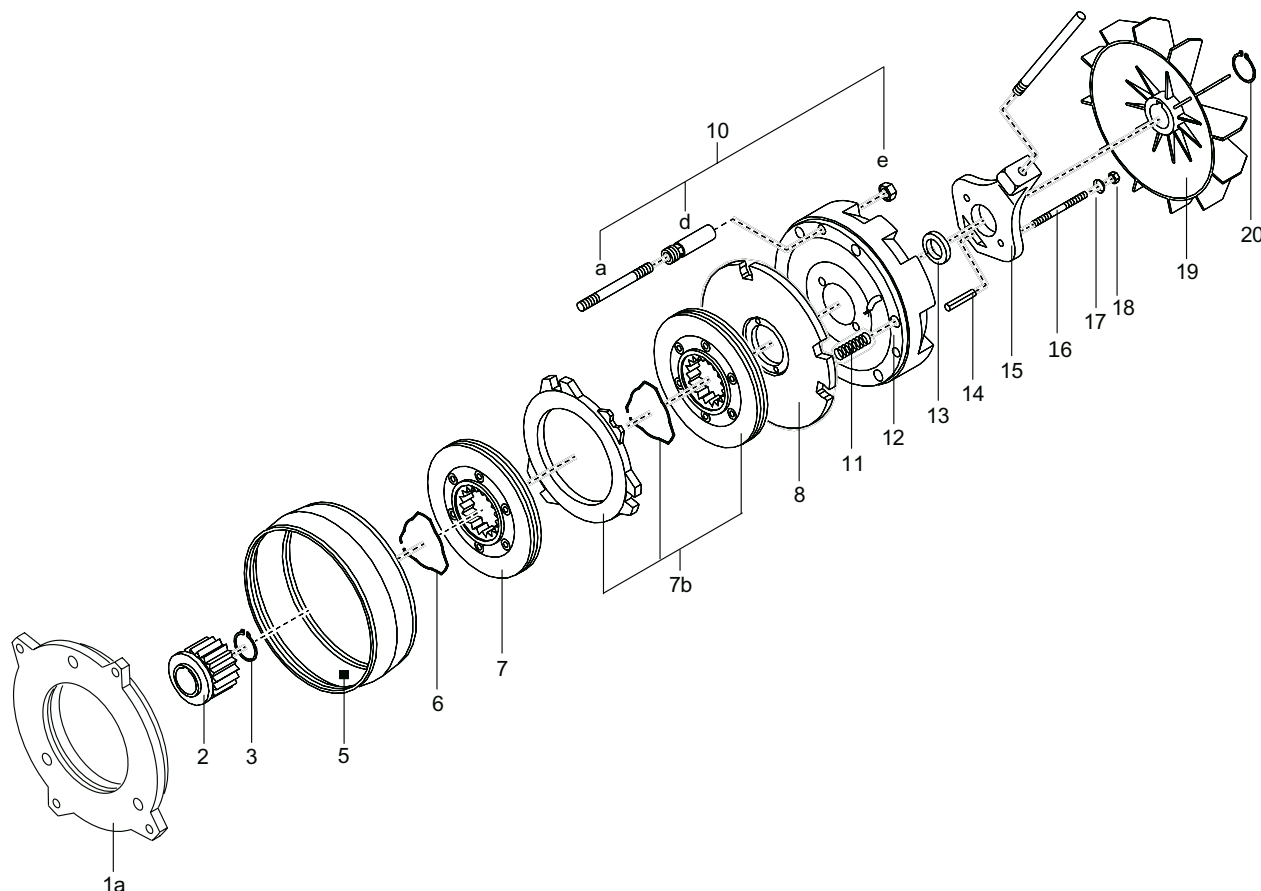
Brake BM(G)05-15



50923AXX



Brakes BM30-62, BMG61/122



50920AXX

Key

- | | | |
|---|----------------------------------|----------------------------------|
| 1 Motor with brake bearing bracket | 8 Pressure plate | 14 Dowel pin |
| 1a BMG61/122 only:
Intermediate flange | 9 Damping plate (BMG only) | 15 Release lever with hand lever |
| 2 Carrier | 10a Stud (3 pcs.) | 16 Stud (2 pcs.) |
| 3 Circlip | 10b Counter spring | 17 Conical coil spring |
| 4 Niro disk (BMG only) | 10c Pressure ring | 18 Setting nut |
| 5 Rubber sealing collar | 10d Setting sleeve | 19 Fan |
| 6 Annular spring | 10e Hex nut | 20 Circlip |
| 7 Brake disk | 11 Brake spring | 21 Fan guard |
| 7b BM 32, 62 only:
Brake stationary disk, annular
spring,
Brake disk | 12 Brake coil body | 22 Housing screw |
| | 13 In BMG: Seal
In BM: V-ring | |



Setting the working air gap in brakes BMG05-8 / BM15-62 / BMG61/122 without encoder mounting



1. Isolate the motor and brake from the power supply, safeguarding them against unintentional power-up!
2. Remove the following:
 - If installed, forced cooling fan, tacho/encoder (→ Sec. 'Preliminary work for motor and brake maintenance').
 - Flange cover or fan guard (21).
3. Push the rubber sealing collar (5) aside:
 - Release the clip to do this, if necessary.
 - Extract the abraded matter.
4. Measure the brake disk (7, 7b):

If the brake disk is

 - ≤ 9 mm on brake motors up to size 100
 - ≤ 10 mm on brake motors up to size 112
 - ≤ 12 mm on brake motors up to size 250

replace the brake disk (→ Sec. 'Changing the brake disk BMG 05-8, BM 15-62').
5. **In BM30-62 / BMG61/122 without encoder mounting:**

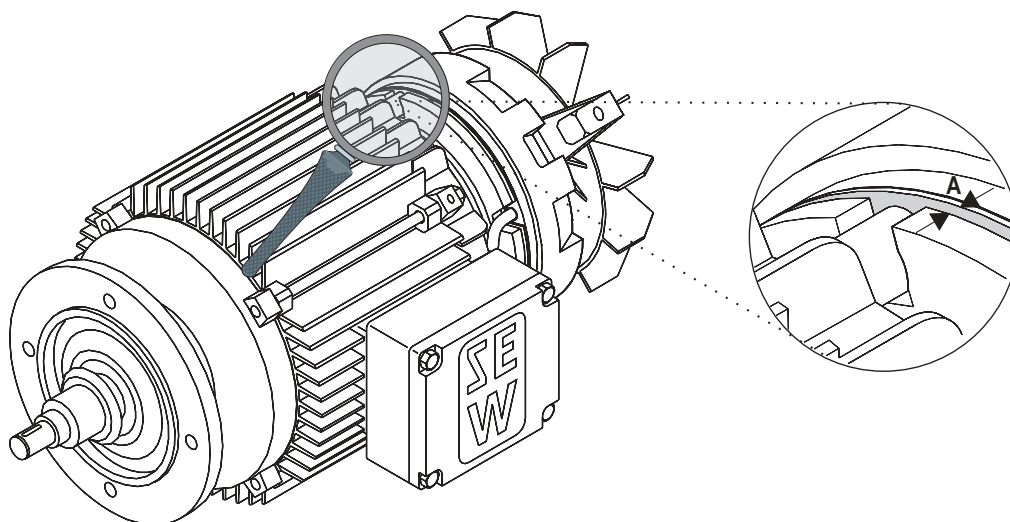
Loosen the setting sleeve (10d) by turning it towards the bearing bracket.
6. Measure the working air gap A (→ following figure)

(use a feeler gauge and measure at three points offset by 120°)

 - In BM, between the pressure plate (8) and the brake coil body (12).
 - With BMG, between the pressure plate (8) and the damping plate (9).
7. Tighten the hex nuts (10e)
 - until the working air gap is set correctly (→ Sec. 'Technical data').
 - in BM 30-62, until the working air gap is initially 0.25 mm.
8. **In BM30-62 / BMG61/122 without encoder mounting:**

Tighten the setting sleeves

 - against the brake coil body.
 - until the working air gap is set correctly (→ Sec. 'Technical Data').
9. Fit the rubber sealing collar back in place and re-install the removed parts.



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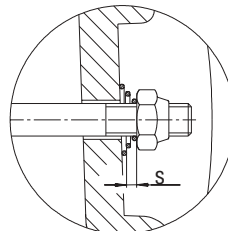


Fitting a new brake disk in brakes BMG05-8 / BM15-62 / BMG61/122 without encoder mounting

When fitting a new brake disk (in BMG05-4 ≤ 9 mm; in BMG62 ≤ 10 mm; in BMG61/122 ≤ 12 mm) inspect the other removed parts as well and fit new ones if necessary.



1. **Isolate the motor and brake from the power supply, safeguarding them against unintentional power-up!**
2. Remove the following:
 - If installed, forced cooling fan, tacho/encoder ($\rightarrow$ Sec. 'Preliminary work for motor and brake maintenance').
 - Flange cover or fan guard (21), circlip (20) and fan (19).
3. Remove the rubber sealing collar (5) and the manual brake release:
 - Setting nuts (18), conical coil springs (17), studs (16), release lever (15), dowel pin (14).
4. Loosen hex nuts (10e), carefully pull off the brake coil body (12) (brake cable!) and take out the brake springs (11).
5. Remove the damping cable (9), pressure plate (8) and brake disk (7, 7b) and clean the brake components.
6. Install new brake disk.
7. Re-install the brake components
 - except for the rubber sealing collar, fan and fan guard, set the working air gap ($\rightarrow$ Sec. 'Inspecting brake BMG 05-8, BM 30-62, setting the working air gap', points 5 to 8).
8. With manual brake release: Use setting nuts to set the floating clearance 's' between the conical coil springs (pressed flat) and the setting nuts ($\rightarrow$ following figure).



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Brake	Floating clearance s [mm]
BMG05-1	1.5
BM15-62, BMG2-8, BMG61/122	2



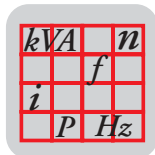
Notes

Important: This floating clearance 's' is necessary so that the pressure plate can move up as the brake lining wears. Otherwise, reliable braking is not guaranteed.

9. Re-install the rubber sealing collar and the removed parts
 - The lockable manual brake release (type HF) is already released if a resistance is encountered when operating the grub screw.
 - The self-reengaging manual brake release (type HR) can be operated with normal hand pressure.



Important: In brake motors with self-reengaging manual brake release, the manual brake release lever must be removed after startup/maintenance! A bracket is provided for storing it on the outside of the motor.



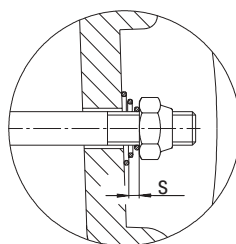
Changing the braking torque of BMG05-122, BM15-62

The braking torque can be changed in steps (→ Sec. 'Technical data')

- by installing different brake springs.
- by changing the number of brake springs.
- by changing the brake coil body:
 - **BMG05:** if the maximum braking torque is not sufficient for the specific application, install the brake coil body (12) of brake BMG1 of the same design in order to ensure safe braking
 - **BMG2:** if the maximum braking torque is not sufficient for the specific application, install the brake coil body (12) of brake BMG4 of the same design in order to ensure safe braking



1. **Isolate the motor and brake from the power supply, safeguarding them against unintentional power-up**
2. Remove the following:
 - If installed, forced cooling fan, tacho/encoder (→ Sec. 'Preliminary work for motor and brake maintenance')
 - Flange cover or fan guard (21), circlip (20) and fan (19)
3. Remove the rubber sealing collar (5) and the manual brake release:
 - Setting nuts (18), conical coil springs (17), studs (16), release lever (15), dowel pin (14).
4. Loosen hex nuts (10e), pull off the coil body (12)
 - by approx. 50 mm (watch the brake cable!).
5. Change or add brake springs (11)
 - Position the brake springs symmetrically.
6. Re-install the brake components
 - except for the rubber sealing collar, fan and fan guard. Set the working air gap (→ Sec. 'Setting the working air gap', points 5 to 8).
7. With manual brake release: Use setting nuts to set the floating clearance 's' between the conical coil springs (pressed flat) and the setting nuts (→ following figure)



01111BXX

Brake	Floating clearance s [mm]
BMG05-1	1.5
BMG2-8, BMG61/122, BM15-62	2

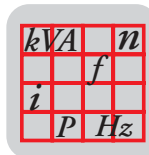
Important: This floating clearance 's' is necessary so that the pressure plate can move up as the brake lining wears. Otherwise, reliable braking is not guaranteed.

8. Re-install the rubber sealing collar and the removed parts.

Replace setting nuts (18) and hexagon nuts (10e) if the removal procedure is repeated!



Note



9 Technical Data

9.1 Work done, braking torques BMG02

Brake type	For motor size	Work done until maintenance [10 ⁶ J]	Thickness of the brake disk [mm]		Braking torque [Nm]
			max.	min.	
BMG02	DT56 ET56	30	6	5.6 5.4	1.2 0.8

9.2 Information for ordering a replacement BMG02

Brake type	Voltage	Braking torque	Brake part number
	[V _{DC}]	[Nm]	
BMG02	24	0.8	0574 319 2
		1.2	0574 323 0
BMG02/HR	24	0.8	0574 327 3
		1.2	0574 331 1

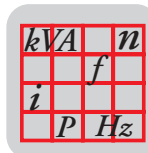
Brake type	Voltage	Braking torque	Brake part number
	[V _{AC}]	[Nm]	
BMG02	230	0.8	0574 320 6
		1.2	0574 324 9
	400	0.8	0574 321 4
		1.2	0574 325 7
	460/500	0.8	0574 322 2
		1.2	0574 326 5
BMG02/HR	230	0.8	0574 328 1
		1.2	0574 332 X
	400	0.8	0574 329 X
		1.2	0574 333 8
	460/500	0.8	0574 330 3
		1.2	0574 334 6



9.3 Work done, working air gap, braking torques BR03, BMG05-8

Brake type	for Motor size	Work done until maintenance [10 ⁶ J]	Working air gap [mm]		Braking torque [Nm]	Braking torque settings Type and no. of springs		Order number of springs	
			min. ¹⁾	max.		Normal	Red	Normal	Red
BR03	63	200	-	0.8	3.2 2.4 1.6 0.8	6 4 3 -	- 2 - 6	185 815 7	185 873 4
BMG05 ²⁾	71	120	0.25	0.6	5.0 4.0 2.5 1.6 1.2	3 2 - - -	- 2 6 4 3	135 017 X	135 018 X
BMG1	80	120			10 7.5 6.0	6 4 3	- 2 3		
BMG2 ³⁾	90 100	260			20 16 10 6.6 5.0	3 2 - - -	- 2 6 4 3	135 150 8	135 151 6
BMG4	100	260			40 30 24	6 4 3	- 2 3		
BMG8	112M 132S	600	0.3	1.2	75 55 45 37 30 19 12.6 9.5	6 4 3 3 2 - - -	- 2 3 - 2 6 4 3	184 845 3	135 570 8

- 1) Please note when checking the working air gap: Parallelism tolerances on the brake disk may give rise to deviations of ± 0.15 mm after a test run.
- 2) BMG05: If the maximum braking torque (5 Nm) is not sufficient, it is possible to install the brake coil body of the BMG1 brake.
- 3) BMG2: If the maximum braking torque (20 Nm) is not sufficient, it is possible to install the brake coil body of the BMG4 brake.

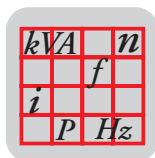


9.4 Work done, working air gap, braking torques BM15 - 62, BMG61/122

Brake type	for Motor size	Work done until maintenance [10 ⁶ J]	Working air gap [mm]		Braking torque [Nm]	Braking torque settings Type and no. of springs		Order number of springs	
			min. ¹⁾	max.		Normal	Red	Normal	Red
BM15	132M, ML 160M	1000	0.3	1.2	150	6	-	184 486 5	184 487 3
					125	4	2		
					100	3	3		
					75	3	-		
					50	-	6		
					35	-	4		
BM30	160L 180	1500	0.3	1.2	25	-	3	136 998 9	136 999 7
					300	8	-		
					250	6	2		
					200	4	4		
					150	4	-		
					125	2	4		
BM31	200 225	1500	0.3	1.2	100	-	8	136 998 9	136 999 7
					75	-	6		
					50	-	4		
					300	4	-		
					250	2	4		
					200	-	8		
BM32 ²⁾	180	1500	0.4	1.2	150	-	6	136 998 9	136 999 7
					100	-	4		
					600	8	-		
					500	6	2		
					400	4	4		
					300	4	-		
BM62 ²⁾	200 225	1500	0.4	1.2	250	2	4	186 838 1	186 839 X
					200	-	8		
					150	-	6		
					100	-	4		
					600	8	-		
					500	6	2		
BMG61	250 280	2500	0.3	1.2	400	4	4	186 838 1	186 839 X
					300	4	-		
					200	-	8		
					1200	8	-		
					1000	6	2		
					800	4	4		
BMG122 ²⁾			0.4	1.2	600	4	-	186 838 1	186 839 X
					400	-	8		

1) Please note when checking the working air gap: Parallelism tolerances on the brake disk may give rise to deviations of ± 0.15 mm after a test run.

2) Double disk brake



9.5 Operating currents

The current values I_H (holding current) specified in the tables are r.m.s. values. Use only r.m.s. instruments for your measurement. The inrush current (accelerator current) I_B only flows for a short time (max. 120 ms) when the brake is released or during voltage dips below 70 % of rated voltage. There is no increased inrush current if the BG brake rectifier is used or if there is a direct DC voltage supply – both are possible with brakes up to size BMG4 only.

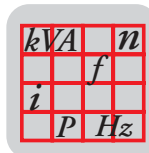
BMG02, BR03 brake

	BMG02	BR03
Motor size	56	63
Max. braking torque [Nm]	1.2	3.2
Braking power [W]	25	25
Inrush current ratio I_B/I_H	-	4

Rated voltage V_N		BMG02		BR03	
V_{AC}	V_{DC}	I_H [A _{AC}]	I_G [A _{DC}]	I_H [A _{AC}]	I_G [A _{DC}]
	24	-	0.72	-	0.72
24 (23-26)	10	-	-	1.5	1.80
42 (40-45)	18	-	-	0.81	1.01
48 (46-50)	20	-	-	0.72	0.90
53 (51-56)	22	-	-	0.64	0.80
60 (57-63)	24	-	-	0.57	0.72
67 (64-70)	27	-	-	0.50	0.64
73 (71-78)	30	-	-	0.45	0.57
85 (79-87)	36	-	-	0.40	0.51
92 (88-98)	40	-	-	0.35	0.45
110 (99-110)	44	-	-	0.31	0.40
120 (111-123)	48	-	-	0.28	0.36
133 (124-138)	54	-	-	0.25	0.32
147 (139-154)	60	-	-	0.22	0.29
160 (155-173)	68	-	-	0.20	0.25
184 (174-193)	75	-	-	0.17	0.23
208 (194-217)	85	-	-	0.16	0.20
230 (218-243)	96	0.14	0.18	0.14	0.18
254 (244-273)	110	-	-	0.12	0.16
290 (274-306)	125	-	-	0.11	0.14
318 (307-343)	140	-	-	0.10	0.13
360 (344-379)	150	-	-	0.09	0.11
400 (380-431)	170	0.08	0.10	0.08	0.10
460 (432-500)	190	0.07	0.09	0.07	0.09

Key

- I_B Accelerator current – brief inrush current
- I_H Holding current r.m.s. value in the connecting harness to the SEW brake rectifier
- I_G Direct current with direct DC voltage supply with rated voltage V_N
- V_N Rated voltage (rated voltage range)



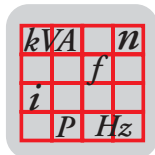
Brake BMG 05 - BMG 4

	BMG05	BMG1	BMG2	BMG4
Motor size	71/80	80	90/100	100
Max. braking torque [Nm]	5	10	20	40
Braking power [W]	32	36	40	50
Inrush current ratio I_B/I_H	4	4	4	4

Rated voltage V_N		BMG05		BMG 1		BMG 2		BMG 4	
V_{AC}	V_{DC}	I_H [A _{AC}]	I_G [A _{DC}]	I_H [A _{AC}]	I_G [A _{DC}]	I_H [A _{AC}]	I_G [A _{DC}]	I_H [A _{AC}]	I_G [A _{DC}]
	24		1.38		1.54		1.77		2.20
24 (23-25)	10	2.0	3.3	2.4	3.7	-	-	-	-
42 (40-46)	18	1.14	1.74	1.37	1.94	1.46	2.25	1.80	2.80
48 (47-52)	20	1.02	1.55	1.22	1.73	1.30	2.00	1.60	2.50
56 (53-58)	24	0.90	1.38	1.09	1.54	1.16	1.77	1.43	2.20
60 (59-66)	27	0.81	1.23	0.97	1.37	1.03	1.58	1.27	2.00
73 (67-73)	30	0.72	1.10	0.86	1.23	0.92	1.41	1.14	1.76
77 (74-82)	33	0.64	0.98	0.77	1.09	0.82	1.25	1.00	1.57
88 (83-92)	36	0.57	0.87	0.69	0.97	0.73	1.12	0.90	1.40
97 (93-104)	40	0.51	0.78	0.61	0.87	0.65	1.00	0.80	1.25
110 (105-116)	48	0.45	0.69	0.54	0.77	0.58	0.90	0.72	1.11
125 (117-131)	52	0.40	0.62	0.48	0.69	0.52	0.80	0.64	1.00
139 (132-147)	60	0.36	0.55	0.43	0.61	0.46	0.70	0.57	0.88
153 (148-164)	66	0.32	0.49	0.39	0.55	0.41	0.63	0.51	0.79
175 (165-185)	72	0.29	0.44	0.34	0.49	0.37	0.56	0.45	0.70
200 (186-207)	80	0.26	0.39	0.31	0.43	0.33	0.50	0.40	0.62
230 (208-233)	96	0.23	0.35	0.27	0.39	0.29	0.44	0.36	0.56
240 (234-261)	110	0.20	0.31	0.24	0.35	0.26	0.40	0.32	0.50
290 (262-293)	117	0.18	0.28	0.22	0.31	0.23	0.35	0.29	0.44
318 (294-329)	125	0.16	0.25	0.19	0.27	0.21	0.31	0.25	0.39
346 (330-369)	147	0.14	0.22	0.17	0.24	0.18	0.28	0.23	0.35
400 (370-414)	167	0.13	0.20	0.15	0.22	0.16	0.25	0.20	0.31
440 (415-464)	185	0.11	0.17	0.14	0.19	0.15	0.22	0.18	0.28
500 (465-522)	208	0.10	0.15	0.12	0.17	0.13	0.20	0.16	0.25

Key

- I_B Accelerator current – brief inrush current
- I_H Holding current r.m.s. value in the connecting harness to the SEW brake rectifier
- I_G Direct current with direct DC voltage supply
- V_N Rated voltage (rated voltage range)


**Brake BMG 8 -
BM 32/62**

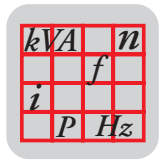
	BMG8	BM 15	BM30/31; BM32/62
Motor size	112/132S	132M-160M	160L-225
Max. braking torque [Nm]	75	150	600
Braking power [W]	65	95	130
Inrush current ratio I_B/I_H	6.3	7.5	8.5

Rated voltage V_N		BMG8	BM 15	BM 30/31; BM 32/62
V_{AC}	V_{DC}	I_H [A _{AC}]	I_H [A _{AC}]	I_H [A _{AC}]
	24	2.77 ¹⁾	4.15 ¹⁾	5.00 ¹⁾
42 (40-46)	-	2.31	3.35	-
48 (47-52)	-	2.10	2.95	-
56 (53-58)	-	1.84	2.65	-
60 (59-66)	-	1.64	2.35	-
73 (67-73)	-	1.46	2.10	-
77 (74-82)	-	1.30	1.87	-
88 (83-92)	-	1.16	1.67	-
97 (93-104)	-	1.04	1.49	-
110 (105-116)	-	0.93	1.32	1.78
125 (117-131)	-	0.82	1.18	1.60
139 (132-147)	-	0.73	1.05	1.43
153 (148-164)	-	0.66	0.94	1.27
175 (165-185)	-	0.59	0.84	1.13
200 (186-207)	-	0.52	0.74	1.00
230 (208-233)	-	0.46	0.66	0.90
240 (234-261)	-	0.41	0.59	0.80
290 (262-293)	-	0.36	0.53	0.71
318 (294-329)	-	0.33	0.47	0.63
346 (330-369)	-	0.29	0.42	0.57
400 (370-414)	-	0.26	0.37	0.50
440 (415-464)	-	0.24	0.33	0.44
500 (465-522)	-	0.20	0.30	0.40

1) Direct current in BSG operation

Key

- I_H Holding current r.m.s. value in the connecting harness to the SEW brake rectifier
- I_B Accelerator current – brief inrush current
- I_G Direct current with direct DC voltage supply
- V_N Rated voltage (rated voltage range)

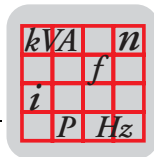

BMG61, BMG122
brake

	BMG61	BMG122
Motor size	250M...280S	
Max. braking torque [Nm]	600	1200
Braking power [W]	200	
Inrush current ratio I_B/I_H	6	

Rated voltage V_N V_{AC}	BMG61/122 I_H [A _{AC}]
208 (194-217)	1.50
230 (218-243)	1.35
254 (244-273)	1.20
290 (274-306)	1.10
318 (307-343)	1.00
360 (344-379)	0.85
400 (380-431)	0.75
460 (432-484)	0.65
500 (485-500)	0.60

Key

- I_B Accelerator current – brief inrush current
 I_H Holding current r.m.s. value in the connecting harness to the SEW brake rectifier
 V_N Rated voltage (rated voltage range)



9.6 Permitted ball bearing types

Motor type	Driving end A bearing (AC motor, brake motor)			Non-driving end B bearing (Foot/flange-mounted, gearmotors)	
	Flange-mounted motor	Gearmotor	Foot-mounted motor	AC motor	Brake motor
DT56	-	6302-Z-J	-	6001-2RS-J	6001-2RS-J
DFR63	6203-Z-J	6303-Z-J	-	6202-2C-J	6202-2RS-J-C3
DT71-DT80	6204-Z-J	6303-Z-J	6204-Z-J	6203-J	6203-RS-J-C3
DT(E)90 - DV(E)100		6306-Z-J		6205-J-2Z	6205-RS-J-C3
DV(E)112 - 132S	6208-Z-J	6307-Z-J	6208-Z-J	6207-J-2Z	6207-RS-J-C3
DV(E)132M - 160M		6309-Z-J-C3		6209-2Z-J-C3	
DV(E)160L - 180L		6312-Z-J-C3		6213-2Z-J-C3	
DV(E)200 - 225		6314-Z-J-C3		6314-Z-J-C3	
DV250 - 280		6316-Z-J-C3		6315-Z-J-C3	

9.7 Lubricant table for anti-friction bearings of SEW motors



The motor anti-friction bearings are factory-filled with the greases listed below. SEW-EURODRIVE recommends filling one third of the cavities between the rolling elements with grease when regreasing.

The bearings with the lubricants from Klüber are designed as sealed bearings 2Z or 2RS..

	Ambient temperature	Manufacturer	Type
Anti-friction bearing in motor	-25 °C ... +80 °C	Esso	Unirex EQ3 ¹⁾
	+80 °C ... +100 °C	Klüber	Barrierta L55/2 ²⁾
	-45 °C ... +60 °C	Klüber	Asonic GHYF2 ²⁾

1) Mineral lubricant (= mineral-based anti-friction bearing grease)

2) Synthetic lubricant (= synthetic anti-friction bearing grease)

10 Revision Status

The following additions and changes have been made to the previous edition of the AC Motors operating instructions (publication number: 1055 3517, edition 10/2002):

- | | |
|--|--|
| General | <ul style="list-style-type: none">• The energy saving motors DTE / DVE and encoder EH1. have been added to the operating instructions. |
| Electrical Installation section | <ul style="list-style-type: none">• The energy saving motor information has been added in the table "Encoder overview." |
| Startup section | <ul style="list-style-type: none">• The energy saving motors information has been added to the table in the paragraph "Altering the block direction on motors with backstop." |
| Inspection/ Maintenance section | <ul style="list-style-type: none">• The paragraphs "Preliminary work for motor and brake maintenance" and "Removing the incremental encoders ES1. / ES2. / EH1." have been revised.• The table "Lubrication of the backstop" now includes motor type 250/280. |
| Technical Data section | <ul style="list-style-type: none">• The table "Permitted ball bearing types" has been revised. |



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Production Assembly Sales Service	Greenville	SEW-EURODRIVE INC. 1295 Old Spartanburg Highway P.O. Box 518 Lyman, S.C. 29365	Tel. +1 (0) 864 4 39 75 37 Fax Sales +1 (0) 864 439-78 30 Fax Manuf. +1 (0) 864 4 39-99 48 Fax Ass. +1 (0) 864 4 39-05 66 Telex 805 550 http://www.seweurodrive.com cslyman@seweurodrive.com



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Assembly Sales Service	San Francisco	SEW-EURODRIVE INC. 30599 San Antonio St. Hayward, California 94544-7101	Tel. +1 (0) 510 4 87-35 60 Fax +1 (0) 510 4 87-63 81 cshayward@seweurodrive.com
	Philadelphia/PA	SEW-EURODRIVE INC. Pureland Ind. Complex 2107 High Hill Road, P.O. Box 481 Bridgeport, New Jersey 08014	Tel. +1 (0) 856 4 67-22 77 Fax +1 (0) 856 4 67-37 92 csbridgeport@seweurodrive.com
	Dayton	SEW-EURODRIVE INC. 2001 West Main Street Troy, Ohio 45373	Tel. +1 (0) 9 37 3 35-00 36 Fax +1 (0) 9 37 4 40-37 99 cstroy@seweurodrive.com
	Dallas	SEW-EURODRIVE INC. 3950 Platinum Way Dallas, Texas 75237	Tel. +1 (0) 214 3 30-48 24 Fax +1 (0) 214 3 30-47 24 csdallas@seweurodrive.com
	Additional addresses for service in the USA provided on request!		
Venezuela			
Assembly Sales Service	Valencia	SEW-EURODRIVE Venezuela S.A. Av. Norte Sur No. 3, Galpon 84-319 Zona Industrial Municipal Norte Valencia, Estado Carabobo	Tel. +58 (0) 241 8 32 98 04 Fax +58 (0) 241 8 38 62 75 sewventas@cantv.net sewfinanzas@cantv.net

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